

Chemistry and Morphology of Polymer Thin Films for Electro-Optical Application

A thesis submitted in fulfilment of the requirements for the degree of
Doctor of philosophy

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Declaration

I certify that except where due acknowledgement has been made, the work is that of the author alone; the work has not been submitted previously, in whole or in part, to qualify for any other academic award; the content of the thesis is the result of work which has been carried out since the official commencement date of the approved research program; and, any editorial work, paid or unpaid, carried out by a third party is acknowledged.

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Darren Simon

/ / 2005

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List of Abbreviation

Å	Angstrom
A	Absorbance
AFM	Atomic force microscope
BA	N-butyl acetate
cm ⁻¹	Wave number
DTA	Differential thermal analysis
DR1	Disperse Red 1
DSC	Differential scanning calorimetry
E-O	Electro-optical
FTIR	Fourier transform infrared
HDPE	high density polyethylene
ITO	Indium-tin-oxide substrate
ITO/glass	Indium-tin-oxide on glass substrate
IBE	Ion beam etching
IR	Infrared
LiNbO ₃	lithium niobate
MW	Molecular weight
n	Refractive index
NIR	Near infrared
NLO	Non-linear optical
PCL	Polycaprolactone
PEEK	Poly(aryl ether ether ketone)
PLA	Poly(L-Lactic acid)
PEG	Poly(ethylene glycol)
PES	Poly(ethylene succinate)
PET	Poly(ethylene terephthalate)
PMMA	Poly(methyl methacrylate)
PC	Polycarbonate
PVME	poly(vinyl methyl ether)
PVP	poly(4-vinylphenol)
Ra	Surface roughness
SBS	Poly(styrene-b-butadiene-b-styrene)

SBR	Styrene-butadiene rubber
SIS	Poly(styrene-b-isoprene-b-styrene)
T _c	Crystallisation temperature
T _b	Boiling temperature
TE	Transverse electric mode
T _g	Glass transition temperature
THF	Tetrahydrofuran
TM	Transverse magnetic mode
T _p	The programmed temperature of DSC
T _m	Melting temperature
T _r	The reference temperature in DSC
T _s	The substance temperature in DSC
T	Transmittance
rms	Root mean square
WAXD	wide angle X-ray diffraction
UU	Urethane-urea copolymer
UV	Ultraviolet
UV15	Epoxy acrylate
XP-2	Stylus XP-2 [Ambios Profilometer]; thickness and roughness measurement technique
ΔH	Enthalpy change
ΔT _s	The difference between the substance and programme temperatures
ΔT _r	The difference between the reference and programme temperatures
λ _{max}	Maximum wavelength

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Summary

Thin polymer films and their properties have received great attention due to the availability of sensitive and accurate polymer characterisation tools, in chemical, physical and thermal property analysis. The characteristics of crystalline polymers according to film thickness have been improved using different crystalline polymers such as polycaprolactone (PCL). The melting enthalpy of PCL has increased when the film thickness decreased and the peak melting temperature showed no significant changes with film thickness. Film thickness variation influenced surface roughness. Crystal size changed with film thickness, thicker films had smaller size crystal and optical microscope images showed the rougher surface of thicker films. The spinning time has shown no influence on film thickness and no significant changes to surface roughness.

Thin films of block copolymers were used in the surface modification study; films studied included poly(styrene-b-butadiene-b-styrene) (SBS) and poly(styrene-b-isoprene-b-styrene) (SIS). Surface modifications of block copolymers have been controlled using different treatment methods. Films of SIS heated at different temperatures have shown different surface texture and roughness. Films treated at low temperature (45 °C) had smooth surfaces when compared with films heated at high temperature (120 °C and 160 °C). Phase separation of heated block copolymer caused bulges of different sizes to cover the surface of SIS films heated at high temperature (120 °C and 160 °C). The height and width of the bulges showed variation with film thickness and heat treatment temperature. Substrate interaction with SBS and SIS block copolymer films showed different surface texture when spun using the same type of substrate whereas films of different texture were obtained when SBS block copolymer solutions were spun onto different substrates. It has been demonstrated that using different

solvents in copolymer preparation caused different texture on the block copolymer surface depending on the solvent evaporation rate.

Amorphous polymers were used to investigate thermal and surface property variations with film thickness. Surface roughness of poly(methyl methacrylate) PMMA and disperse red 1-poly(methyl methacrylate) DR1-PMMA, PMMA has improved using thickness variation. Glass transition temperature measurement of poly(L-Lactic acid) PLA has increased when film thickness was increased..

The glass transition temperature (T_g) and surface roughness of UV15 UV-curable coating polymer has been modified using UV curing and heating methods. T_g variation was observed when curing time and curing intensity were changed causing the optical properties of the polymer to be more variable. The heating caused by the plasma in a plasma etcher caused wrinkles to occur on the surface of unheated UV15 when the polymer was covered with a mask layer the T_g of UV 15 increased when curing time increased. The Fourier Transform Infrared Spectroscopy (FTIR) spectra of cured UV15 film have shown peak variations of the ester and carbon double bond regions over the range of $1850\text{-}1700\text{ cm}^{-1}$.

Urethane-Urea (UU) Polymer thin films were used in this study to investigate optical properties and develop an optical waveguide. Absorption and transmission properties of light using non-linear optical (NLO) polymer such as UU was investigated and used in optical waveguide fabrication. Refractive indices were measured to examine UU films at two different wavelengths using a Metricon prism coupler. A UU film of $1\text{ }\mu\text{m}$ thickness caused a maximum absorption at $\lambda_{\text{max}} = 471\text{ nm}$ and absorption of light was also obtained at 810 nm wavelength. Many methods of fabrication were used; photolithography, plasma etching in a

barrel reactor and thin film deposition using sputtering and evaporation. Etched depths from 1 μm to 100 μm were obtained by variations of the fabrication technique.

An optical waveguide has been prepared using plasma etching of a cured UV15 as a cladding layer on a silicon substrate. The guiding layer was spun onto the cladding layer using PMMA into the etched channels. Smooth interfaces were formed using a barrel etcher.

Chapter 1

1.1 Introduction

Recently there has been increased interest in thin polymer films and the study of their properties. This has arisen due to fundamental interest, and the availability of more sensitive techniques for characterisation and commercial interest in thinner films. Polymers have extensive application as films for packaging, adhesives, and coatings and as passive and active components of optical and electronic systems. In addition to their applications, films are a convenient form in which to measure many properties such as melting temperature, glass transition temperature, melting enthalpy, and to characterise their surface morphology. The polymer could be linear, branched or crosslinked [1]. When a polymer is formed into a thin film on a substrate the properties of the film are influenced, or even controlled by the substrate. This is particularly the case with thin polymer films, considered as being 0.1-16 μm . In a thin film, molecules close to the surface are adsorbed onto the substrate, thereby decreasing their mobility. The loss of molecular (segmental) mobility may increase glass transition temperature and decrease the change in specific heat at the glass transition, ΔC_p . The substrate may delay crystallisation due to surface immobilisation or nucleate crystallisation via an epitaxial mechanism. Previous research has shown that crystallisation is changed when a polymer formed a thin film on a substrate [2, 3].

In this project, techniques for preparing and characterising thin films are investigated. In the preparation of thin films it is important to be able to control the thickness of the films and minimise surface roughness. Spin coating is used to obtain films of different thickness from polymer solutions depending on the solvent evaporation and this must occur without any precipitation. This is a stringent requirement, since the polymer must be completely dissolved

at all concentrations until the last trace of solvent has evaporated. The film thickness must be controlled by spinning speed. The other important issue is the surface uniformity and surface roughness. Two factors control the surface morphology is investigated (i) the drying process and (ii) the crystallisation process [4] and the rates of change of these contributions determine the morphology of the polymer on the surface. Therefore the crystallisation and melting processes modify thin films, and furthermore they depend on substrate surface adsorption and nucleation. The glass transition temperature changes as the film becomes thinner and more molecules are immobilised on the substrate. The influence of film thickness on the glass transition temperature is investigated. Note that for thinner films there is less material to respond to measurements.

The polymers studied are comprised of three groups; the first group is semi-crystalline polymers such as PLA, poly(ethylene succinate) and poly(caprolactone) (PCL). These polymers are all aliphatic polyesters that show slow crystallisation and facilitate study using polarised optical microscopy. The second group is amorphous polymers such as polycarbonate and PMMA and a mixture of disperse red 1 and PMMA. These polymers are glassy so the changes in their glass transition temperature (T_g) are part of this investigation. Films of uniform density, with smooth surfaces and high optical transmission are required so treating the glassy polymer surface is part of the surface modification and surface morphology investigation. Other polymers in this second group are coating polymers such as epoxy resins. Epoxy acrylate-based resins or vinyl esters are used in many industrial applications e.g. as adhesives, dentistry and for their resistance towards water [5]. Finally the third group investigated is block copolymers such as; poly(styrene-*b*-butadiene-*b*-styrene) (SBS) and poly(styrene-*b*-isoprene-*b*-styrene) (SIS). In block copolymers two different types of monomers are joined in the same polymer chain. Block copolymers are thermoplastic; they

are polymerised in a reaction in which there is no termination, and the polymer chains continue to grow until monomer one is consumed, then a second monomer is added to form the second block; and finally a third monomer, usually the same as the first, is added to make block three.

The glass transition temperature is a measure of the temperature above which segmental motions of the polymer chains can occur. This can be considered as a type of segmental Brownian motion. The glass transition temperature of a polymer is investigated when the polymer is a thin film adsorbed onto a substrate [5]. Adsorption on a surface is expected to immobilise the polymer, restricting nucleation, diffusion and segmental motions. In this study the correlation between T_g and film thickness is investigated and the glass transition temperature variation of epoxy acrylate system corresponding to UV curing and baking processes is also investigated.

An emerging application of polymer films (amorphous) is their use in optical waveguide devices and in particular to provide modulation or switching functions in optical communication devices [6]. A modulator requires the velocity of light in the polymer to be changed in response to a change in an applied voltage. Electro optical polymer (E-O) active polymers are those that are used to transmit light such as optical fibres, while their transmission characteristics respond to changes in the polymer density or ordering. The glass transition temperature of the polymer or glassy elasticity may provide the small changes required for a modulator. The glass transition has the characteristics of a second order transition which is a thermal transition that involves a change in heat capacity, but does not have a latent heat. To replace electrical methods of power transformation by photonic methods is achieved using chromophore materials to provide increased non-linear optical

response to the polymers. Polymers can be synthesised or chemically modified to develop various systems for optical characteristic requirements to integrate optical devices [7].

Many second-order non-linear optical (NLO) polymers have been modified to provide electro-optical properties comparable to inorganic NLO materials such as lithium niobate [8]. In order to obtain second-order NLO behaviour, an amorphous centro-symmetric polymer must be exposed to an electric field for poling to provide orientation of the chromophores in the polymer [9, 10]. E-O chromophores can be bonded into the polymer backbone in the main chain or as substituents in side-chain systems [11]. The NLO activity will be enhanced by formation of inter-chain chemical bonds that will decrease the mobility of chromophores in the matrix [12]. Cross-linking will increase the glass transition temperature (crosslinking makes the polymer harder). In many applications using polymers it is necessary to reduce thermal effects. This may happen through crosslinking polymers which are usually shaped before being crosslinked. The polymer obtains its permanent shape by casting from solution, and crosslinked as a film [13]. Thin polymer films are known to exhibit unique thermal, optical and mechanical properties as compared with bulk materials [14]. Studies have shown that the thermal and mechanical properties are changed due to a decrease in the mobility of polymer chains in thin films [15].

The study of crystallinity of thin polymer films is used to determine the property changes during the crystal growth at different thicknesses. Heating the films above their melting temperature and cooling them to room temperature were used to create a thermal history [16]. Crystallisation depends on many factors such as substrate type and the surface properties [17].

1.2 Aim

Preparation of thin films using spin coating techniques from a selection of semicrystalline polymers, amorphous polymers and coating polymers and block copolymers. The films are made to different thickness on different substrates such as glass and silicon wafer. The property variations were investigating corresponding to film thickness and concentration including, surface roughness, morphology, nucleation, crystallisation, endothermic and exothermic transmission.

1.3 Objectives

1. Investigation of crystallisation, crystal size, melting and glass transition temperature of polymers such as PCL and PLA and their properties variation with film thickness.
2. Investigate the surface morphology variation of block copolymer films and (their phase separation) corresponding to film thickness, solvent variation, and substrate and surface temperature treatment variation.
3. Investigation of glass transition temperature variation corresponding to post curing and baking time of epoxy acrylate-based film and its mechanical properties with different film thicknesses including optical polymers to improve their suitability for optical devices.
4. To characterise the surface morphology variation of thin films using different types of polymers such as semi-crystalline, amorphous, coating polymers and block copolymers on various substrates where the macromolecules are absorbed on the substrate and developing methods for the analyses of the data obtained for the topography of the surface.
5. Fabrication of polymers using different methods of curing such as ultraviolet light (UV), heating and etching using ion beam and plasma treatments.

This proposed research raises some questions that are important for meeting the objectives.

The following questions need to be resolved;

- o What are the effects of spin coating process parameters (speed of spinning, time of spinning, and the concentration of the polymer) on the thickness and the surface roughness of polymer films?
- o Will there be any change in the crystal morphology when the thickness of a semicrystalline polymer film is decreased?
- o What will happen to the thermal properties, such as glass transition temperature when film thickness decreases?
- o What are the effects of decreasing the thickness on the mechanical properties of the thin film of the selected polymers?
- o What changes will arise in the optical and dielectric properties of the thin films when the thickness is decreased?
- o How will the optical waveguide properties change with thickness, density and presence of chromophores?

The aim of this project is introduced in this chapter and presented in seven chapters as follows; this chapter covers the introduction to the project. Chapter two presents a review of relevant published literature of interest for this project. Chapter three covers the experimental part of the research and includes the instruments that are used in the research measurements. Chapter four elucidates a study made of semi-crystalline polymer and the investigation of the variation of its properties corresponding to film thickness variation. This polymer was PCL. Chapter five is an investigation of copolymer film properties such as poly(styrene-b-

butadiene-b-styrene) and poly(styrene-b-isoprene-b-styrene). This includes surface morphology variations corresponding to surface treatments including changes of temperature, surface roughness and segregation that occurred according to selection of solvents and substrates. Chapter six describes the investigation of the amorphous polymer properties and their changes corresponding to the surface treatments. Polymers investigated included poly(methyl methacrylate) (PMMA), poly(methyl methacrylate) disperse red 1 (PMMA-DR1). This chapter includes investigation of glass transition temperature variation with film thickness using PLA. Chapter seven covers investigation made on fabrication of waveguide using optical polymer such as PMMA and protected with a polymeric cladding layer. Chapter eight is the conclusion of the thesis and discussion of future investigation.

Chapter 2

Literature Review

2.1 Introduction

Thin polymer films have attracted much interest in recent years. Like other thin film materials, their properties can vary compared with the material's bulk properties. Crystallisation is changed when a polymer is formed into a thin film on a substrate compared with the bulk polymer [2]. Adsorption on a polymer surface can immobilise a polymer, restricting nucleation, diffusion and segmental motions and decreased molecular mobility causes increased glass transition temperature [16]. The progress of analytical techniques for thin polymer films and the different surface properties occurring for films less than 1 μm in thickness compared with bulk material are of interest. Film structure can be controlled by melting, crystallisation and wetting [4]. Crystallinity has been shown to control mechanical properties [17, 18, 19]. Many factors influence the film properties e.g. selection of solvent, polymer solution concentration, evaporation of the solvent, substrate surface energy, heating, processing effects and spin coating speed. It has been shown that crystallisation influences the surface uniformity of polymers [4, 20].

This literature review is arranged in four main sections. The first section covers thin film properties including crystalline and amorphous thermoplastics preparation of polymer thin film from different solutions, spinning or casting films of different thicknesses, surface properties, and interactions with substrates and wetting of surfaces. Section 2 covers review thermoset property variations of the glass transition temperature of coating polymers using curing and heating technique. The third section covers surface morphology variations of copolymer using heating at different temperatures, different solvent for preparation and

different substrates. Finally, section four presents discusses optical polymers and their usage in waveguide fabrication. The main focus is on the film thickness, surface roughness and surface texture variation using different methods of preparation and comparing this variation for different polymers.

2.2 Polymer Thin Films

Polymer films of thickness less than 1 μm are shown to have different properties compared with the bulk polymer such as increase in birefringence with decreasing film thickness in amorphous polymers, the degree of crystallinity and the rate of crystallization were reduced in thin films of poly(di-n-hexylsilane) [3, 21] and large and thicker lamellae of polyethylene (PE) were observed. Uniform films can be deposited on a glass substrate [3] [4] [21]. Generally polymer thin films frequently exhibit change with film thickness. A fundamental understanding of the properties of polymer thin films is critical for the successful use of soft materials in nano-scale technologies.

The melting temperature and the range of temperature over which polymers melt depend on the lamella thickness distribution [22]. The T_g of polymers, in a thin film adsorbed on a substrate surface, may arise from restriction of molecular mobility [23]. Destabilizing long-range intermolecular interactions can affect the lateral uniformity and stability of thin films below a certain thickness [24].

Carnell found that under the conditions of polymer film formation, the reproducibility of the film thickness and the surface uniformity was higher when the thickness of the film decreased to less than 1 μm . The factors involved are the evaporation rate, temperature and concentration of the polymer [25]

2.2.1 Polymer properties

Polymer-based materials are widely used in many advanced technologies such as coating semiconductors and optics. The increasing demands of using polymers in advanced technologies depend on the progress and development of the physical, chemical and mechanical properties of polymers [3]. Thin films are a resource to facilitate surface analysis, which are important in areas such as biocompatibility and adhesion. Semi-crystalline polymers including nylon and isotactic polypropylene exhibit strength and ductility at room temperature and they become brittle at low temperature or high strain rates [26].

Measurements melting and crystallization temperature and mechanical properties characterization allow understanding of the influence of thin film polymer films on their properties that are beneficial for product development using these thin films. Researchers in the field of polymer thin films for optical devices have interests in developing devices such as waveguides using optically transparent polymer layers protected with another polymeric cladding layer [10, 28, 27].

One of the standard techniques for T_g measurement is differential scanning calorimetry. DSC technique is sensitive in dealing with small amounts of polymer, so thin films can be characterised [29]. Some other techniques have been employed for measuring the polymer mechanical properties with highly accurate results including methods such as thermomechanical analysis [30]. DSC can be used to measure actual thermodynamic properties, for example specific heat capacity and enthalpy [31]. Mechanical properties variation of polymer thin film can be compared by measuring the T_g. The T_g of PMMA on silicon dioxide increased by 7 °C above the bulk value for film having thickness of 18 nm [32].

Many studies of polymer thin films showed different interactions between the polymer and the substrate [32]. Polymer thin films play an increasingly important role in a range of technological applications, including coatings, adhesives, and lithography, and organic light emitting diodes, sensors, such as electronic noses and whole cell adhesion sensors, and organic photodiodes [33].

These applications require polymers to meet diverse performance criteria, ranging from electronics, optical adhesive and mechanical performance. For organic light emitting diode applications the electronic and optical properties of the polymer are important whereas for thin film coating and lubrication applications, solution viscosity, structural stability and other mechanical properties are critical [34]. A major advantage of using polymer thin films over non-polymeric materials in some technologies is that polymers are of lower cost and easy to fabricate [35].

2.2.2 Polymer characterization

Characterization of polymers is an important task for researchers when the polymer properties are modified by adsorption on a substrate, thermal annealing or curing. FTIR and DSC are useful techniques for detecting chemical structural changes and microstructures. Optical microscopy or surface profilometry can be used to measure changes in the surface texture after heating block copolymers [36]. When an epoxy resin is cured changes to the chemical structure may occur and these changes can be measured using FTIR. DSC can also be used to measure any changes that occur to the T_g due to curing the epoxy resin or due to absorption onto a substrate.

2.2.3 Polymers modification

Modification of a polymer can be brought about by introducing additives which lead to molecular rearrangement. A nucleation agent is an additive that can control the crystal morphology [37] and the size of spherulites. Anisotropy arises from molecular orientation phenomena [38] that can result from moulding conditions. Anisotropy influences optical properties including birefringence [39, 40].

A polymer blended with other polymers has shown modifications to its crystallisation properties such as PCL blended with poly(vinyl methyl ether) (PVME). DSC shows cold crystallization peaks during reheating of a thin film of such a blend (Figure 2.1, curve 2 and 3). A typical isothermal crystallization for the PCL-PVME blend (Figure 2.1, curve 1) is shown with one melting peak at 56.7 °C since PCL exhibits a higher crystallinity during the isothermal crystallization process. In curve 2, crystallization was slowed in PCL-PVME blends during fast quenching from the melt. The reheating curve shows a cold crystallization peak with an enthalpy approximately equal to that of the following melting peak. The heating scan (curve 3) obtained after annealing the 60:40 blend at 40 °C displays a cold crystallization peak and a larger melting peak suggesting that most of the crystallinity developed during heating the blends at 40 °C [20].

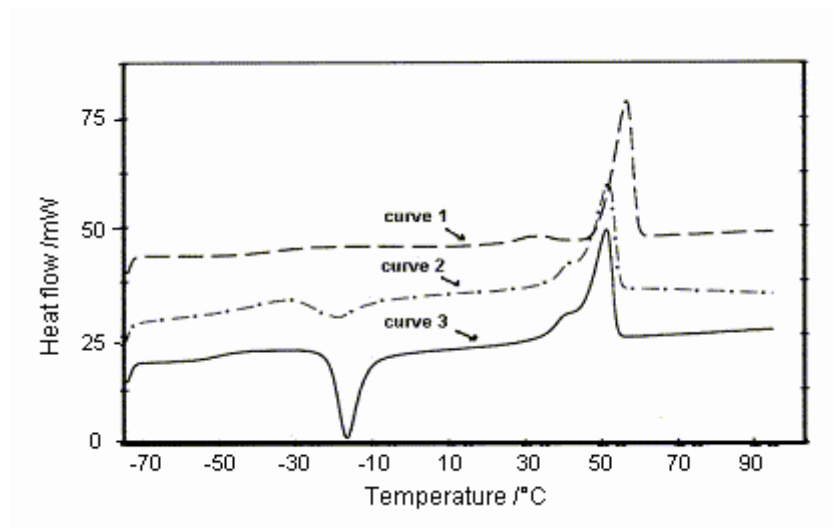


Figure 2. 1: DSC curves of reheating cycles for the PCL/PVME 60/40 blend [20]

Graft copolymers are commonly used as compatibilisers in polymer blends to stabilize the morphology of two immiscible polymers by reducing the interfacial tension and increase the adhesion between their phases. One of the interactions that happen between compatibiliser and immiscible polymers is the chemical structure of the graft copolymers and the polymers in an incompatible polymer blend [41] (for example the T_g of block copolymer materials) varies with composition as shown in Figure 2.2.

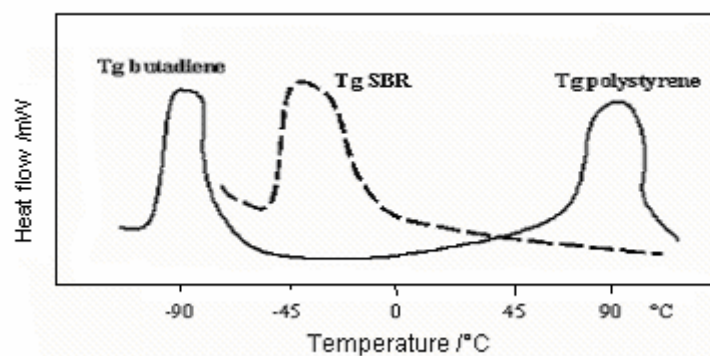


Figure 2.2: Glass transition temperatures (T_g s) of random styrene-butadiene rubber SBR and styrene-butadiene-styrene copolymers (SBS) [42]

Graft and block copolymers exhibit an intrinsic ability to form a range of mesophases. For example, poly(ethylene-co-propylene-co-diene)-graft-polystyrene forms two phase morphology with each phase exhibiting the properties of its consistent grafted polymers. A

block copolymer of styrene and butadiene forms two phase morphology where each phase has T_g corresponding to the homopolymers (polystyrene, semi copolymer and polybutadiene) whereas a random copolymer (SBS) has an intermediate T_g (Figure 2.2) [42].

2.2.4 Thin polymer film variation with temperature

The response of polymer and polymer blend thin films to temperature variations has been studied [4]. Reversible melting of high density polyethylene (HDPE) has been observed where melting-crystalline-melting processes occur in the T_m range [43, 44]. Many films formed at elevated temperatures exhibit cracks and holes [4]. Holes and film rupture have been formed in polystyrene films formed at elevated temperatures [45]. Chapleau et al. have found variations in interfacial tensions with temperature in polymer blend films, between a polyamide (PA) in two HDPE and between a polycarbonate (PC) in a polypropylene (PP) [16] as shown in Figure 2.3.

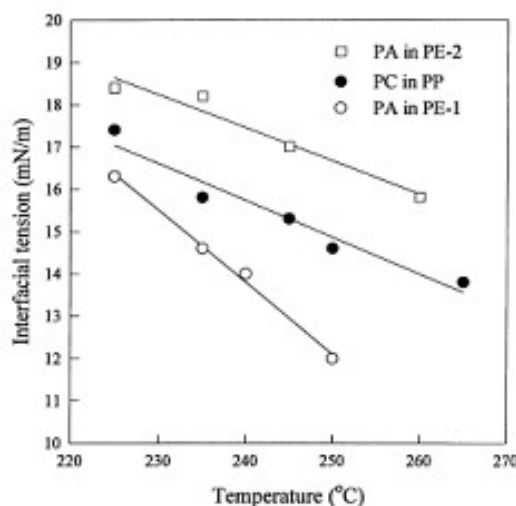


Figure 2. 3: Interfacial tension as a function of temperature for the homopolymers studied [16]

An extensive interest in the T_g has been generated by large shifts in its value for thin films, and these shifts were observed whether the thin films were on substrates or freely standing and for glass forming liquids were confined in nanoscopic pore systems. In practical systems

pure finite size effects are linked to surface and interface effects. The free surface of a supported film may be intrinsically more mobile than the bulk, while strong interactions with a solid substrate may lead to a layer of the molecules of the glass-forming liquid being effectively immobilized. T_g variation occurs when the polymer such as poly(methyl methacrylate) has a strong interaction with a substrate and this variation decreases with weakly interacting substrate [29].

2.2.5 Crystallizing polymers

Blends with poly(styrene-co-acrylonitrile) (SAN) poly(vinyl chloride) (PVC) show slower crystallisation and forms stretching induced crystalline orientation compared with pure PCL under ambient temperature conditions [46]. PCL–PS blends show crystallisation in the PCL phase depending on the thermal treatment. Crystallisation and phase separation phenomena may take place simultaneously, or either crystallisation induced phase separation or phase separation induced crystallisation may dominate. Polymer crystallinity is an important determinant of morphology, modulus, yield strength and toughness [47]. DSC is widely used to characterise crystalline polymers by determining melting characteristics and interpreting crystallisation mechanisms [48]. Different analytical techniques yield different crystallinities such as wide angle X-ray diffraction (WAXD) compared with DSC methods [49]. Kong and Hay have found that the fractional crystallinity of poly(ethylene terephthalate) (PET), has different value to that measured at room temperature using WAXD according to the temperature dependence of crystallisation and fusion enthalpies, when measured from the melting endotherm. Kong and Hay first measured the fractional crystallinity at different temperatures showing no difference in crystallinity and then compared that with results using the WAXD technique, proving that DSC does not measure different averages of crystallinity

since the enthalpies of crystallisation and fusion are temperature dependent as shown in Figure 2.4.

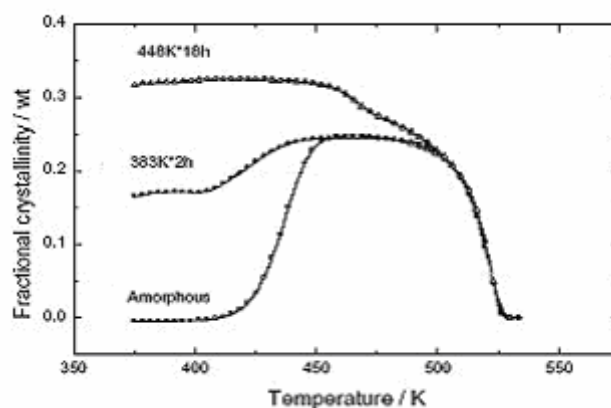


Figure 2. 4: The temperature dependence of the fractional crystallinity of PET on heating amorphous, crystallised at 383 K for 2 h and 448 K for 18 h [49]

Morphology formation of thin films of the semicrystalline PCL, spun on glass substrate, was found to be controlled by two processes, crystallisation and dewetting. The rate of these processes depended on polymer concentration [4]. It was found that the nucleation and crystal growth of HDPE at elevated temperatures over the range 100-180 °C depends on the silicon and gold substrate. Crystal growth was found to depend on both the substrate type and surface properties of the film. For thin films ($<0.5 \mu\text{m}$), the morphology was composed of aggregates on the edge of oriented lamellae, instead of the flattened spherulites observed in thicker films [3].

Crystallisation was found to control PCL film properties on a substrate [4]. There was no crystallisation of PCL in the hybrid system at weight ratios below 40 wt % because of the PCL amorphous state. With increasing PCL weight in hybrid materials the crystallite size of PCL increases [50]. In biaxial drawing of PCL it was found that the degree of crystallinity has no effect on the T_m [4]. PCL has a high rate of crystallisation and its crystals dominate and provide nucleation for other crystals when it is mixed with PES [51]. A copolymer of PES

and PCL showed a decreased degree of crystallinity [51]. It has been found that the blends of PCL/ poly(4-vinylphenol) (PVP) containing less than 50 % of PCL are amorphous and PVP enhances PCL crystallinity [52]. Crystallinity is the main factor that controls the biodegradation rate of PES in block copolymers of PES/PCL. PES 75 mol % in block a copolymer had a higher biodegradation rate than the homopolymer PES [51]. Crystallisation was found to start at the surface of a PET film, with the crystallisation temperature affecting the lamellae and spherulites and there was no evidence of bulk crystallisation occurring when FTIR was used [53]. An addition of a non-crystalline thermosetting polymer of epoxy resin into PCL caused a depression of the crystallisation rate [54]. While DSC allows the study of crystallisation [3, 36, 55], hot stage optical microscopy allows determination of the radial growth rate, such as for growth of PCL spherulites in a blend with poly(vinyl methyl ether). The PCL spherulites decreased with increasing PCL molecular weight [20]. The degradation rate of PCL was found to be controlled by crystallinity when PCL was blended with PES of low molecular weight [51]. Many property changes of polymer films have been found to occur due to crystallinity [37].

Enthalpy is an equilibrium thermodynamic property of low molar mass materials that has been applied to polymers. Measurements of polymer crystallinity from enthalpy change may give inaccurate results [56].

Jonas et al compared the measurement of crystallinity of poly(aryl ether ketone) (PEEK) using DSC and specific gravity [57] and they found that recrystallization formation was occurring during a heating scan of 10 °C min⁻¹, suggesting that DSC was not a suitable technique to measure PEEK crystallinity. [57].

Polymer crystallisation was incomplete due to kinetics rather than thermodynamics, when highly viscous melts were involved. Dynamic chain entanglements may prevent some

molecular segments from crystallising, as observed in crystal morphology and lamellar thickness. The rate of heating may alter the observed enthalpy of fusion and the temperature range over which melting is observed [45], although for low molecular weight materials, crystals melting over a wide range of temperature and the temperature dependence of the enthalpy of fusion cannot be ignored. Differences between the effect of temperature and time on the crystallization of PET was found during the crystallization of amorphous PET using DSC to examine the occurrence of double melting peaks. It was found that the higher temperature peak was due to a recrystallization and melting process during the scan [58].

2.2.6 Glass transition temperature

Amorphous polymers have a disordered molecular arrangement with considerable chain entanglement. A glass transition temperature is exhibited in amorphous and partially crystalline polymers. A polymer at a temperature above its T_g is in the rubbery state, and is flexible and elastic. A polymer below its T_g is in the glassy state, is brittle and is characterised by stiffness, hardness and often high transparency. The T_g can be affected by thermal history and polymer structure, including cross-linking and copolymerization [59].

The T_g is determined by the mobility of polymer chain segments and is related to the degree of freedom available to the molecules. When molecules are adsorbed onto a surface their motion or degrees of freedom will be reduced. Therefore an increase in T_g is expected. More of the molecules will be close to the surface and immobilised in thin film, so an increase in T_g will be observed. As the film becomes thicker the bulk molecules will dominate the observed T_g . An analogous increase in T_g occurs with increase in molecular weight since segmental mobility is decreased when there are less chain ends [60].

The change of the molecular mass of PS produces changes in the dynamics of the co-operative conformational rearrangements of the chain segments due to the change in the concentration of end-groups. The volume element in which the rearrangement takes place without affecting the rest of the material is called a co-operative rearranging region [60].

Elastomers such as polyisoprene and polyisobutylene are soft and flexible (or a rubbery state) above their T_g . Some elastomers like PS and PMMA are glassy below their T_g . The T_g of a freely standing polystyrene film for example, decreases with decreasing film thickness compared with films still on the substrate [61]. T_g variations of thin polymer films have investigated by many researchers. These include the T_g of thin films of polystyrene decreasing for weakly interacting surfaces (glass) [62] and T_g of PMMA increasing on strongly attractive surfaces (silicon) [63]. The T_g of PVC, functionalised by thiophenol, increased with the degree of modification in thin layer geometry when compared to T_g in the bulk [61].

2.2.7 Measuring the glass transition temperature

There has been much research on glass transition temperature in thin polymer films. DSC is a suitable technique to measure temperature, glass transition temperatures, and associated changes in heat capacity. This procedure is usually used to measure the change in heat capacity of a polymer subject to heating, but the glass transition can be measured using a cooling program. Many factors may affect the glass transition temperature of thin polymer films cast on a surface such as energetic interactions with the interface, molecular ordering at the interface, and inherent stresses [63]. These effects are of interest in applications in which polymers are subjected to confined geometries such as when the thin polymer layers are used in electronic components. Beaucage et al have found during measurement of T_g , using the temperature dependence of refractive index that T_g was significantly shifted between heating

and cooling for polystyrene films, and they suggested the shift was due to strains induced in the cooling films [64]. T_g is dependent on heating and cooling rates. The T_g depends on the thermal history of the glass. T_g determined by heating the glass through the transition, may not differ from the T_g from a cooling measurement. T_g of As_2Se_3 and As_2S_3 bulk glasses determined from the thermodynamic component, a step-wise DSC technique, was found to be independent of both the heating or cooling rate, and the thermal history of the glass. T_g depended only on the chemical composition of the glass [65]. A strong correlation was found between the density of PMMA and surface interactions and the T_g at the interface. PMMA with a high interfacial interaction exhibited increased T_g near the interface, whereas PMMA with lower interfacial bonded segments exhibited a T_g depression [61].

Some polymers will have both amorphous and crystalline domains, so the polymer sample can show a melting temperature and a glass transition temperature. An important difference between melting and the glass transition is that when a crystalline polymer is heated at a constant rate, the heat required does not increase at a constant rate [66]. The heat capacity of a material is the heat required for raising the temperature of one gram of the material by one degree Celsius. Uniform heating can be used to uniformly raise the temperature until the material reaches its melting temperature, where the amount of heat required to raise the materials temperature will increase due to the heat required to melt the material. The additional heat is called the latent heat of melting [66].

A crystalline polymer absorbs heat for melting (the latent heat of melting) and it goes through a change in its heat capacity. When an amorphous polymer is heated to the glass transition temperature, no latent heat is required for the transition but there is an increase in heat capacity during the transition [67]. Two plots in Figure 2.5 show the amount of heat added to the polymer on the y-axis and the temperature on the x-axis.

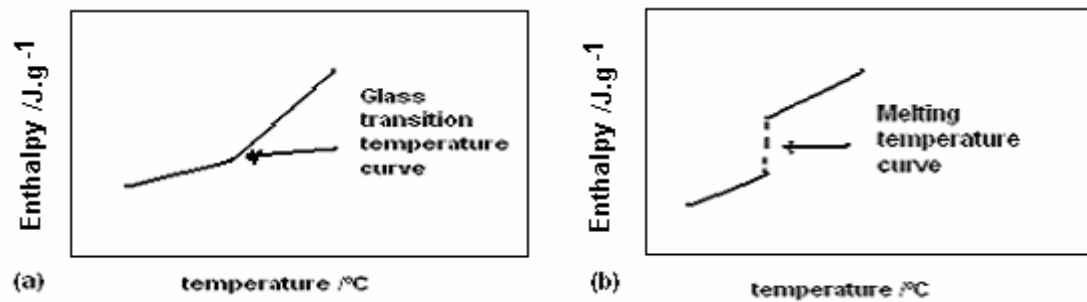


Figure 2. 5: A specific heat versus temperature plots to compare for (a) glass transition temperature of amorphous polymer and (b) melting temperature of non-amorphous polymers [67]

The plot for heating and melting a crystalline polymer shows that heat is added without changing the temperature when the latent heat of melting is involved. The heat capacity can be calculated from the slope of this plot, because the increase in steepness corresponds to the increase in heat capacity above the melting temperature.

The change at the glass transition temperature is an increase in the slope of the heat flow curve, caused by an increase in heat capacity. The heat capacity has changed at the T_g , as the inflection in the curve shows in Figure 2.5(a). On the other hand Figure 2.5(b) shows a shift in enthalpy at the melting temperature of a crystalline polymer [68] [69].

2.2.8 Substrate influence on thin film properties

Thin polymer films on substrates have received attention from researchers, as polymer films are broadly used in several applications such as micro-electronics, printing technology and membrane technology [70]. Investigations have focused mainly on the film rupturing mechanisms that can occur during solvent evaporation. Some studies showed a relation of crystal nucleation growth, decomposition with film thickness, and rupturing mechanism for films of thickness < 100 nm, such as films shown in Figure 2.6 [71].

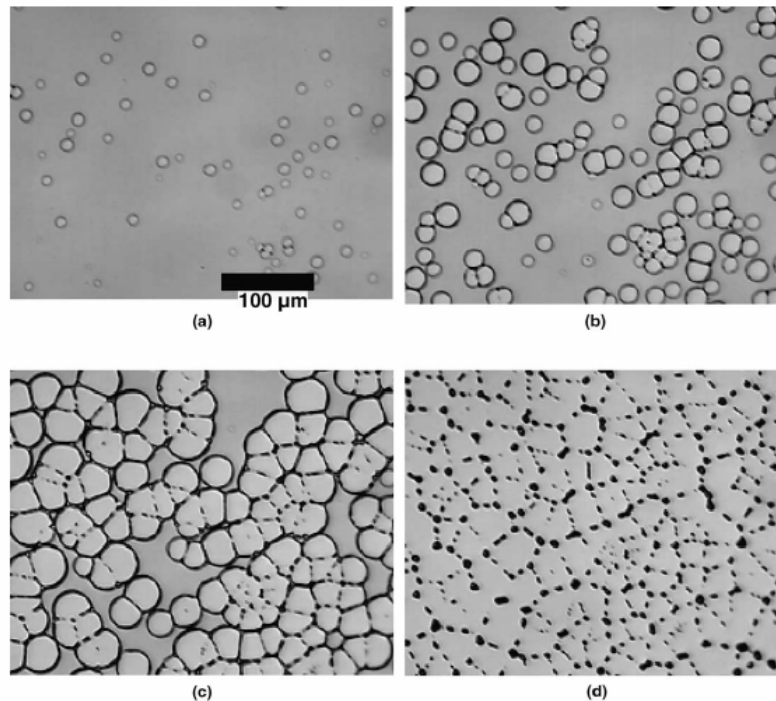


Figure 2. 6: Drying process of polystyrene (PS) on Si, upon heating above the T_g . (a) The film ruptures to tiny dry patches that expose Si substrate to the air. (b) Dry patches expand laterally and (c) coalesce. (d) The polymer mass is accumulated in ridges which result in droplets. PS film thickness is = 17 nm [71].

Some studies tested the growth of dry holes on substrates and many investigations focused on prevention of these holes by avoiding dry surfaces [72, 73, 74]. Winn and Lyons showed that when cadmium telluride CdTe thin films were produced by evaporation onto glass substrates, between 40 °C and 460 °C, the sudden changes of physical properties such as resistivity changes and a clear increase in crystallinity and absorption edge sharpness were noted [75], in contrast to Sarmah and Das [76] who observed a decrease in crystallinity. However, Hayashi et al. [77] reported an increase in the conductivity to a maximum that was followed by a decrease to a minimum, followed by an increase again as the substrate temperature was increased. Winesett et al reported on the effect of different casting surfaces during a development and evolution study of the morphology in as-cast and annealed films of polystyrene blends on silicon, cobalt and gold substrates using scanning transmission X-ray

microscopy. The cast films on different substrates showed changes in the morphologies of polystyrene films as shown in Figure 2.7 [78].

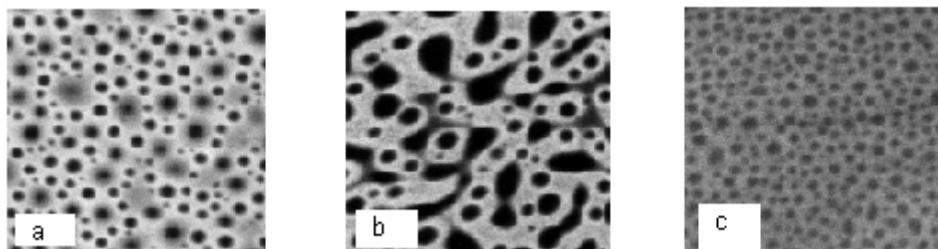


Figure 2. 7: Polystyrene film cast on different substrate annealed at 165 °C, (a) on silicon, (b) on cobalt, (c) on gold [78]

2.3 Thermoset Coating Polymers

Epoxy acrylate is used as coating polymers in electronic devices, such as waveguide fabrication. Epoxy acrylates, such as UV15 provide thermal and dimensional stability, high tensile strength and modulus, exceptional chemical and corrosion resistance, and ease of processing, with wide applications in electronic industries as adhesives [79]. The properties of cured epoxy polymers largely depend on the nature of the chemical structure of the starting resins. Hence, they can be adapted and synthesized through adding moieties into the backbone of molecules for particular requirements. Clive and Adrian found when toughening tetrafunctional epoxy resins using polyetherimide, Young's modulus at 23 °C showed a modest reduction from 3.6 to 3.5 GPa over the same composition range [80]. Some technical efforts have been spent on the development of epoxy resins that improve heat resistance and electric resistance for use in electrical and microelectronic devices. The physical properties of the resulting polymers observed by nKai et al when epoxy resin was synthesised, containing both a naphthalene moiety and a cycloaliphatic group by chemical bonding in the same molecule conducted starting from naphthol and limonene, the results were evaluated using different techniques, such as dynamic thermal mechanical spectroscopy, thermal mechanical analysis, and X-ray diffraction.

A cured polymer of the novel epoxy resin such as bearing naphthyl and limonene moieties showed higher glass transition temperature, lower coefficient of thermal expansion, higher thermal stability, better moisture resistance and dielectric property. The resulting polymers have shown relatively high T_g but lower cracking-resistance when subjected to high temperature and a moist environment, which may restrict their application [81]. The optical activity of all polymeric derivatives has been shown to be much higher than those of monomer. Low molecular weight model compound is highly dependent on their average molecular weight at low values of polymerization degree. [82].

The physical properties of polymeric films mainly depend on coating formulation and type of radiation encountered, such as UV and electron beam (EB). The materials are curable by UV or EB, and they are used in inks, adhesives and coatings oligomers, monomers, photoinitiators, pigments and additives. Thermoset films are produced upon polymerisation and cross-linking under irradiation processes. [83].

2.3.1 UV curable epoxy coating polymer

Epoxy resins are characterised by more than one 1, 2-epoxy groups per molecule, which are the active centres of the resin. They provide a high aromatic content and crosslink sites and give a cured resin with high heat resistance. The glass transition temperature of high performance resins is typically 30–50 °C higher than bisphenol-A epoxy resins. Curing is polymerisation initiated by UV or EB radiation. The curing changes the epoxy from a liquid into solid in less than 1 s using ionising radiation such as UV, visible, IR, microwave and radio frequency. UV curing, however, creates polymers from monomers of low-molecular weight oligomers initially present. [81].

Most epoxy resins are about 90% based on the reaction between bisphenol-A (2, 2-bis(4'-hydroxyphenyl) propane) and epichlorohydrin. The resulting glycidyl terminated monomers can contain one or more bisphenol-A groups, depending on the required applications (Figure 2.8) [84].

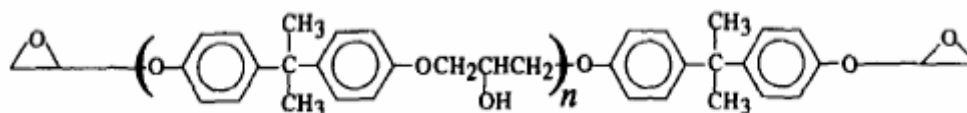


Figure 2. 8: Epoxy resins containing bisphenol-A [84]

Curing can result in a tridimensional network with properties depending on the extent and density of crosslinking. Aromatic amines generally lead to thermosets with higher Tgs, superior thermal resistance and mechanical performances. When aliphatic acids are used to curing agents with a more flexible material then this will result with lower Tg [85].

2.3.2 Etching influence on optical polymer surface

Etching of epoxy resins for optical purposes has been widely utilized for fabrication of electronic and optoelectronic devices; these materials are resistant against wet chemical etchants. Gas systems for dry etching using, Cl- and F-based gases are usually used to obtain high etch rates. Methane-based gas mixtures are neither corrosive nor toxic, but they have low etch rates. They are attractive for controlled damage-free etching for nanostructure formation [86, 87].

The influence of nitrogen on methane-based ECR plasma etching of GaN has been reported and roughness of 2.6 nm was obtained when etched by Ar gas without N₂. The etch rate and roughness decreased with an increase of N₂ gas flow. Zhi et al suggested that the decrease of etch rate was due to the change in plasma conditions and Ar-ion sputtering effects. The

addition of N_2 gas flow resulted in a smoother etched surface. The loss of nitrogen from the surface can be compensated when the N_2 gas was added. The suppression of loss of N (by introducing a higher N gas flow) is the main reason for improvement (reduced roughness) of the etched surface morphology as shown in Figure 2.9 [88].

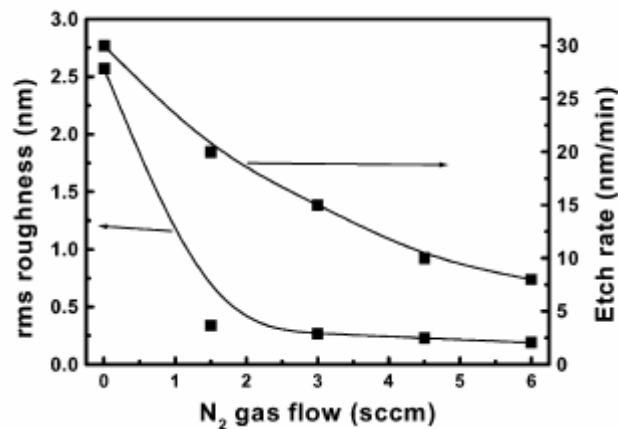


Figure 2. 9: Etch rate and rms roughness as a function of N_2 gas flow [88]

Effects of the bias power on the surface roughness of silicon carbide (SiC) films, etched in a C_2F_6 inductively coupled plasma has been reported. The experimental data showed that the sensitivity of the surface roughness to the variations in the bias power was dependent on the plasma conditions. The surface roughness was correlated to the DC bias. At low pressure, the stability was linked to the variations in the source power as shown in Figure 2.10 and 2.11, [89].

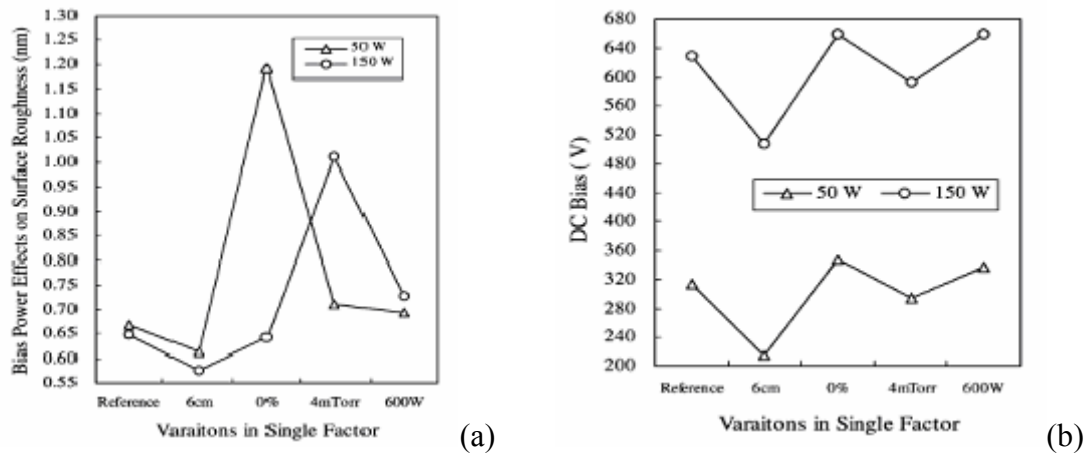


Figure 2. 10: Surface roughness as a function of the bias power at (a) various plasma conditions. The reference plasma was set to 900 W source power, 16 m torr pressure, 80 % O₂ fraction, and 12 cm gap. (b) DC bias as a function of process parameters at fixed bias power. [89]

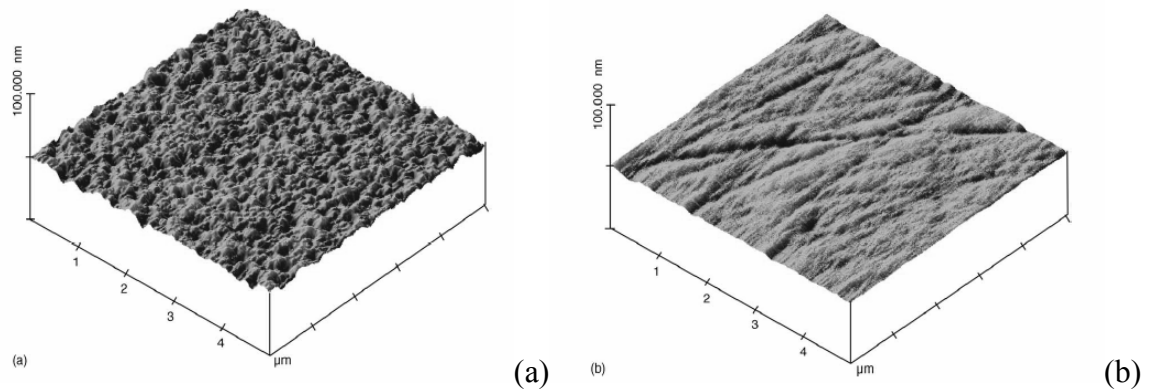


Figure 2. 11: AFM image taken at: (a) 900 W source power, 50 W bias power, 16 m torr pressure, 0 % O₂ fraction, and 12 cm gap; (b) 900 W source power, 150 W bias power, 16 m torr pressure, 0 % O₂ fraction, and 12 cm gap[89]

Dry plasma etching is more suitable than wet chemical etching for forming well defined trenches in the UV15 polymer. Plasma can consist of both energetic ions (used for mechanical etching) and reactive free radicals (used for reactive etching). When the etch process is only by reactive etching much smoother polymer surfaces can be obtained [90].

Reactive ion etching (RIE) is the most widely used plasma etching system in the semiconductor industry and etching polymers for waveguides has been reported by Cheng et al [91]. RIE has a significant component of high-energy ions bombarding the polymer surface

but typically has a more significant reactive etching component. Energetic ions can cause increased surface roughness on the sample surface. De Ridder et al [6] suggest that the waveguide layer should not be directly etched to form a rib in order to prevent damage. As well as the damage caused by very energetic ions, another disadvantage of the Ion Beam Etching (IBE) system is a low throughput per etching and long period process compared with the Barrel etcher (low energy ions and mainly etches by reactive etching) because; a high vacuum pumping stage is required for the IBE system. The Barrel etcher works at higher pressure (e.g. 0.5 torr) and only a roughing pump is required [92].

2.3.3 Cure rate and dry etch patterning of thermoset polymers

The fabrication methods associated with integrated optical circuits and components are varied and they depend on the substrate with which the optical device is fabricated. Among the materials suitable for integrated photonic technology, polymers occupy a special position as they exhibit some useful physical properties electro-optic, piezo-electric and non-electric effects Electro-optic coefficients can be even higher than LiNbO_3 (lithium niobate) crystals [93], the dominant electro-optic device material. Polymers are advantageous because they can be formed as thin films, typically by spin coating, on any substrate. Other materials that have desired properties for optical devices and other micro fabricated technologies are not as convenient to use because they cannot be conveniently formed as thin films, e.g. complete substrates of lithium niobate have to be used to realize devices made from such materials [94].

Thermoset polymers are convenient to use, being a one component material supplied as a liquid and cured using UV radiation. The thermoset polymer used in this study was Master Bond UV15, which is a cycloaliphatic type epoxy that uses effective cationic catalyst curing [95]. The polymerisation process is photoinitiated and fastest cures are obtained with light sources emitted at wavelengths in the 250 – 350 nm UV range. Further UV radiation and

thermal post-curing treatment increases the materials chemical resistance, as well as flexibility to residual stresses and strains. UV cationic curing is recognized for its outstanding adhesion to substrates; the absence of oxygen inhibition; low shrinkage on curing; and high chemical and solvent resistance when combined with post curing. The photoinitiators used in cationic curing are typically aryl sulfonium salts or iodonium salts [96]. Both are used with a variety of counter ions. On irradiation with UV light these photoinitiators generate strong acids that cause a rapid ring-opening whereby cycloaliphatic epoxies crosslink with each other and with hydroxyl compounds, if there are present. Epoxide resin systems cured by a cationic mechanism continue to propagate after the removal of the radiation source (commonly referred to as dark cure) as a result a more thorough cure is achieved. Furthermore, post UV cure heating will cause additional polymerization for systems exposed to low levels of UV [97].

The cationic initiation step is begun by incident light radiation, and to a slighter extent, by heat. The chain reaction is typically endothermic, and is therefore accelerated by addition of heat to the reaction mixture. Chain transfer and termination steps are promoted by heat addition; however the termination step is a function of the nucleophilicity of the anionic species [97].

A technique for quantifying the polymerization of a polymer is to measure the cross-link density with FTIR [98]. FTIR produces an infrared absorption spectrum, so that the chemical bond vibrations and deformations can be identified. As well as FTIR, DSC can be used to measure the T_g [99] by measuring the T_g, a qualitative assessment can be made of the materials hardness. Many different paths can be taken when etching a thermoset polymer, simply by controlling the variables in each phase of the fabrication process (Figure 2.12).

Little is known about how the path affects the quality of etching, and the relationship between the cross-link density of the polymer with the quality and depth of the etch rate [100].

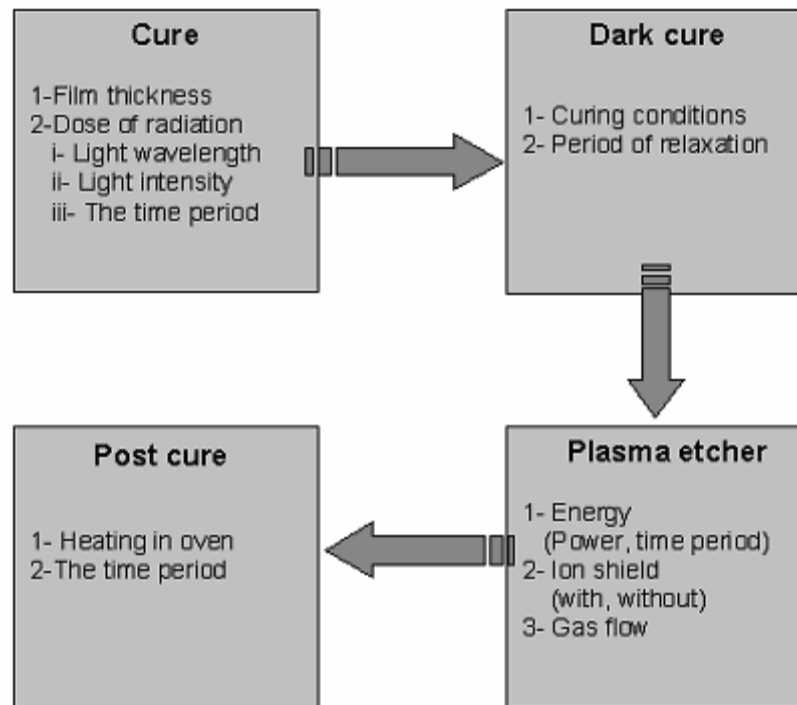


Figure 2. 12: Figure 1: Fabrication Process for the patterning of a thermoset polymer [101]

2.4 Surface Morphology

It is well known that many physical properties of polymers such as wettability, adhesion and their modification by chemical reactions depend on surface properties. [102]. The existence of cracks, pores and holes, which determine surface roughness of polymer films for example, have an enormous influence on the diffusion of oxygen or reactive particles such as radicals into the polymer matrix, so they play a crucial role in photo-oxidative degradation of plastics [103] and reactive plasma etching also.

Using the grazing-incidence small-angle scattering (GISAXS) technique for morphology investigation of polymer films spin-coated on different silicon substrates, Pietsch et al have found that the morphology of relatively thick polymer films is controlled by the surface

morphology of the substrate and when the substrates are covered with a metallic film, the surface shows a rough structure. The appearance of small metal clusters after evaporation is known to occur [104]. Pietsch et al linked the surface morphology that was adopted by the polymer, to density inhomogeneities during spin-coating. On pure silicon, the films appear homogeneous and flat as the substrate. Heating the polymer above T_g caused the segment mobility to be much increased and can homogenize the polymer density [105].

Toa and et al have found, in their study on the evolution of the surface morphology with annealing time for thin films of poly(styrene-*b*-2-ferrocenylethyl methacrylate) prepared from different solvents on silicon substrate that the influence of the solvent on the film surface morphology continues even after 24 hours of annealing. Films spin coated from tetrahydrofuran (THF) solution, showed a rough surface and lower adhesion to the substrate and films that were prepared from toluene and xylene showed a smooth surface before and after annealing. The conclusion that Toa and co-workers made on surface roughness of films prepared from the THF solution was attributed to the higher vapour pressure of THF [106]. It was also found that the substrate surface cleaning procedure had significant influence on the morphology and the number of defects in the films [107].

Many researchers have studied the dewetting of polymer thin films, since thin polymer films that have a high level of stability are needed in many applications of technology such as microelectronics, coatings, and adhesives [108].

The stability of the coating depends on (i) the surface energy of the substrate, (ii) coated-substrate interfacial energy and on (iii) the film thickness. For example, films less than $0.1\mu\text{m}$ thick allow van der Waals forces to dominate, causing rupture to the polymer film. Because of the importance to many technological areas, it is necessary to investigate the factors influencing the wetting and dewetting of polymer thin films [109].

In many applications such as electronics and optical technology, polymer thin film surface morphology has influenced the optical, mechanical, thermal, or electrical characteristics. The roughness of these films is often crucial to their applications, and in particular, roughness influences optical applications [110].

Surface roughness can be controlled by variation of the solvent evaporation rate; using solvents of lower boiling temperature results in higher surface roughness [111].

Phase separation in blended polymers has influenced surface morphology of polymers, which then affected the mechanical and electrical properties of a film [112]. The initial morphology of spin coated thin polymer films depends on factors such as the blend composition, film thickness, solvent variation and annealing temperature [113-116].

2.4.1 Copolymer

Polymer mixtures are called copolymers when they are made from more than one kind of monomer. When a polymer is made by linking only one type of small molecule, or monomer, together, it is called a homopolymer. When two different types of monomers are joined in the same polymer chain, the polymer is called a copolymer. If two monomers are arranged in alternating ways, this arrangement is called an alternating copolymer such as, for styrene and butadiene, S-B-S-B-. When the arrangement is a random copolymer, the two monomers may follow in any order such as, -S-S-B-B-S-B-S-S-B-. In a block copolymer, all of one type of monomer is grouped together, and all of the other is grouped together. A block copolymer can be two homopolymers joined together at the ends -S-S-S-B-B-B- [117, 118, 119].

2.4.2 Block copolymer

Block copolymers are excellent self-assembling materials for many reasons such as specific control over microstructure length scale. Block copolymer microstructures have domain dimensions that can be varied continuously up to a few tens of nanometres, through molecular weight variation. Block copolymers have recently received much attention because of their ability to be scaled in microstructures in the range of tens of nanometres. Their various chemical and physical properties such as etching rates, convenient size and shape of microdomains occur simply by changing their molecular weights and compositions [120]. Self-assembling of block copolymers is used to control block copolymer morphology. Different equilibrium symmetries of block copolymer morphology have been reported; such as lamellae, hexagonal cylinders, bicontinuous gyroid, and body-centered cubic arrays of spherical micelles [121].

Surface morphology is adjusted by carefully choosing a right selection of mechanisms and conditions to achieve a desired microstructure and this achievement can be important in practical applications employing block copolymers and their ability to form desired microstructures so that they can undergo operating conditions other than the equilibrium conditions [122].

Addition of a homopolymer or solvent to a block copolymer provides extra degrees of freedom for controlling morphology and properties. Solvents are already present in many applications employing self-ordered block copolymers, either in the final product or as an integral part of processing. Solvents can facilitate the fixation of a desired or metastable microstructure through methods such as solvent-phase polymerization crosslinking or rapid solvent evaporation [123].

In the bulk state the selection of morphology is mainly by composition, however temperature and changes in architecture can be used to modulate the equilibrium structure. Control over domain functionality and properties through advances in synthetic techniques have allowed freedom in selecting the polymers for each block [117]. Quantitative prediction of the structures can provide accurate composition profiles for the range of ordered states of polymeric materials [124].

The main ideas dealing with block copolymers are concerned with those that are composed of two or more chemically distinct blocks joined together end-to-end. [37] [125]. The surface nature of the polymer can be modified either by absorbing a different polymer or by a chemical reaction. Surface modification depends on the occurrence of surface segregation such as functional groups or polymer chains with low surface energy affecting free energy on the surface [126]

Low surface energy will cause segregation of non-functional groups. It has been found that segregation on the surface varies directly with end-functional groups or indirectly with end-functional groups. End-functional polymers are a class of chemically heterogeneous materials. End-functional polymers are convenient as a model system that reflects the general behaviour of all functional polymers and suitable for modifying surface properties without causing substantial changes in the bulk properties [126]. Different techniques such as heating the copolymers and using different solvent have been used in studies of the copolymer morphology such as nanometer scale patterning that is based on self-assembly. These patterning techniques have been considered as alternatives to replace high resolution lithographic technologies using X-ray, electron beam and interference lithography to enable a variety of nanotechnologies. [127]. Measurements of phase shifts using tapping-mode AFM have shown images exhibiting a high contrast [128], Magonov et al have found that the tip

sample force interaction can be probed by performing AFM measurements at different tip-sample forces using thin films of a styrene butadiene styrene (SBS) triblock copolymer. Figure 2.13 shows the effect of tapping mode imaging at different forces of probing and the contrast is due to the nanoscale microstructures in the film [129].

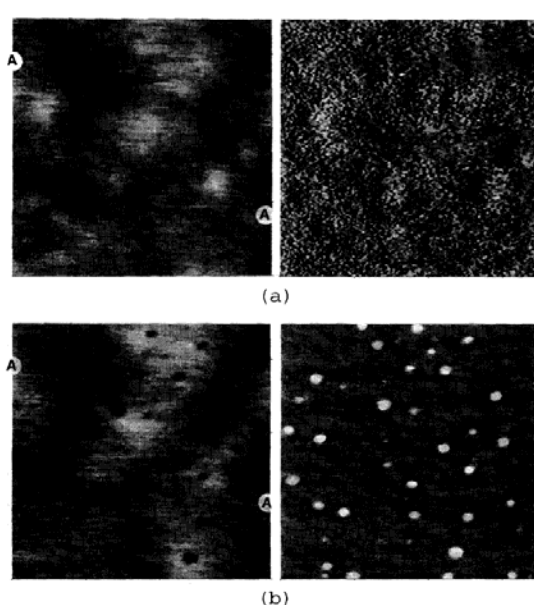


Figure 2. 13: Set of height (left) and phase (right) images obtained simultaneously by tapping-mode AFM. The contrast covers height variations in the 0-5 nm range in all height images, and phase variations. The scan area is 440 nm x 440 nm [129]

Experiments on block copolymers show that they exhibit different patterns, depending on film thickness, as shown in Figure 2.14. The figure shows, the thinner film of 35 nm thickness has spontaneously ruptured with circular holes appearing on the surface. Whereas discrete channels appeared in slightly thicker films, then these patterns were repeated for the thicker films. Greena and Limary suggested that the instabilities in these films can be caused by the effect of long and short-range intermolecular forces on thin films.

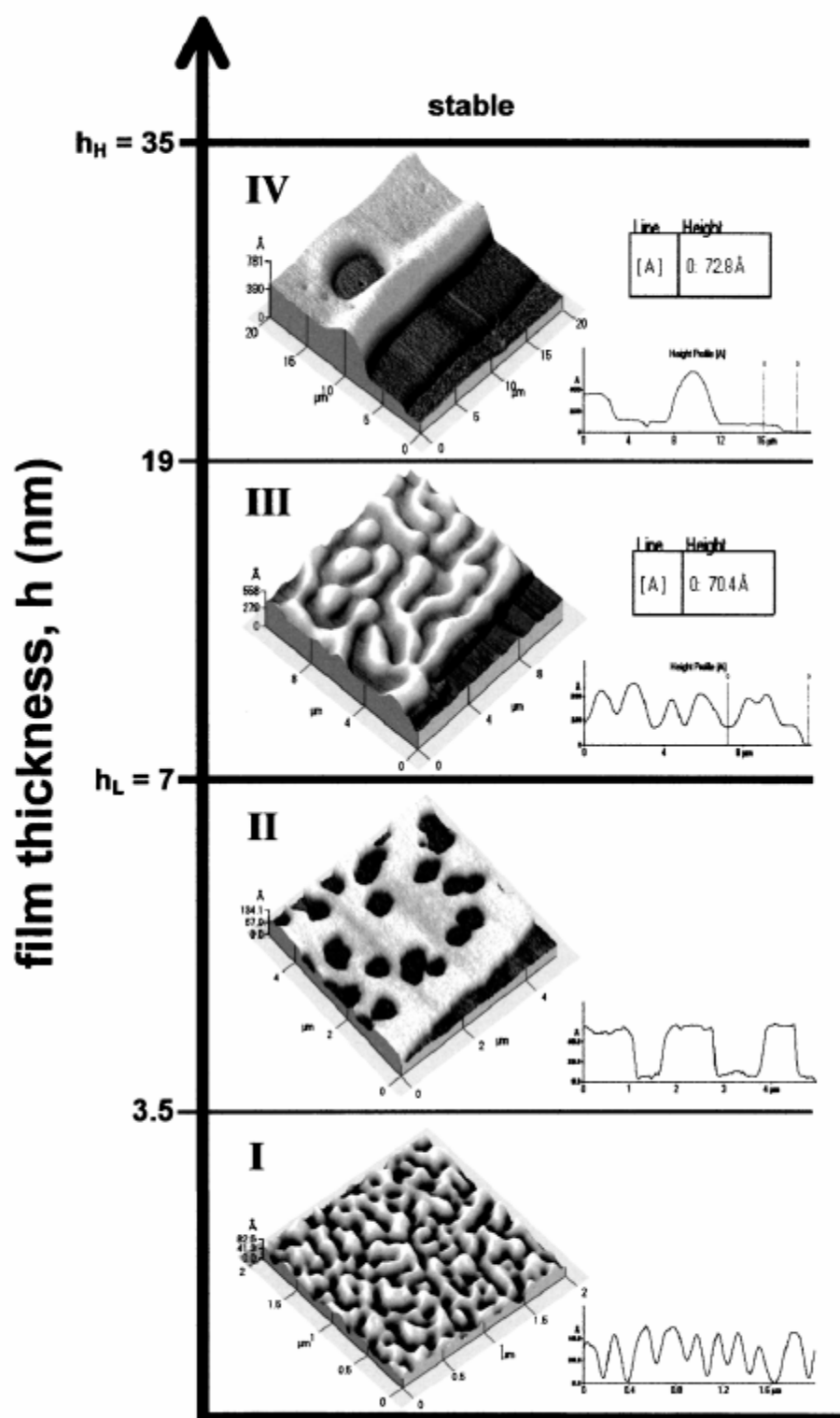


Figure 2.14 AFM images of topographical features showing a pattern formation in a symmetric polystyrene and poly(methylmethacrylate) copolymer [130]

2.4.3 Glass transition temperature of copolymers

Random copolymers show a single phase morphology with a single composition dependent T_g . The morphology of a two phase block copolymer depends on the block composition and many exhibit spheres or rods of the dispersed component in the matrix of the dominant component [131].

Any influence on the glass transition due to interfacial interactions is usually neglected. The major reason for such ‘free’ boundary conditions originates from classical mean-field theories, or molecular dynamic simulations [132]. It is assumed that interfacial interactions are completely within a distance corresponding to the resolution depth of the polymer film. The interfacial interaction has clear confinement effects such as pinning of molecules at substrate surfaces, covering it up to 1 nm diameter [132]. This distance suggests that the effect occurs during the film coating process, with fast solvent evaporation [133]. Usually spin coating has many parameters that have an influence on the polymer coating procedure [134]. It has been found that interfacial interactions can affect the polymer film up to 200 nm distance from the substrate interface [133, 134]. Two influencing interfaces have been suggested for thin films: the interface with the substrate, and the air or liquid-polymer interface [135,136,137].

2.5 Electro-Optical Polymer and Waveguide Devices

There is increasing interest in the design and fabrication of precision, yet low cost optical waveguides using polymers. Their applications are in photonics, sensors, semiconductors, liquid crystal displays, microfluidics and holographics. The electro-optic effect is where a polymer refractive index changes with applied electric field. Waveguides consist of a higher density transparent medium through which light can travel and be contained by total internal

reflection. Polymeric optical fibres have been prepared using a core and cladding of polymers with the core of higher refractive index. Polymeric slab waveguides may be prepared by creating a two-layer polymer sandwich. A typical polymer waveguide structure consists of a polymer thin film with a raised rib with [138] the optical path created by forming a channel in one of the layers as shown in Figure 2.15.

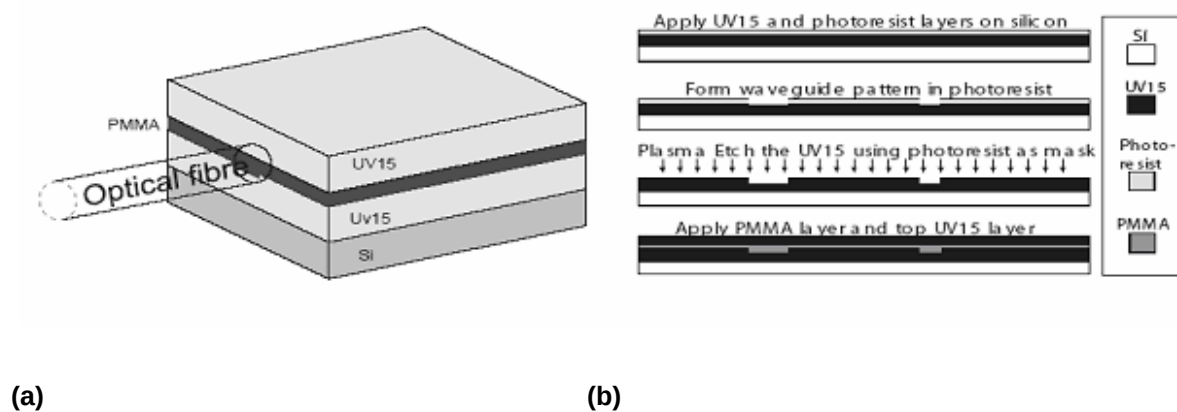


Figure 2. 15: (a) Typical polymer waveguide layers (b) fabrication procedure [138]

Maybe this should be in colour, to see the layers clearly.

A higher density polymer is used to fill the channel. The two layers must retain a distinct interface so that the first cladding layer should preferably be crosslinked before application of the second guiding layer to prevent interdiffusion. The guiding layer must have the higher refractive index so that total internal reflectance occurs at the interface and the light is confined to the guiding layer [6] [140-149].

Polymer electro-optic (EO) waveguide devices have been considered and there have been a range of experimental results to reveal the qualities of EO polymer devices including high-speed modulators and optical waveguides [150,151], multilevel-stacked integrated optical circuits [152,153], integration of electronic circuits with polymer waveguides [154,155], and polarization controlling waveguide devices [156].

Typical EO polymers have higher absorption loss than LiNbO₃, particularly at 1550 nm wavelength, due to C-H vibration overtones. There has been recent work which shows the losses can be decreased by substitution of fluorine or chlorine for the hydrogens [157]. A high T_g backbone polymer, a side-chain chromophore, and a crosslinked polymer matrix are utilized to improve thermal stability [158, 159]. EO polymers show both high EO coefficient and excellent poled order stability at 85 °C [160]. It was found that some EO polymer materials degrade due to the chemical reaction with excited oxygen [161]. It has been found that the photostability depends on the structure of the EO polymer and its backbone [162].

EO polymers can be developed to suite many commercial devices and material synthesis research has been directed toward the optimization of several requirements. These include reduced insertion loss, long-term thermal stability at 85 °C, and long term photo-stability at 10 mW input power with a high electro-optic coefficient. Some devices are based on a guest-host polymer structure with a highly nonlinear chromophore such as phenyltetraene bridge (CLD) and amorphous polycarbonate (APC) [163]. The chromophore was mixed in a PMMA host to achieve a push-pull modulator [164]. A nonlinear chromophore such as APC has improved its high thermal stability and low loss at 1550 nm wavelength. Other EO polymers of CLD and APC mixtures have shown short term thermal stability at 120 °C with loss in light transmission at 1550 nm [165].

Chapter 3:

Experimental Methods

3.1 Introduction

Specific techniques are necessary for preparation of thin polymer films and to study crystalline and amorphous property variation according to film thickness. Film preparation was carried out using available techniques such as spin coating for thickness limitation, optical microscopy (OM) for imaging surface profile and with hot-stage optical microscopy (HSOM) for melting and crystallisation. DSC was used for glass transition temperature (T_g) melting temperature (T_m) and transition enthalpy measurements, a profilometer was used for thickness and surface roughness measurements and FTIR was used to measure structure variation.

A spin coating device was used to spin solutions of selected commercial polymers and mixtures of polymers with chromophores, at different speeds and for different spinning times, Spin coating at high speed was used to make thinner films of polymers on glass and silicon substrates. Spin coating requires cleanliness of all equipment and materials. Solutions are required to be micro-filtered and usually dilute, typically $0.1\text{--}1.0\text{ g.L}^{-1}$. Thermoset polymers often have suitably low viscosity for spinning without added solvent.

A temperature programmable hot stage was used to precisely control the temperature and study crystallisation processes. The hot-stage was connected to an optical microscope for observation of the polymer melting processes and imaging the stages of crystallisation.

Optical microscopy with a digital camera was used to image film surfaces and roughness variation as changes in colour shading corresponding to surface treatment. Spherulite and

crystal morphology of the crystalline polymers is imaged after the surface segregation of block copolymers corresponding to different solvent, substrate and temperature.

DSC is used to characterise thermal transitions of different polymer films with measurement of treatments such as melting enthalpy, T_g and crystallisation temperature corresponding to the film formation conditions and film thickness.

Scanning electron microscopy (SEM) was used to record the microstructure of the thin films after etching and baking. Plasma and ion-beam etching were used to pattern the polymer films.

Qualitative FTIR spectroscopy is useful for characterising the chemical structure of polymer films. Absorption frequencies are characteristic of the structure due to spectral relationships with molecular vibrations and deformations. The shape and pattern of the spectrum must be characterised by applying the information obtained from standard references of group frequency data and compared along with other chemical and physical data for the material.

FTIR spectroscopy is used to compare the polymer structural changes with treatments such as curing and baking. The interpretation of infrared spectra involves correlation of absorption bands in the spectrum of an unknown compound with the known absorption frequencies for types of bonds. Intensity is important for identification of the source of an absorption band, in particular whether the intensity is weak or strong and its shape.

3.2 Project Methodology

3.2.1 Spin coating technique

Spin coating is one of the most common techniques for applying thin films onto substrates such as glass, silicon wafers and aluminium as shown in Figure 3.1a. The process involves

fluid flow and evaporation behaviours that generally give relatively uniform coatings. However, difficulties arise with complex coating solutions that often to prevent perfectly flat and high-quality, coatings.

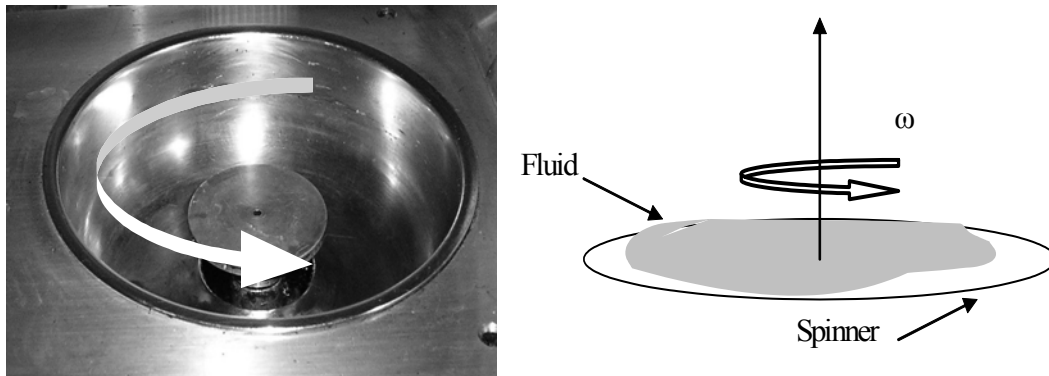


Figure 3.1: (a) Spin coating apparatus, (b) schematic of spin coating rotation technique for fluids

Spin coating is used for many applications where relatively flat substrates are required to be coated with thin layers of substance [26]. The substance must be dissolved or dispersed in a solvent prior to the spin coating process leaving a uniform layer. Solution flow on a flat spinning substrate is an important physical process involved in spin coating [166]. A solution of the substance is required to cover the substrate surface to determine a film of particular thickness when coating the surface of the substrate. Wetting the surface is another important factor for coating the surface during the spinning process. Evaporation of the solvent has an influence on the final coating thickness, and the environment must be controlled during solvent evaporation, such as temperature and humidity during the spinning process [167].

The thermal conductivity of the substrate has an influence on solvent evaporation and stability of the surface uniformity of the film during the spinning process [168]. Any difference in temperature between parts of the film surface during the spinning process may impact on the thickness and roughness of the film. Hence any lower temperature in some areas leads to a lower evaporation rate of the solvent [169].

The environment around the spinning process has an influence on the film quality. Many factors may be involved in this event such as evaporation rate. At high evaporation rate and high humidity, moisture condensation may occur [170]. The coating quality is apparent as coating roughness and micro cracking of the coating are formed upon drying. Vacuum evaporation of solvent may not allow time for film equilibration.

Another factor may affect the coating of film on the substrate; the boundaries of the substrate may influence the spreading. Surface tension effects counter the centripetal force and limit radial flow causing a film with a thicker edge [3]. The shape of the substrate has an influence on the coating process, for example a substrate with a rectangular shape (glass slide) will affect airflow over the substrate surface and beyond the edge, resulting in non-uniformity of coating thickness in some areas. The edge effects are often seen because the solution flows uniformly outward, and some droplets at the edge are spilled. This event generally depends on factors such as viscosity and rotation rate [171].

3.2.2 Speed of rotation

The formation of film depends on the spreading of solution on the substrate by rotational motion as shown in Figure 3.1b. The solution on the substrate surface will flow radially outward by centripetal force during the spinning process. There must be enough solution to form a complete film and minimise the differences in thickness across the surface depending on the viscous directed towards the centre, i.e. opposing outward motion. When the substrate is spinning at a constant rate, the viscosity will control the film thickness and uniformity. This will be observed as a coloured interference pattern because the film will be of the order of 1 μm thick [172].

Solvent evaporation is a critical issue in the final thickness and uniformity of the film. Evaporation occurs from the top surface, and only some of the solvent in the solution can evaporate freely to the environment [173]. The remaining solvent must diffuse through the surface polymer film while the viscosity of the solution will be increasing [174].

3.3 Optical Microscopy with Hot-Stage

A Mettler FP82HT hot-stage with Mettler FP90 processor was used. Films of polymers including PCL, PES and PLA were melted by inserting a glass microscope slide with the polymer film into the hot-stage chamber.

The aim of using hot-stage microscopy was to create a thermal history for the polymer film and observe the thermal transitions through heating and cooling. A Nikon Labophot 2 optical microscope was used for observing the polymer films with and without a polariser. Images were captured using a Nikon digital camera (Coolpix 5000, 5 Mpixels per image).

3.4 Differential Scanning Calorimetry

DSC is a powerful technique to measure thermal properties of polymers in particular for quantitative analysis. The power compensation DSC technique is based on measuring the electrical power required to balance the sample and reference temperatures. The arrangement of the sample and reference configurations in the DSC instrument is shown in Figure 3.2. The arrangement shows the comparison between the sample and reference temperatures.

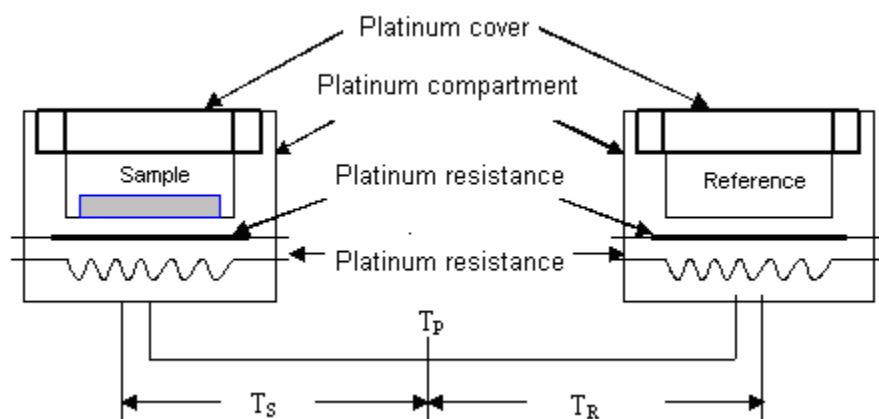


Figure 3.2: Schematic diagram of the power compensation DSC calorimeter

A Perkin-Elmer Pyris1 DSC with Intracooler 2P (Figure 3.3) was used after calibration using indium and zinc standards for temperature and indium for enthalpy.

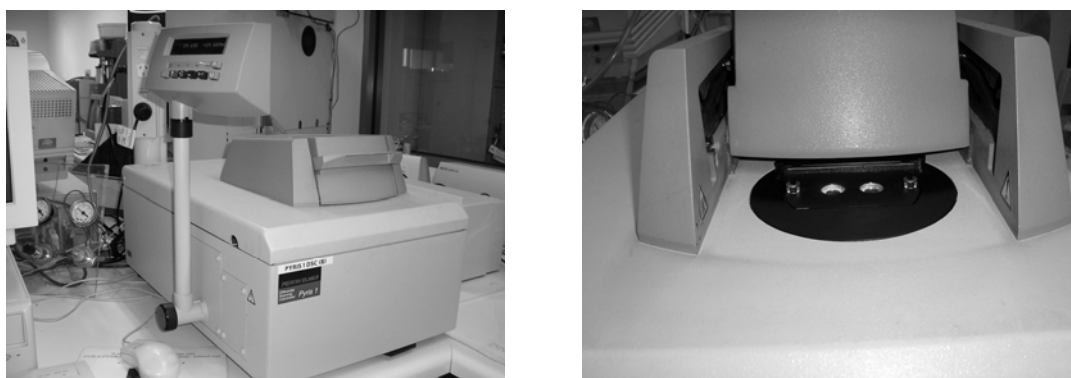


Figure 3. 3: Perkin-Elmer Pyris 1 DSC instrument and cell arrangement

The polymer was weighted and placed in an aluminium pan with cover and then hermetically sealed. Within the DSC instrument there are two cell locations, one for the sample and one for a reference. The reference cell contained an identical empty pan to balance the heat capacity of the sample pan.

The DSC instrument is used for quantitative analysis through balancing of the sample and reference temperatures and measuring the differential power required. Figure 3.4 shows the data display page with the options for control and monitoring of the DSC. The data display page shows a DSC curve of heat flow endo up (mW) versus temperature.

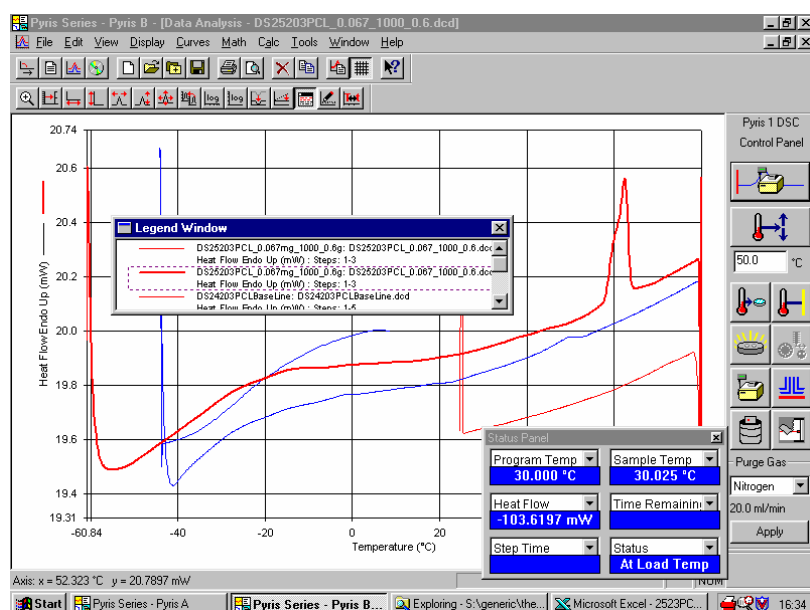


Figure 3. 4: DSC data display page with display and analysis options

DSC of polymer films of different thickness were measured as shown in Figure 3.5. The cooling curve was used to investigate the crystallisation (exothermic) process and the heating curve was used to investigate the melting (endothermic) process.

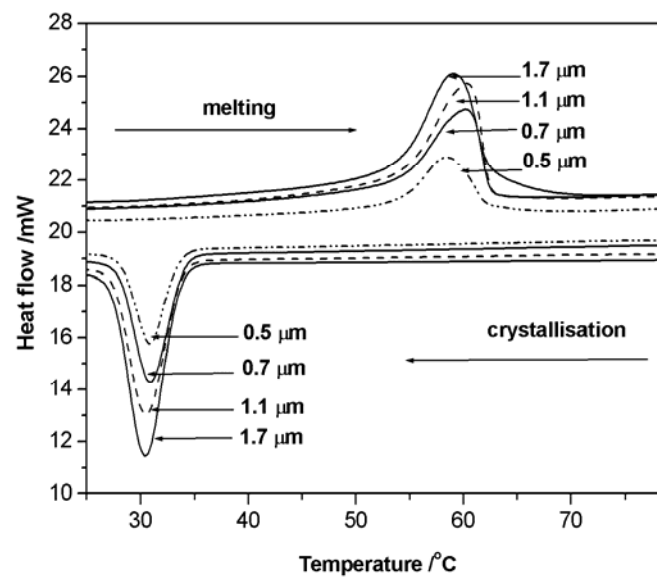


Figure 3. 5: Endothermal and exothermal analysis cycles of PCL films with four different thicknesses

3.5 Stylus Profilometer Technique

3.5.1 Exploring sample surface using Ambios XP-2

An alpha-Step 250 Tencor profilometer was the first technique used in measuring the surface roughness and morphology (Hirbour Technic) performed with: Linear-scan stage technology and precision optical smooth (smooth to 0.6 nm.mm^{-1}), ensuring high precision measurements. Step height repeatability was $\pm 1.0 \text{ nm}$ or 0.1% of full range [175].

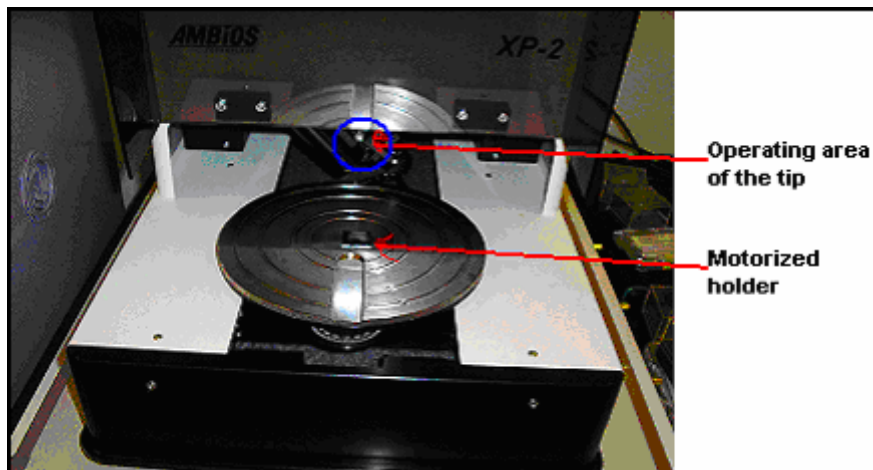


Figure 3. 6: Ambios XP-2 Stylus profilometer instrument used to measure polymer film thickness and to scan the surface roughness, XP-2 has an acoustic isolation enclosure

3.5.2 Characterisation with the Ambios XP-2 system

A more advanced XP-2 Ambios profilometer is shown in Figure 3.6. The Ambios XP-2 Profilometer was used with more facilities for exploring surface roughness and morphology. The system has a linear-scan stage. Step height repeatability and high accuracy measurements were made using the standard 200 mm vacuum stage with motorized holder for sample translation to the area of measurement under the tip (Figure 3.7) 150 mm x 178 mm travel and manual 360 ° rotation.

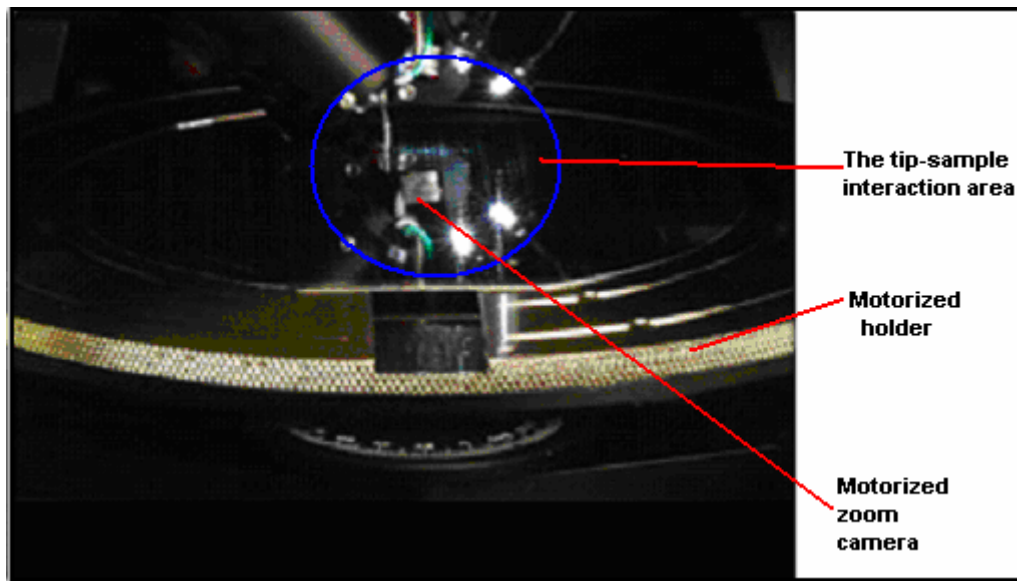


Figure 3. 7: Ambios XP-2 standard 200 mm vacuum stage with motorized sample translation

The Ambios XP-2 has a low mass and inertia stylus assembly to minimize tip-sample interaction with stylus loads as low as 0.05 mg. A motorized colour zoom camera with standard 40-160 X magnification, up to 60,000 data per scan line, high tip resolution (regardless of scan length) and data collection system.

3.5.3 Ambios XP-2 software facilities

The window from the XP-2 software is shown in Figure 3.8. It allows using the advantage of the instrument motorized stage to make automated step and repeat measurements, move to as many as 256 predetermined locations, or move to specific X and Y locations.

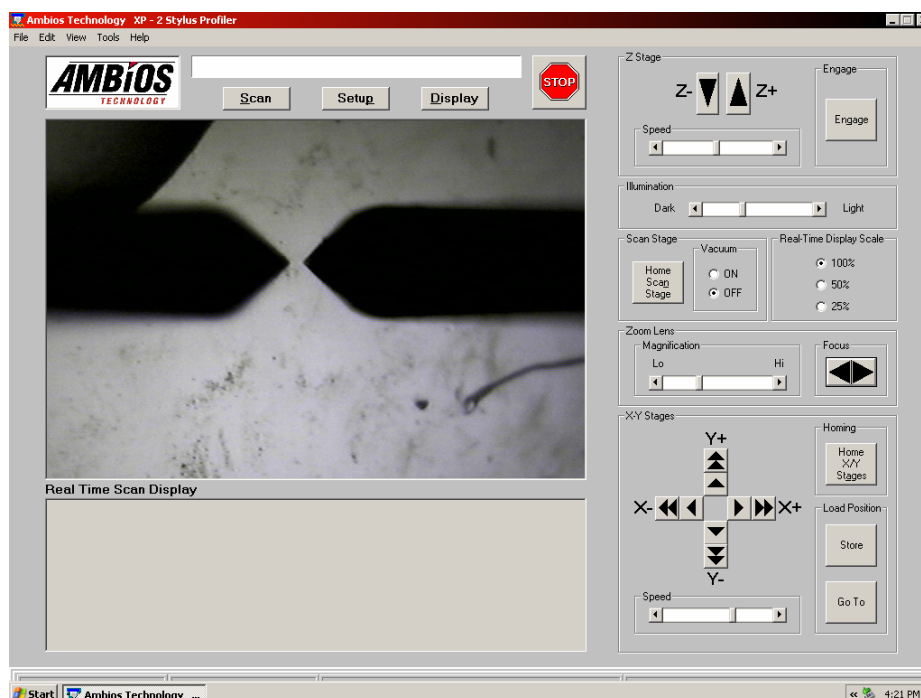


Figure 3. 8: Ambios XP-2 Stage navigation screen (motorized colour zoom camera with standard 40-160 X magnification and 1-4 mm field of view)

This technique can be operated in either the instrument coordinate system or with an arbitrary coordinate system defined in relation to the specimen. The specimen stage translation capability of 20 mm x 80 mm in this system was manually available making sample placement easy and fast. The software used to process the data from a scan was used to compare pre- and post-processing curves.

The facility for exploring a polymer thin film surface and thickness was an important aspect in using the profilometer stylus for measurement of surface roughness. The system allowed data analysis of surface roughness of the polymers after different surface treatments, such as temperature conditioning. Figure 3.9 shows the data display page with options used in measuring polymer film thickness and surface roughness. The data display page shows the depth of the film and the roughness across the film surface in unit of angstrom Å.

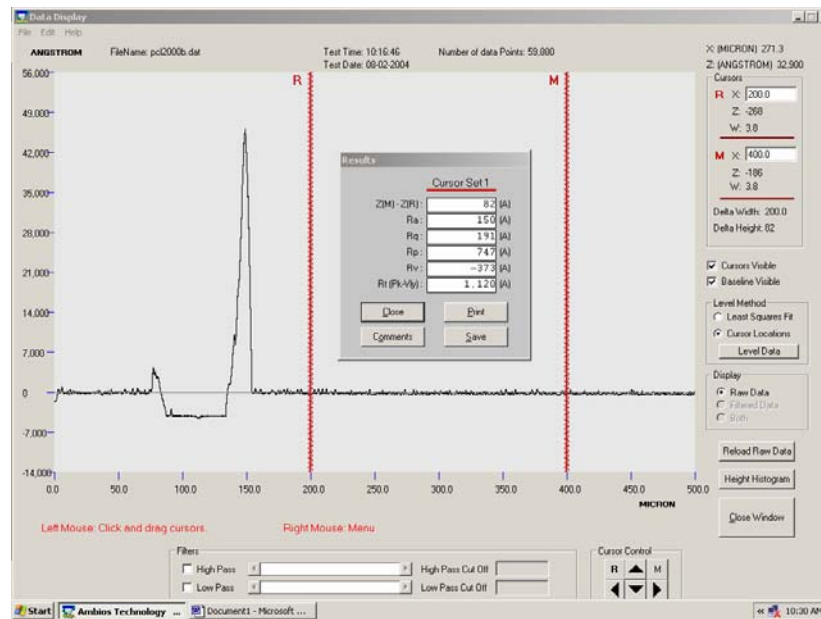


Figure 3. 9: The stylus profilometer data display page with options used in measuring polymer film thickness and surface roughness

The two peaks in the left side of figure 3.9 around the scratch are the residual of the polymer after being scratched and pushed to the sides of the scratch. Figure 3.10 shows a film of crystallised PCL with a scratch made across the film to measure the film thickness using the stylus profilometer.

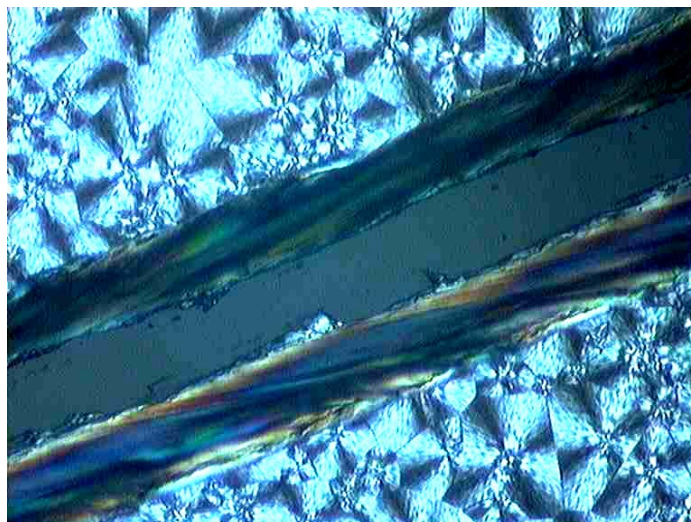


Figure 3. 10: PCL crystallised film with a scratch made across the film for thickness measurement

A surface contour plot of the film shown in Figure 3.10 is shown in Figure 3.11. The width of the tunnel and the scan translation across the film surface were measured in angstrom and converted to μm .

The profilometer addresses surface finish metrology measurement for surface roughness, and quality, surface topography via surface scanning, surface texture management, surface imaging and surface technology as a surface roughness gauge.

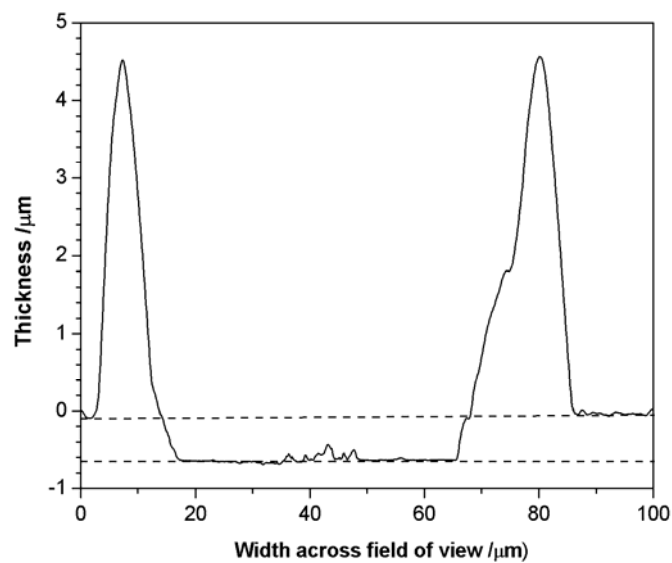


Figure 3. 11: Thickness measurements of polymer film were made by scanning the polymer film surface across a scratch made using a sharp blade

The stylus profilometer was a useful technique to measure surface topography for comparison of grain structures that change with the structures, such as ridges, folds, bumps, and patterns. Figure 3.12 shows the 2D surface morphology of SIS with changes appearing on the surface because of segregation occurring with the blocks of SIS. The SIS was treated at 120 °C and Figure 3.13 shows a graph of surface morphology scanned using the stylus profilometer.

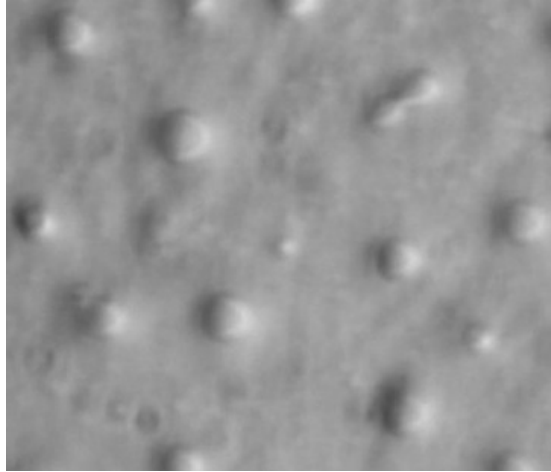


Figure 3. 12: 2D surface morphology of styrene-isoprene-styrene block copolymer

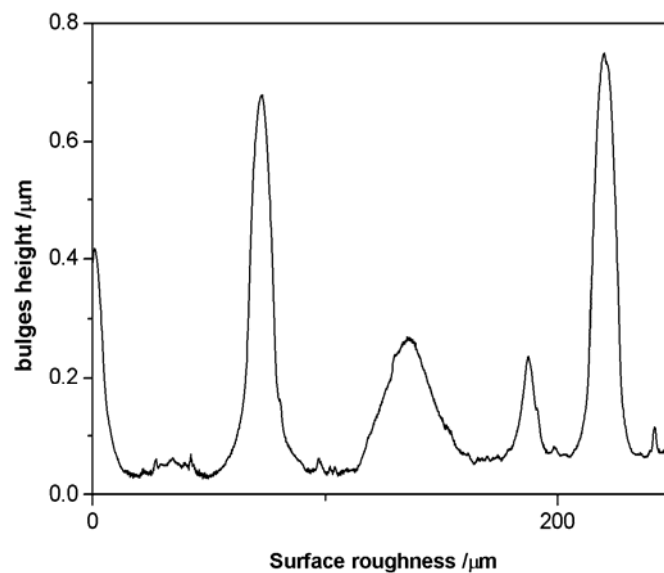


Figure 3. 13: Morphology of SIS block copolymer scanned using stylus profilometer

3.5.4 Ambios Tip radius

Making measurements using a contact stylus profiler required a special tip. The best stylus tip radius for an application was in the range from 2-20 μm . Choice of stylus tip for optimum resolution is important for an application, as is the force that can be used without damaging the specimen surface.

3.5.5 Imaging a polymer surface

The smallest feature that can be resolved by a given tip radius is approximately 0.5 to 0.1 of the tip radius as shown in Figure 3.14. For features approaching the size of the tip, the surface resolution is limited by the tip shape. On flat surfaces where only the top of the tip is being used to resolve the surface the resolution is better and can be an order of magnitude smaller than the tip radius.

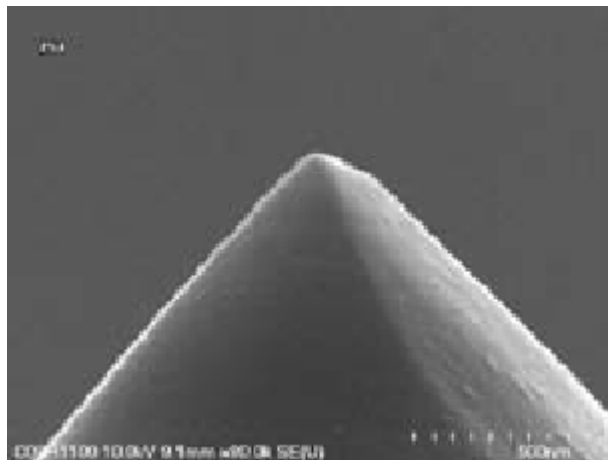


Figure 3. 14: Image of a typical profilometer tip

It is important not to choose the smallest radius tip to gain the best resolution. The surface pressure increases with the square of the tip radius. As the surface pressure increases, the potential for damage to the surface is more likely to increase [176]

The technique has the capability for metrology calibration for height measurement and surface roughness as shown in Figure 3.15. PCL film of 0.9 μm thickness and maximum roughness of 0.12 μm , is shown in Figure 3.15 after the film was scanned using the stylus profilometer

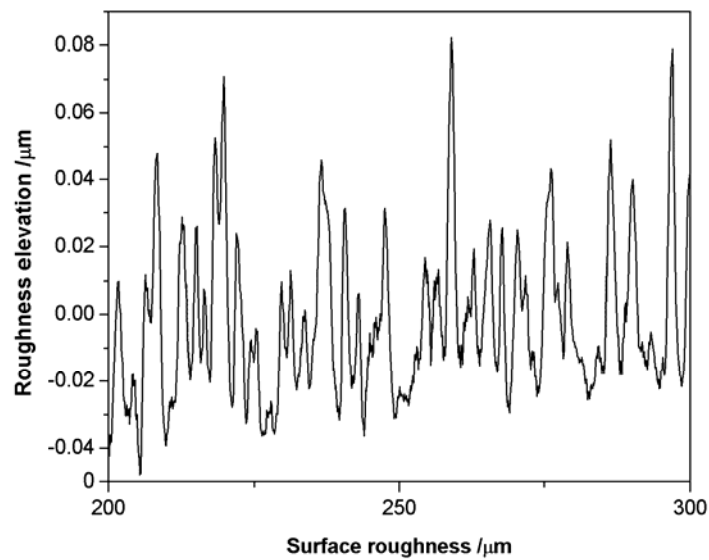


Figure 3. 15: Surface roughness of PCL film

3.5.6 Three dimensional image measurements using profilometer

The stylus profilometer has the capability to display surface topography data in a wide variety of displays using Ambios Technology 3D Image Software for the system. The software allows interactively rotating, scaling, panning, and zooming of the view of the data.

3.6 Three Dimensional Measurements of the Polymer Film Surface

Three dimensional methods of measurements were used to distinguish between the polymer surfaces before and after any treatment. The measurements were based on surface measurements using the stylus profilometer. Other measurements were based on a method of converting images from the optical microscope to numerical data then making a three dimension graph using Surfer software. The technique was used to convert two dimensional surface graphs to three dimensional images as shown in Figure 3.16. The dark areas represent lower levels and the bright areas represent higher levels.

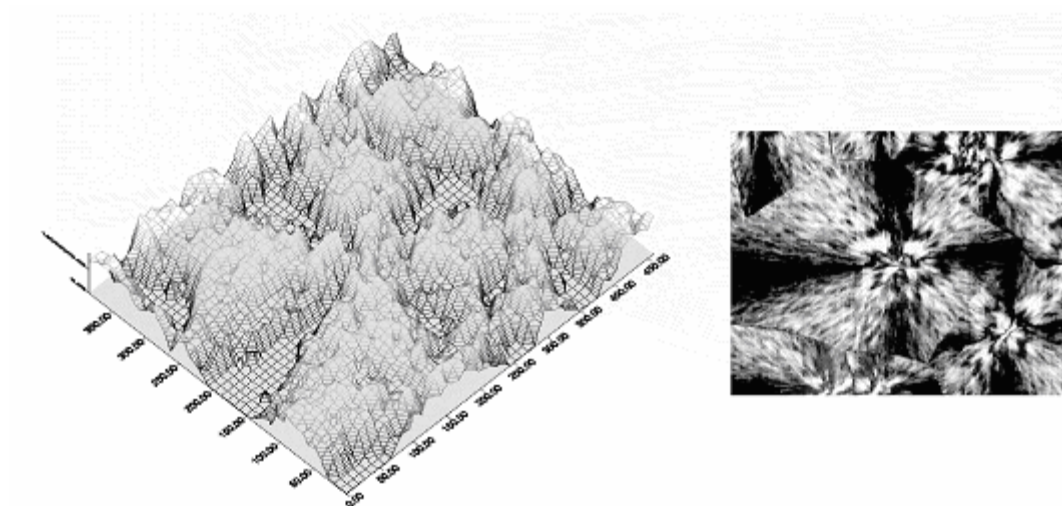


Figure 3. 16: PCL crystal selected from film thickness 16 μm

3.7 Fourier Transform Infra-Red Spectroscopy

FTIR spectroscopy is a powerful analytical technique for qualitative and quantitative analysis [177]. The Perkin-Elmer spectra 2000 FTIR spectrometer shown in Figure 3.17 was used to record spectra of polymer films before and after curing with UV light.



Figure 3. 17: Perkin-Elmer FTIR spectrometer

FTIR has been broadly used in the study of polymer, structure, modification and chemical behaviour, qualitatively and quantitatively. It provides a convenient and dependable

instrument for the investigation of polymers. The differences in vibrational modes can be used to differentiate polymer chemical functionality [178] [179].

Fourier-transform spectrometers simultaneously expose a film to the full range of infrared frequencies. The beam of infrared radiation passes through the film and reaches the detector and the resulting intensity data is used to construct an interferogram. This will be converted to a spectrum by performing a Fourier transformation [180]. Figure 3.18 shows a typical spectrum of UV15 polymer before and after curing.

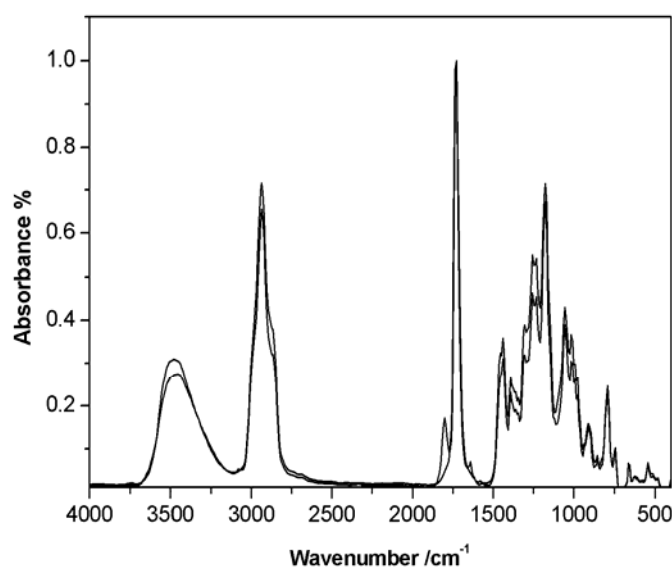


Figure 3. 18: FTIR absorbance spectra of UV15 before and after curing with UV light

3.7.1 FTIR spectroscopy frequency range

An infrared spectrum is characteristic of a compound. The infrared region covers the wavenumber range from 400-4000 cm^{-1} . The wavelength of infrared radiation is usually expressed in terms of wavenumbers that are the reciprocal of wavelength. The absorbed radiations by the film will give information as a spectrum of absorbance versus wavenumber. Functional groups absorb infrared light at a unique frequency [181].

3.7.2 Spectral analysis

The study of the characteristic absorption bands arising from various types of hydrocarbon chemical functional groups has produced a number of reliable correlations between the band positions and the type of structural groups present. C-H stretching and bending modes are the most common and the C-C frequencies for many types of branched-chain structures are available. The C-H stretching mode absorptions are useful in differentiating saturated and unsaturated materials [182].

Infrared spectroscopy is very useful for qualitative analysis to identify chemical compounds since it produces unique spectrum for every substance, with peaks resulting from distinct structural features [183][81].

The strength of the bonds determine their stretching frequency, such as a carbonyl group absorbs infrared radiation in a specific frequency range because the bond between the carbon atoms is constantly stretching and contracting within a range of bond lengths[184].

3.7.3 Film thickness influence on IR spectroscopy measurement

Chemical information about the substance under investigation can be obtained from any material, such as plastics, paper, fibres and coatings. Measurements of infrared spectra using transmission mode require a thin film to avoid total absorption at any wavenumber. Film thickness required depends upon structure, but generally about 10 μm or less is suitable [184].

3.7.4 FTIR measurement of a cured polymer

FTIR measurements were carried out to investigate a cured coating polymer such as UV15. A transparent pellet of KBr was used as a substrate. A thin film of UV15 was cast by depositing a drop of the polymer solution onto a KBr pellet. The polymer film onto the KBr pellet was

cured using UV light before any FTIR examination. The FTIR measurement was made by mounting a KBr pellet with cast film of the polymer into the spectrometer. Glass is not a useful material in infrared spectroscopy because of the strong absorptions due to the Si-O group. The spectra were recoded and used in this investigation by comparing the amplitude of the functional group peaks.

Chapter 4

Thin Film Effects on Properties of Polycaprolactone

4.1 Introduction

An investigation of the properties and surface uniformity of PCL thin films was carried out and presented in this chapter. The characteristics of PCL have attracted its use in many applications [185]. The investigation is based on preparation of thin films using a spin coating technique. Film thickness and roughness are correlated with variation in concentration, spinning speed and spinning time. DSC is used to investigate the crystallisation and melting processes. There is much interest in thin films and their thermal property variation [17]. Measurements are carried out to investigate enthalpy, melting temperature, crystal size and surface roughness variation with film thickness. Images and surface roughness profiles of PCL crystals and lamella are characterised using optical microscopy and profilometry respectively, and they are correlated with film thickness

4.2 Experimental

4.2.1 Materials

Poly(ϵ -caprolactone) (PCL) $\text{H}(\text{OCH}_2(\text{CH}_2)_3\text{CH}_2\text{CO})_n\text{OH}$, $M_w = 80\,000\text{ g.mol}^{-1}$, Sigma-Aldrich Chemicals $T_m = 60\text{ }^\circ\text{C}$ was used to prepare thin films of 0.07-16.0 μm thickness. Solutions of poly(ϵ -caprolactone) (0.2 g.mL^{-1} and 2 g.mL^{-1}) were prepared by dissolving PCL in tetrachloroethane $[\text{Cl}_2\text{CHCHCl}_2]$, obtained from Sigma-Aldrich, $[T_b = 144\text{-}148\text{ }^\circ\text{C}]$ at room temperature.

4.2.2 Solution preparation

PCL solutions were prepared by mixing a selected amount of PCL in 10 mL of tetrachloroethane, to provide a series of concentrations shown in Table 1. The solutions were filtered through a 0.2 μm syringe filter. Prior to the experiments, glass slides were cleaned with acetone and rinsed with deionised water. Finally they were dried under a nitrogen stream and stored in a clean room before use in the spinning process.

4.3 Procedures and Instrumentation

4.3.2 Spin coating process and thickness measurements

The spin coating technique was used to spin solutions of PCL using different speeds at a constant spinning time. PCL was spin coated at different speeds to obtain thin films on glass substrates in a clean room at ambient temperature. Drops of the solution were deposited on the glass substrate using a pipette after setting the spinning speed to a specific value in the range of 1000 - 7000 rpm. Excess solution spilled instantly from the substrate surface while spinning, leaving a thin film covering the glass. After the spinning process, the coated specimens were kept in a fume cupboard in a clean room while allowing the solvent to evaporate. The films were dried in a vacuum oven at 40 °C for 24 h in order to remove any residual solvent. Finally, the films were cooled to room temperature in the clean room. After the film stabilised at ambient conditions, the films were scratched using a sharp blade and the thickness was measured across the cut made on the surface. The surface roughness was also measured.

Films of different thickness were prepared by spinning PCL solution of different concentration to make a range of thicknesses. A particular concentration of the polymer cannot cover the range of thickness required, although the spinner has a range of speeds from

200 rpm to 7000 rpm; for example the low concentration of 0.4 g.mL^{-1} cannot provide thickness less than $0.1 \text{ }\mu\text{m}$ or greater than $1 \text{ }\mu\text{m}$.

4.3.3 Hot stage with optical microscopy

A hot stage was used to precisely control the temperature while visualising crystallisation in the polymer films. Heating the films above melting temperature and cooling them to room temperature was used to create a thermal history [16]. For the heating stage, the film was heated from $20 \text{ }^{\circ}\text{C}$ to $80 \text{ }^{\circ}\text{C}$ at a rate $10 \text{ }^{\circ}\text{C}\cdot\text{min}^{-1}$, and then held for 3 min at $80 \text{ }^{\circ}\text{C}$. For the cooling stage, the film was cooled from $80 \text{ }^{\circ}\text{C}$ to $20 \text{ }^{\circ}\text{C}$ at rate $10 \text{ }^{\circ}\text{C}\cdot\text{min}^{-1}$. Optical microscopy was used to observe the PCL crystallisation, surface morphology and crystal morphology [17]. These images were employed to measure crystal size and investigate surface morphology using the same magnification with all films and a standard reticle slide for calibration.

4.3.4 Differential scanning calorimetry measurements

DSC was used to characterise the crystallisation and melting process of the PCL films. A thermal history was created for all films by heating the PCL film to $80 \text{ }^{\circ}\text{C}$ for 3 min and cooling at $10 \text{ }^{\circ}\text{C}\cdot\text{min}^{-1}$. Films of mass 1-3 mg were accurately weighed, after peeling from the glass substrate, and hermetically sealed in $10 \text{ }\mu\text{L}$ aluminium pans. It was difficult to peel films that had a thickness less than $0.5 \text{ }\mu\text{m}$, from the glass substrate, although other reports describing PCL did not indicate strong nucleation on the glass surface with thickness less than $0.5 \text{ }\mu\text{m}$. [186, 187] The melting temperature and enthalpy were measured from a second heating scan at $10 \text{ }^{\circ}\text{C}\cdot\text{min}^{-1}$.

4.4 Results and Discussion

4.4.1 Correlation between film thickness and spinning speed of PCL

In this investigation the influence of spinning speed on the film thickness was carried out. Films of different thickness were prepared by spinning the PCL solutions. Spinning speed was the first variable used in thin film preparation. Selections of speed were used to spin the polymer solution so as to make films of different thickness. The data for film thickness corresponding to spinning speed are shown in Table 4.1

Table 4. 1: Thickness versus concentration for different spinning speed

Concentration /g.mL ⁻¹	Spinning speed /rpm			
	1000	2000	4000	6000
	Thickness /μm			
0.2	0.2	0.1	0.09	0.07
0.4	0.9	1.7	0.30	0.10
0.6	1.7	1.1	0.70	0.50
1.0	3.3	2.3	1.60	1.50
1.6	7.2	6.3	3.70	3.30
2.0	16.3	9.6	6.60	4.10

The correlation between film thickness and spinning speed for different concentrations was a foundation for investigations in this study. Curves of this correlation are plotted using the data from Table 4.1 and these curves are shown in Figure 4.1. The figure shows four curves for thickness of PCL films that were prepared from solutions of different concentration. This range of measurement was considered as a reference for preparing films. Thinner films prepared from solutions of low concentration had less variation in thickness within the range of thickness.

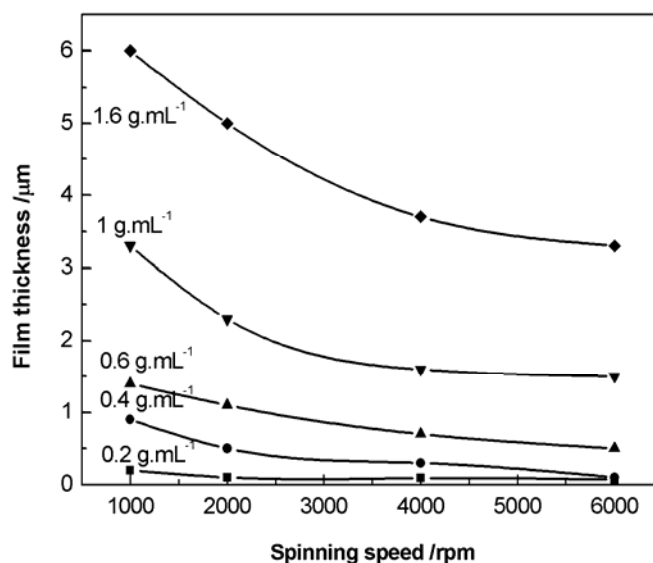


Figure 4. 1: Correlation between film thickness and spinning speed at various concentrations

The final coating thickness of the films spun under ambient conditions was probably influenced by evaporation of the solvent during the spinning process and the thermal conductivity of the substrate influence on the solvent evaporation [168].

4.4.2 Correlations between film thickness and PCL concentration

The influence of PCL concentration on film thickness was investigated using films made from solutions of different concentration. Polymer concentration was the second variable in this investigation. The data for film thickness measurements are shown in Table 4.1. The correlation between the film thickness and PCL concentration is shown in Figure 4.2. The thickness measurements showed dependency on the concentration of the polymer. The figure shows films that were made from solutions of higher concentration have greater thickness.

The film thickness was less sensitive to concentration at low concentration. This is due to the viscosity of the polymer changing with concentration and entanglement concentration controlling film thickness [188].

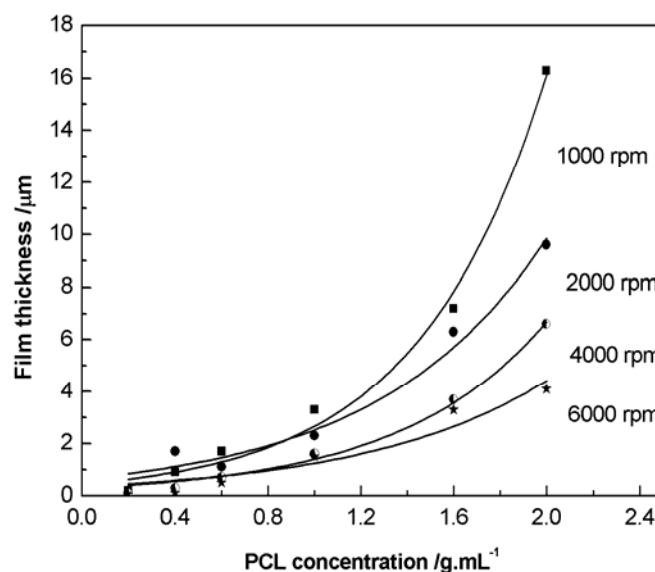


Figure 4. 2: Correlation between film thickness and concentration of PCL at different spinning speeds

4.4.3 Correlation between surface roughness and film thickness

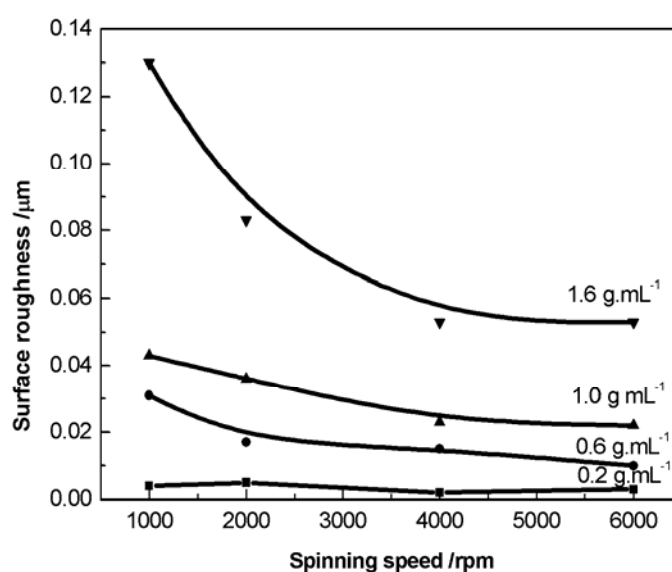
Selections of polymer concentration were used to investigate the influence of concentration on the polymer surface roughness. The measurements of the surface roughness of PCL films for each solution of different concentration were carried out using profilometry. The data for the surface roughness are shown in Table 4.2. The correlation between PCL concentrations versus film thickness is shown in Figure 4.3.

It was important to allow enough time for solvent evaporation under ambient conditions, which leads to low roughness or more uniform surface. It was important to have slow evaporation of the solvent near ambient or at low temperature to minimise macroscopic deficiency and to minimise any instability of the centre of the film. Increasing and decreasing pressure in a vacuum oven was useful to avoid film with holes and probably that occurred during heating the film to allow enough time for solvent evaporation because the film was thin enough to allow some holes to occur.

Table 4. 2: Roughness versus spinning speed for different concentration

Spinning speed /rpm	Concentration /g.mL ⁻¹			
	0.2	0.6	1	1.6
Roughness / μm				
1000	0.004	0.031	0.043	0.130
2000	0.005	0.017	0.036	0.083
4000	0.002	0.030	0.023	0.053
6000	0.003	0.010	0.022	0.053

The correlation in Figure 4.3 demonstrated the surface roughness variation of the thinner films of PCL (low concentration of polymer) had the lowest roughness.

**Figure 4. 3: Correlation between surface roughness and spinning speed at different Concentration**

The PCL films were prepared under ambient conditions then after the solvent evaporation at room temperature the films were heated using a vacuum oven at a temperature lower than the melting temperature. This did not affect the thermal history of the crystals. The surface uniformity (roughness variations) for different thicknesses depended on the many factors such

as the conditions under which the films were made, substrate nature and the nature of the solvent including boiling temperature, solvent evaporation rate, the nature of the substrate and its relation with the polymer and polymer concentration are factors involved in shaping the surface uniformity without any further treatment to the surface just under ambient conditions. Birnie *et al* suggested thermal conductivity of the substrate has influence on the stability of the surface uniformity of the film during the spinning process [169]

4.4.4 Relation between film thickness and period of spinning

The correlation between spinning speed and spinning time was carried out to investigate the influence of time of spinning on the film thickness in this study. Selections of a period of spinning were used to make films by spinning solutions of two different concentrations and with the same spinning speed (2000 rpm). The data for thickness measurements are shown in Table 4.3.

Table 4. 3: Thickness versus spinning time for two different concentrations

Spinning Speed (2000 rpm)		
Concentration	0.4 g.mL ⁻¹	0.6 g.mL ⁻¹
Time /s	Thickness / μ m	Thickness / μ m
20	0.6	1.5
40	0.4	1.3
60	0.5	1.2
80	0.4	1.3

The relation between the PCL film thickness and spinning time is shown in Figure 4.4. The figure includes two curves collected from thickness measurements of films spun from PCL solutions of different concentration. The figure showed slight changes to the film thickness with spinning time.

In Table 4.3, the difference between the film thicknesses in either set of measurements at different concentration was about $0.2\ \mu\text{m}$ per 60 s. The longer spinning time probably allowed more time for solvent evaporation during spinning.

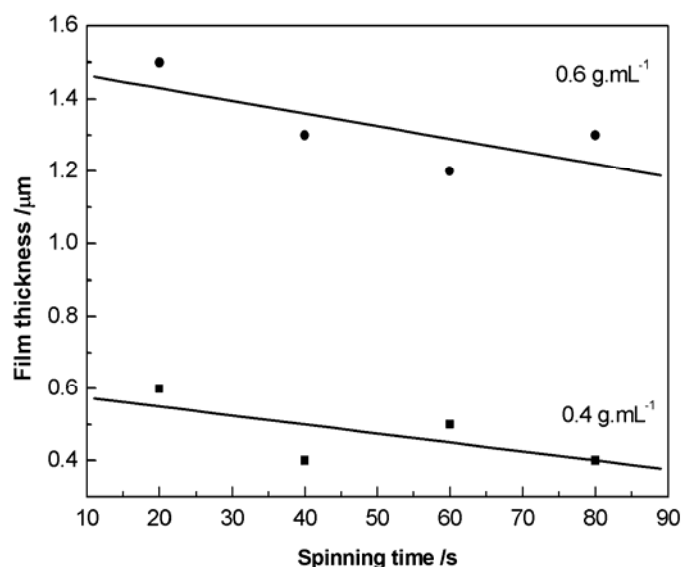


Figure 4. 4: Correlation between film thickness and spinning time at two concentrations

4.5 Thermal Property Measurements

The quantitative data of melting temperature and enthalpy corresponding to film thickness were investigated. DSC was used to measure melting temperature and enthalpy. PCL films of different thickness were used in these measurements. DSC measurements were carried out using PCL film samples of low weight.

4.5.1 Enthalpy variation corresponding to film thickness

Enthalpy variation corresponding to film thickness based on DSC measurement was investigated for films of different thicknesses. The enthalpy data are shown in Table 4.4. PCL film thicknesses were prepared using solutions of different polymers concentration.

Table 4. 4: Melting enthalpy versus thickness

Film thickness	Enthalpy (ΔH)	Film thickness	Enthalpy (ΔH)
/μm	/J.g ⁻¹	/μm	/J.g ⁻¹
0.1	81.2	3.3	65.2
0.7	69.3	4.1	59.7
1.1	72.3	6.6	59.0
1.6	66.1	9.6	59.2
1.7	70.4	16.3	59.6
2.3	66.3		

This correlation of enthalpy variation corresponding to film thickness is shown in Figure 4.5. The figure shows that enthalpy increases as film thickness decreases. PCL crystal lamellae affect the thermal properties [49]. Enthalpy had significantly less variation when PCL was greater than 3.3 μm thick. DSC measurements show greater variation in the enthalpy when thinner film is used in this investigation.

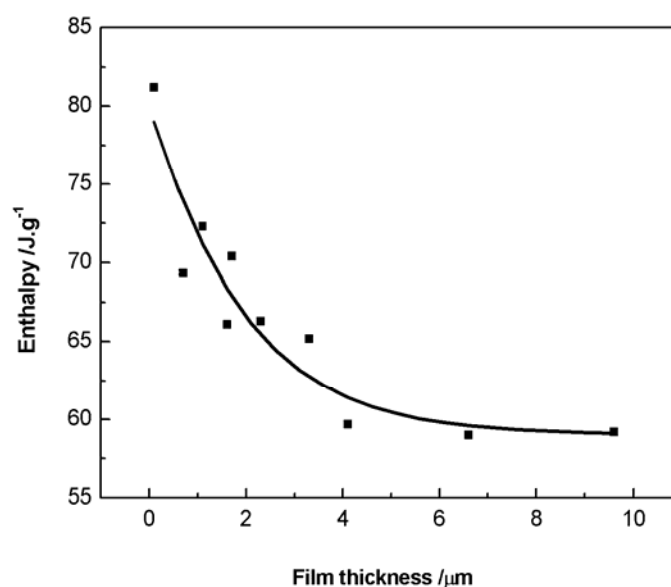


Figure 4. 5: Correlation between PCL melting enthalpy and film thickness

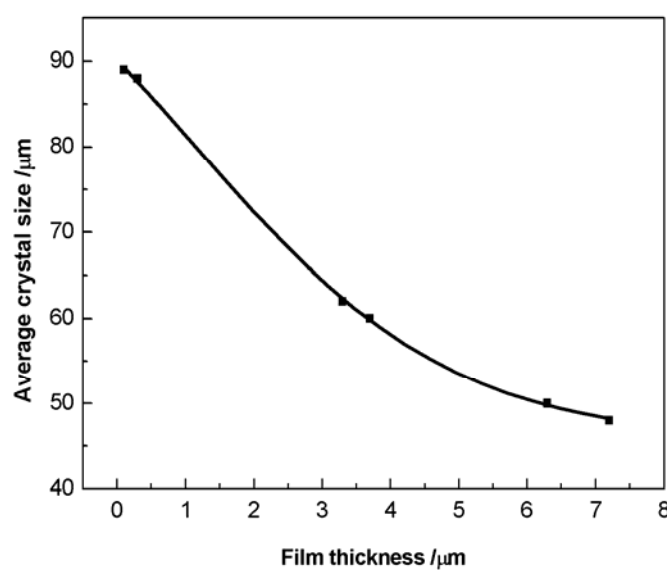
4.5.2 Crystal size variation corresponding to film thickness

Crystal size measurements were carried out for PCL crystallised films. The crystal size measurements were performed using optical microscopy images of crystalline films where the size of each crystal in the image was measured. Average size of all crystals obtained from each image of a particular thickness was calculated. Crystal size versus film thickness is shown in Table 4.5

Table 4. 5: Crystal size versus film thickness

Film thickness	Crystal size
/μm	/μm
0.3	88
0.1	89
3.3	62
3.7	60
6.3	50
7.2	48

The correlation of the average crystal size versus film thickness is shown in Figure 4.6.

**Figure 4. 6: Correlation between PCL average crystal size and film thickness**

The crystal size showed significant variation with film thickness. This variation correlated with nucleation and adsorption of molecules and growth rate. The variation is due to adsorption of PCL onto the substrate surface.

4.5.3 Melting temperature variation corresponding to film thickness

DSC melting temperature measurements was investigated using films of different thickness. The melting temperature for a range of film thickness is shown in Table 4.6 and the graph of melting temperature versus film thickness is shown in Figure 4.7.

Table 4. 6: Melting temperatures versus film thickness

Film thickness	Melting Temperature	Film thickness	Melting Temperature
/μm	/°C	/μm	/°C
0.1	59.5	1.7	59.1
0.5	59.0	3.3	59.7
0.7	60.2	4.1	59.7
0.9	58.8	6.6	59.0
1.1	60.2	9.6	59.2
1.5	58.8	16.3	59.6

Figure 4.7 shows that there is no correlation, the average melting temperature is constant. There is scatter of the melting temperatures at low thickness because the melting endotherm was low and broad, with less precise detection of the peak.

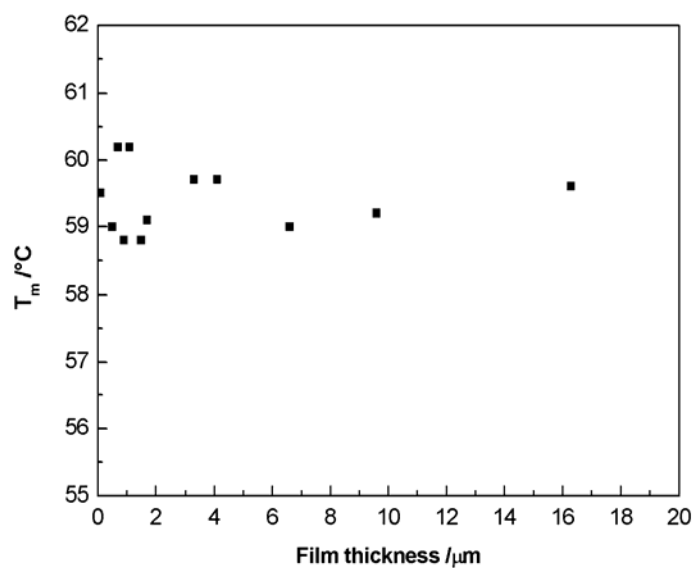


Figure 4. 7: PCL melting temperature versus film thickness

4.5.4 DSC crystallisation measurement of thin PCL film

DSC crystallisation of thin films was investigated. PCL films of four different thickness were prepared. The exothermic process shows that the DSC crystallisation curves for the thicker films such as 3.3 and 1.7 μm to start at lower temperature and with broader range of temperature while the thinner films such as 0.5 and 0.9 μm start at higher temperature with narrower range of the temperature as shown in Figure 4.8.

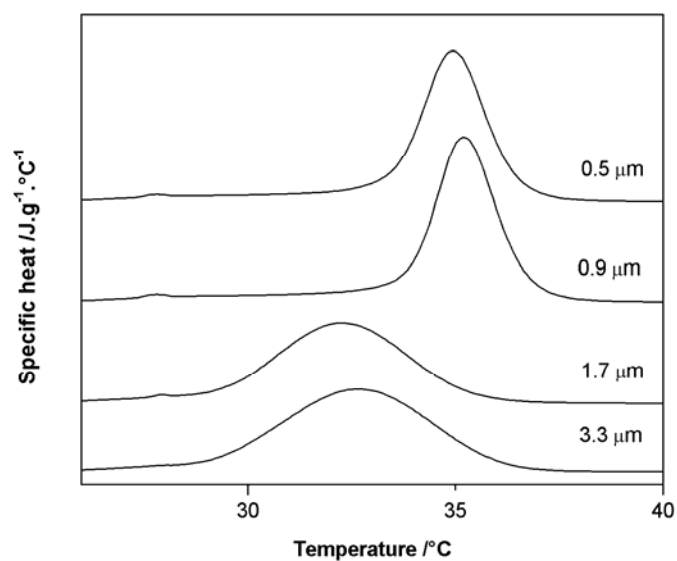


Figure 4. 8: Exothermic curves of PCL films of different thickness

The thinner films may crystallise at higher temperature due to nucleation of adsorbed molecules by the substrate surface. The specific heat curves of the melting endothermic peaks of the same films are shown in Figure 4.9. The peak melting temperature is constant for each of the film thicknesses.

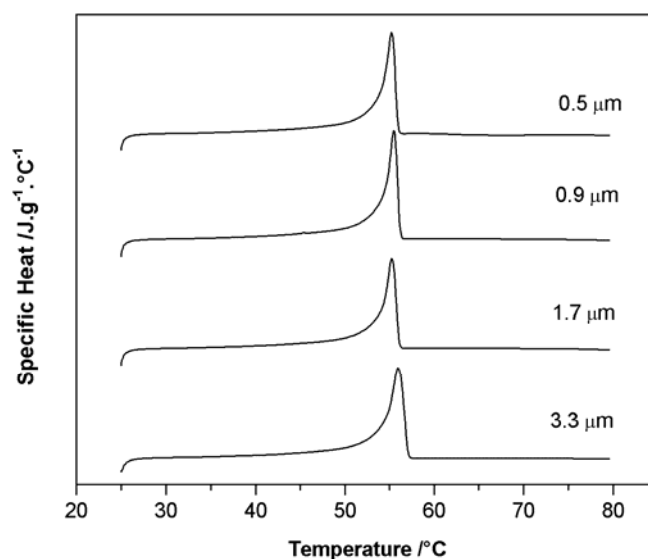
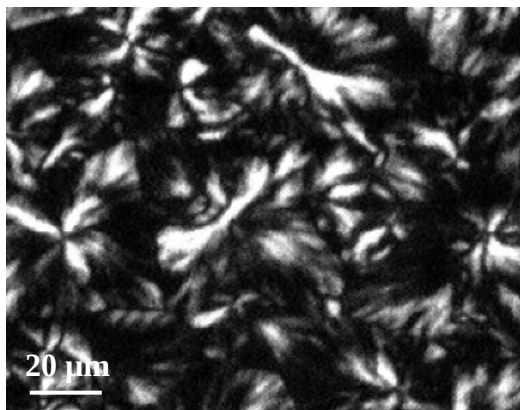


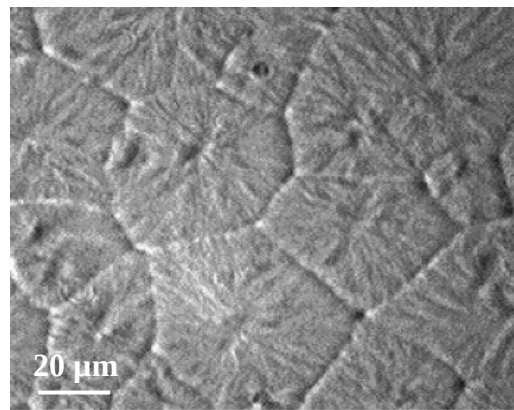
Figure 4. 9: Specific heat curves endothermic (melting) of PCL thin films for four different Thicknesses

4.5.5 Crystal morphology using optical microscopy

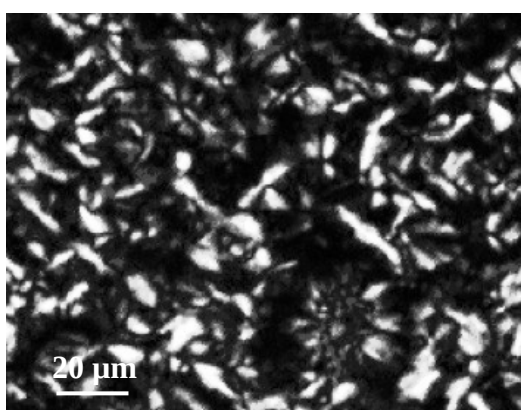
Microscopy of PCL thin films of different thickness for crystals is shown in Figure 4.10. The film thickness range that was used to investigate the crystal morphology of PCL, was between 0.90 - 0.10 μm . However the images of films of different thickness show that the crystal shape of PCL in the thinner films are larger, which is supported by the relation between the crystal size and film thickness shown in Figure 4.6. The optical microscope images of PCL crystals of films of different thickness using a polariser are shown in Figure 4.10 (a, b, c, d) and optical microscope images without a polariser are shown in Figure 4.10 (e, f, g, h).



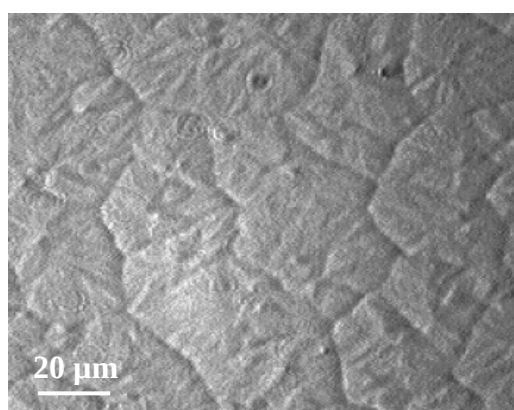
(a) film, thickness = 0.10 μm



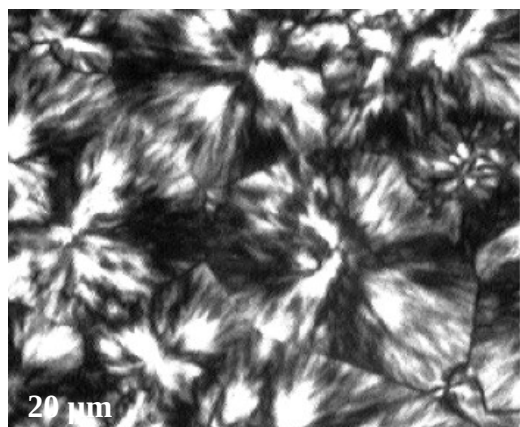
(e) film thickness = 0.10 μm



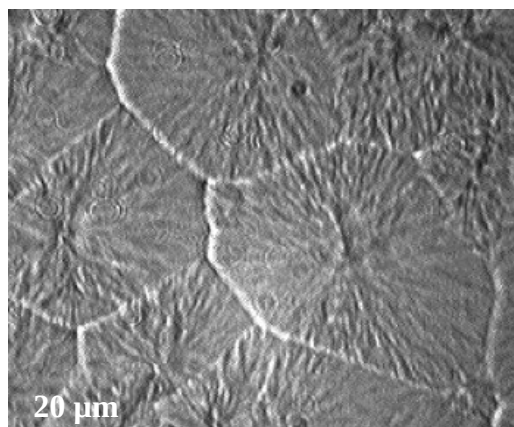
(b) film thickness = 0.30 μm



(f) film thickness = 0.30 μm



(c) film thickness = 0.50 μm



(g) thickness = 0.50 μm

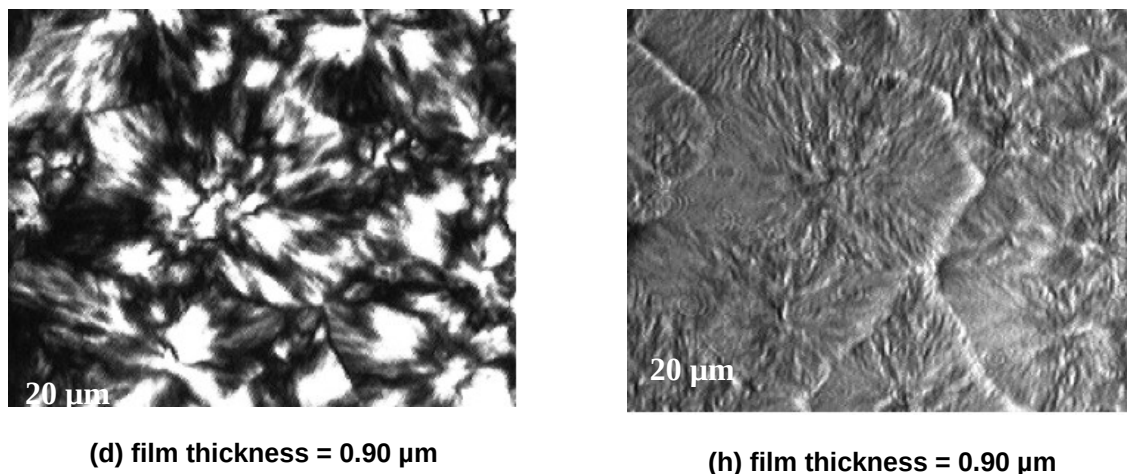


Figure 4. 10: Optical microscope images of PCL thin films, using polariser (left column) and without polariser (right column)

The images show the crystalline PCL films were completely filled with spherulites composed of lamellae spreading from each nucleus. The lamellae of thinner films have a higher nucleation density with less sharp boundaries, more edges meeting spherulites with less ordering and with smooth surface or less roughness. Dark boundaries, probably because of folding, can be seen in the crystals images. It can be seen that thicker cells have rough surface, smaller size, and more dark and bright surface shades because of roughness elevations.

4.5.6 Surface roughness measurements of PCL using surface image patterns

The surface roughness of four thicknesses using crystalline films was investigated. The data of the surface roughness of four different thickness textures of the crystal images were plotted to illustrate the comparison between the surface roughness and the crystal texture as shown in Figure 4.11.

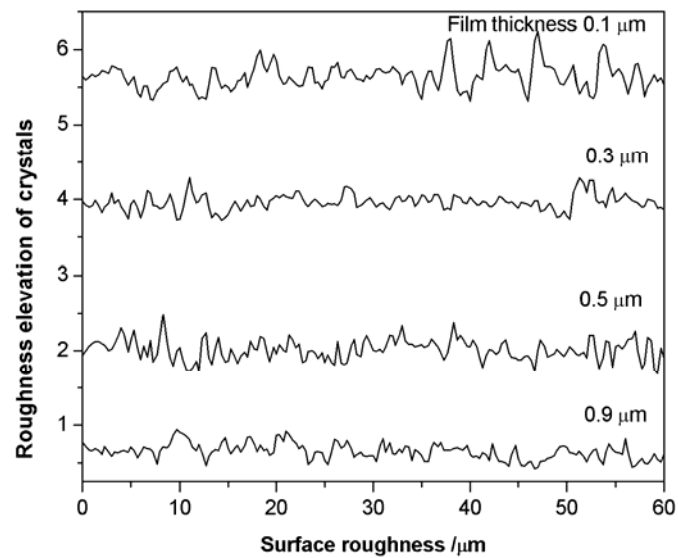


Figure 4. 11: Surface roughness of PCL crystals of four different thicknesses, the curves have been displaced by 2 units consecutively for clarity purpose

Figure 4.11 shows that the 2D profile for the roughness along the film surface of PCL crystal images significantly varied with film thickness. The data for the roughness profile measurement of the crystallised surface are shown in Table 4.7 for all four thicknesses.

Table 4. 7: Surface roughness measurement of PCL using image pattern

Spinning speed /rpm	Thickness / μm	Average surface roughness
1000	0.9	1.2
2000	0.3	1.0
4000	0.5	0.9
6000	0.1	0.6

Figure 4.12 shows a correlation between the roughness and film thickness using four different thicknesses, showing significant variation of roughness as the film thickness increased

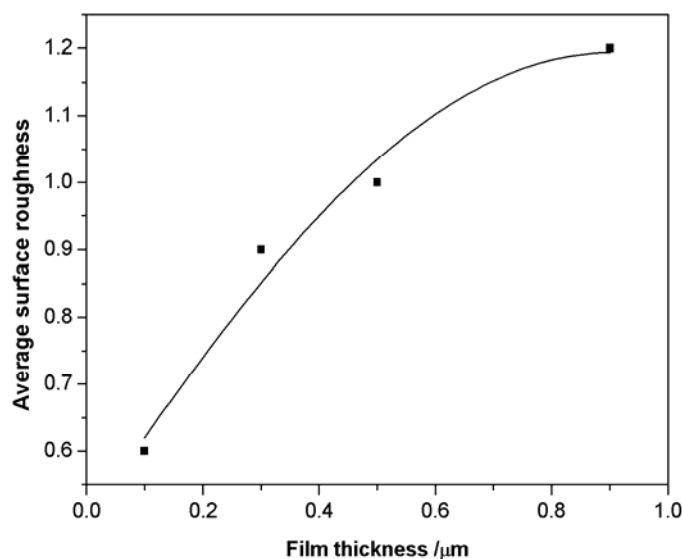


Figure 4.12: Average surface roughness of PCL film using films made from concentration of 0.4 g.mL^{-1}

4.6 Conclusions

Solution concentration and spinning speed were variables used to control film thickness and roughness. A wide range of thickness of PCL films was prepared and the correlations could be useful in future as a reference to make polymer films with specific thickness. Film thickness and roughness were correlated with variation of concentration and spinning speed. Film thickness and roughness showed a progressive decrease when the spinning speed was increased, PCL thickness and roughness considerably increased when the concentration of PCL was increased... Spinning time has caused no change in film thickness. DSC measurements were made to investigate crystallisation and melting variation with film thickness. Thinner films had higher melting enthalpy. Crystal size varied with film thickness. Thinner films had larger crystal size. The texture of PCL crystals caused the surface roughness, and thicker films showed increased roughness.

Chapter 5

Surface Morphology of Block Copolymers Thin Films

5.1 Introduction

Emphasis on nanometer scale entities has become a main topic of research in materials science. Novel approaches have recently appeared for the controlled synthesis and assembly of polymer and copolymer nanoparticles [189]. Independently of numerous potential applications, such polymers and copolymers are ideal candidates to study the effect of changes in their properties on the charge transport properties and linear and non-linear optical responses. Where organic materials and polymers are involved, a much wider range of nanoscale objects of controlled size and shape can be produced, going from the formation of thin films or to the design, synthesis and assembly of complex supramolecular structures such as block copolymers. The development of new analytic techniques has greatly contributed to progress in the study of nanoscale materials [190]. Roughness changes occur because of surface treatment of polymer blocks such as heating at different temperature [191]. Polymers with segregated structures have attracted researchers [190] and the degree of segregation of surface energy has been found to depend on the changes to the surface caused by the environment [192-194]. The properties of block copolymers depend on the composition of the blocks; the block composition may consist of two or more polymers in varying amounts and molecular weights. Any migration of high-energy functional groups to the surface can cause a strong adhesion interaction changing the surface property [195] and functional groups can segregate from the bulk to the surface to decrease the surface tension [196]. The molecular weight of each polymer block has an influence on the surface segregation. The formation of thin films has shown influences on variables such as glass transition temperature and surface energy.

5.2 Experimental

5.2.1 Film preparation

a) Poly(styrene-b-isoprene-b-styrene)

Tri block copolymer SIS a thermoplastic elastomer of formula; $[-CH_2CH(C_6H_5)-]_x[-CH_2CH=C(CH_3)CH_2-]_y[-CH_2CH(C_6H_5)-]_z$. was obtained from Sigma Aldrich, with 22 %wt. styrene. SIS (0.2 g) was mixed with 10 mL of toluene. The mixture was heated in an oil bath at 60 °C for 20 h while stirring using a magnetic stirrer. The apparatus is shown in Figure 5. 1. The solution was spun, using the spin coating technique, onto glass substrates to make films of less than 1 μ m in thickness. The films were left in a clean room for 1h and then they were stored in a vacuum oven at 45 °C for 20 h to allow evaporation of residual solvent. Then they were transferred to the clean room again and left to cool at room temperature.

b) Poly(styrene-b-butadiene-b-styrene) using toluene

Tri block copolymer SBS of formula $[-CH_2CH(C_6H_5)-]_x[-CH_2CH=CHCH_2-]_y[-CH_2CH(C_6H_5)-]_z$ was obtained from Sigma Aldrich, Mw 140,000 g/mol (GPC), with 30 %wt styrene as above. SBS, 0.2 g was mixed with 10 mL of toluene. The mixture was heated at 60 °C using an oil bath and stirred using a magnetic bar for 12 h. The solution after preparation was kept in the stirring setup (continuously stirring) before spin coating onto glass substrates. The spin coated films were dried in a vacuum oven at 45 °C for 24 h to allow evaporation of residual solvent and then left to cool at room temperature..



Figure 5. 1 Setting and instrumentation for preparation of copolymers solution

c) Poly(styrene-*b*-butadiene-*b*-styrene) tetrahydrofuran

SBS tri block copolymer of 0.2g was mixed with 10 mL of tetrahydrofuran (THF) solvent ($T_b = 80\text{ }^{\circ}\text{C}$). The solvent was heated using an oil bath at $50\text{ }^{\circ}\text{C}$ and stirred using a magnetic stirrer for 10 h. The solution was spun onto glass and silicon substrates to make films of different thickness. The films were left in clean room for 1 h. The films were then moved and kept in a vacuum oven at $45\text{ }^{\circ}\text{C}$ for 20 h. The glass slides and silicon wafers were rinsed with acetone and deionised water and then dried with pressurised nitrogen gas prior to spinning the copolymer solution. The spinning speed used for SBS solution was varied to get a series of thickness on each of the glass and silicon wafer substrates.

5.2.2 Instrumentation

Optical microscopy was used for imaging the surface texture after each treatment. Comparisons between surfaces of the same polymer with different treatments are reported with suggested explanations for each texture.

A XP-2 profilometer was used to investigate film thickness and the surface roughness of SBS and SIS films on the different substrates after each treatment. This equipment was used for scanning the surface to obtain measurements of surface roughness and thickness. This provided data on the surface roughness variations and revealed a banded surface with wave shapes throughout. 2D and 3D graphs of roughness measurements were used to quantify the morphology of the treated films of block copolymer. A vacuum oven was used to bake the copolymer films for different periods of time and at different temperatures under the same pressure. This was done to investigate the influence of the baking temperature and the length of baking time on the surface roughness and texture of the copolymer films before and after treatment.

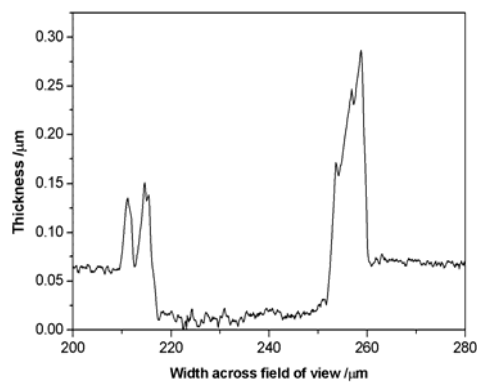
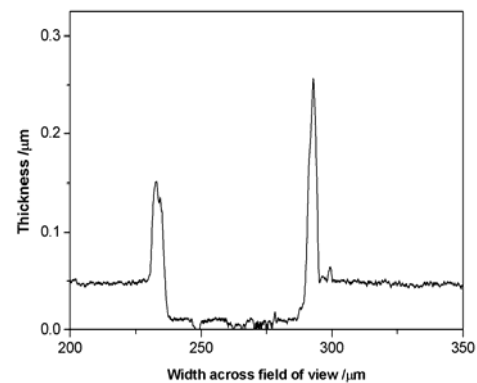
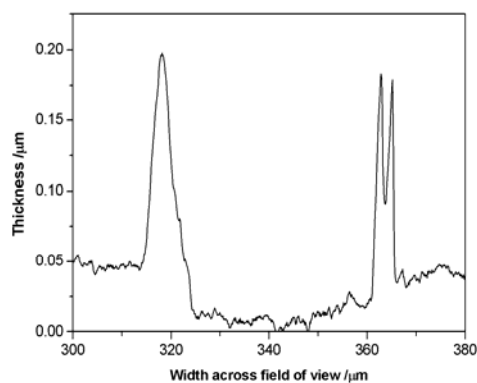
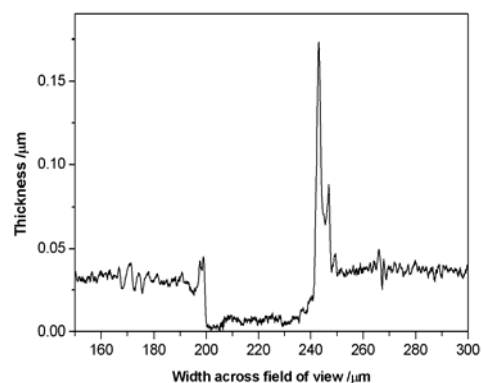
5.3 Results and Discussion

Solution of SIS was used to spin films of different thickness. Thickness measurements of SIS films were prepared for surface roughness and profile variation and investigated according to;

1. film thickness
2. heat treatment

5.3.1 Thickness measurement of SIS film

Thickness measurements of SIS films were carried out using an AMBIOS XP-2 profilometer. Four different thicknesses were selected for the profile variation investigation as shown in Figure 5. 2

(a) Steps on the surface: 0.048 μm (b) Steps on the surface: 0.035 μm (c) Steps on the surface: 0.029 μm (d) Steps on the surface: 0.025 μm **Figure 5. 2 Thickness measurements of four different SIS films****5.3.2 Surface texture of SIS film after heating at 45 °C**

Four different thickness SIS films on glass substrates were used to image surface texture. The films were left for 24 h in clean room before imaging the surface texture. The optical microscope images for SIS films after heating at 45 °C were captured and show no difference in their texture, see Figure 53.

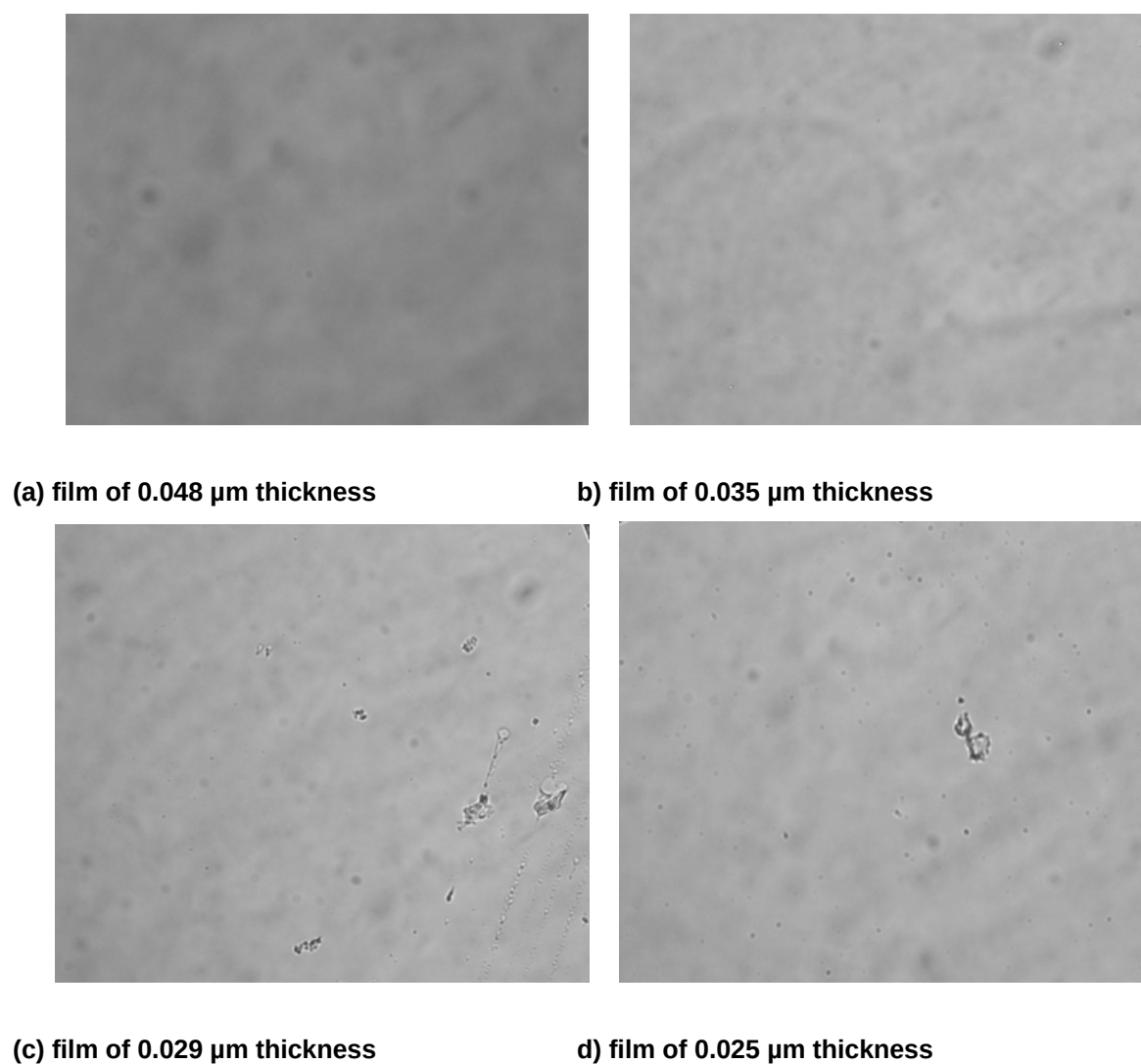
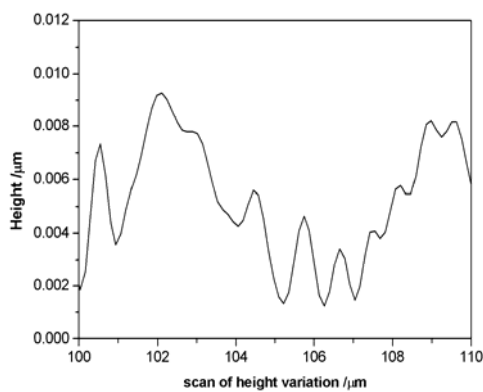
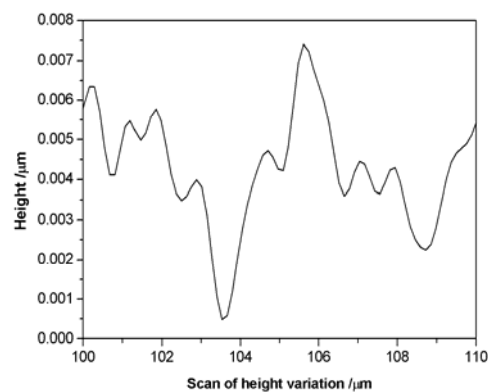


Figure 5. 3: Optical microscopy images of surface texture of SIS films, after heating at 45 °C

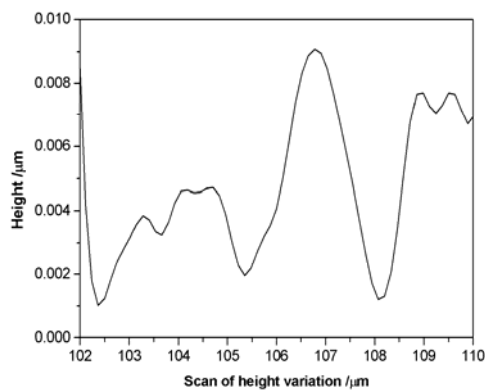
The surface profile of the SIS copolymer films was measured using an Ambios profilometer and is shown in Figure 5.4 for four different thicknesses. The figures show the same texture of the surface profile. The images show no evidence of phase separation since the SIS films were heated at 45 °C and that was below the T_g of polystyrene. So the polyisoprene keeps domain the texture.



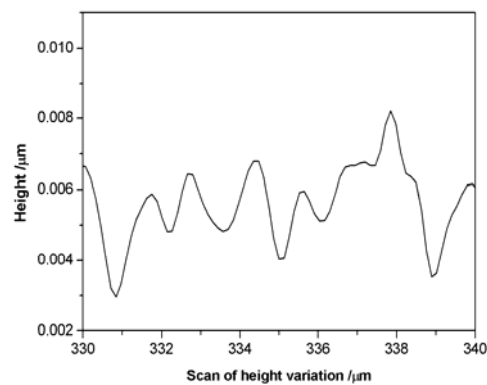
(a) Surface profile of roughness,
Average roughness 0.0122 μm



(b) Surface profile of roughness,
Average roughness 0.0083 μm



(c) Surface profile of roughness
Average roughness 0.0092 μm



(d) Surface profile of roughness
Average roughness 0.0066 μm

Figure 5. 4: Surface Profiles used for determining surface roughness measurement of four SIS films after heating at 45 °C

5.3.3 Surface roughness of SIS film after heating at 45 °C

The thickness and roughness variations with the copolymer concentration and spinning speed are shown in Table 5.1. The effect of heating on SIS surface roughness at 45 °C is shown with surface roughness varying between 0.0122 μm for the thickest film and 0.0066 μm for the thinnest film.

Table 5. 1: SIS measurement of films heated in a vacuum oven at 45 °C for 20 h

Spinning speed	Film thickness	Average film roughness
/rpm	/μm	/μm
1000	0.048	0.0122
2000	0.035	0.0083
4000	0.029	0.0092
6000	0.025	0.0066

The relation between film thickness and roughness with spinning speed for SIS films heated at 45 °C respectively are shown in Figure 5.5. In Figure 5.5 b, the roughness decreased when film thickness (spinning speed) decreased

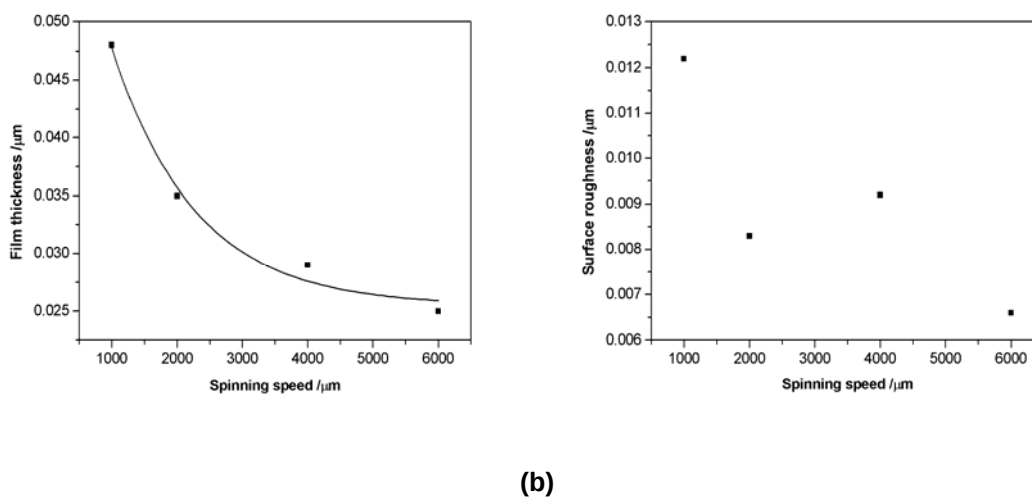


Figure 5. 5: SIS films after heating at 45 °C (a) thickness and (b) roughness measurements versus spinning speed

5.3.4 Surface texture of SIS film after heating at 120 °C

The texture variation of SIS films was investigated after the films were heated at 120 °C for 3 h in a vacuum oven. After heating, the films were cooled to room temperature for 1h. Optical microscope images were captured for films of different thickness after heating at 120 °C and are shown in Figure 5.6. The images of different thickness were found filled with bulges.

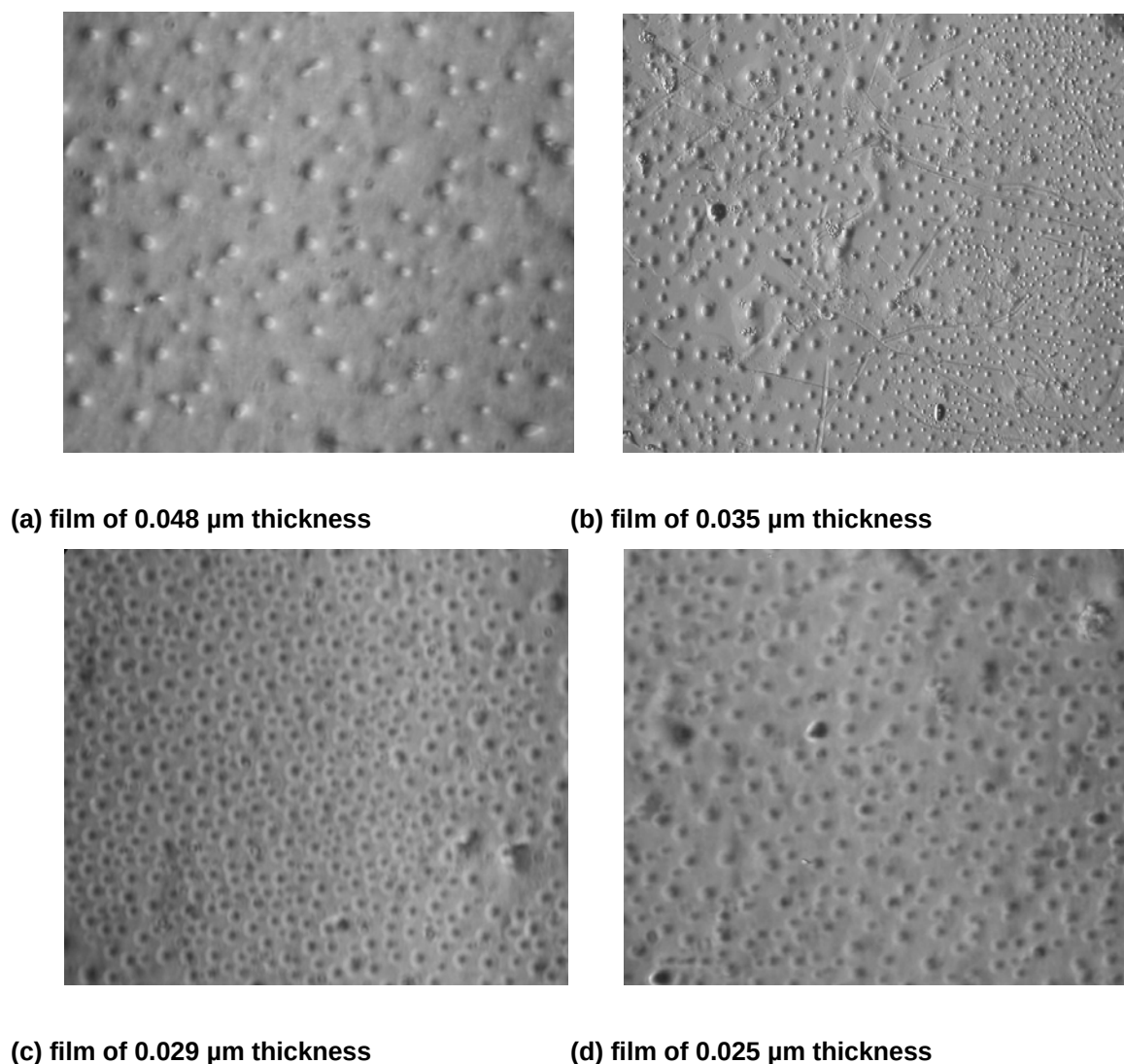


Figure 5. 6: Optical microscope images of SIS surface texture of films having different thickness, after heating at 120 °C

The profile variation with film thickness was investigated by scanning across the film surface and the data of profile variation was plotted in Figure 5.7. The roughness measurements show the bulges which described as higher and wider for thicker films. The bulge height decreased with decreasing film thickness. Heating at 120 °C is above T_g of both polystyrene and polyisoprene phases of the block copolymer. The polystyrene has separated more than the polyisoprene itself producing many surface bulges.

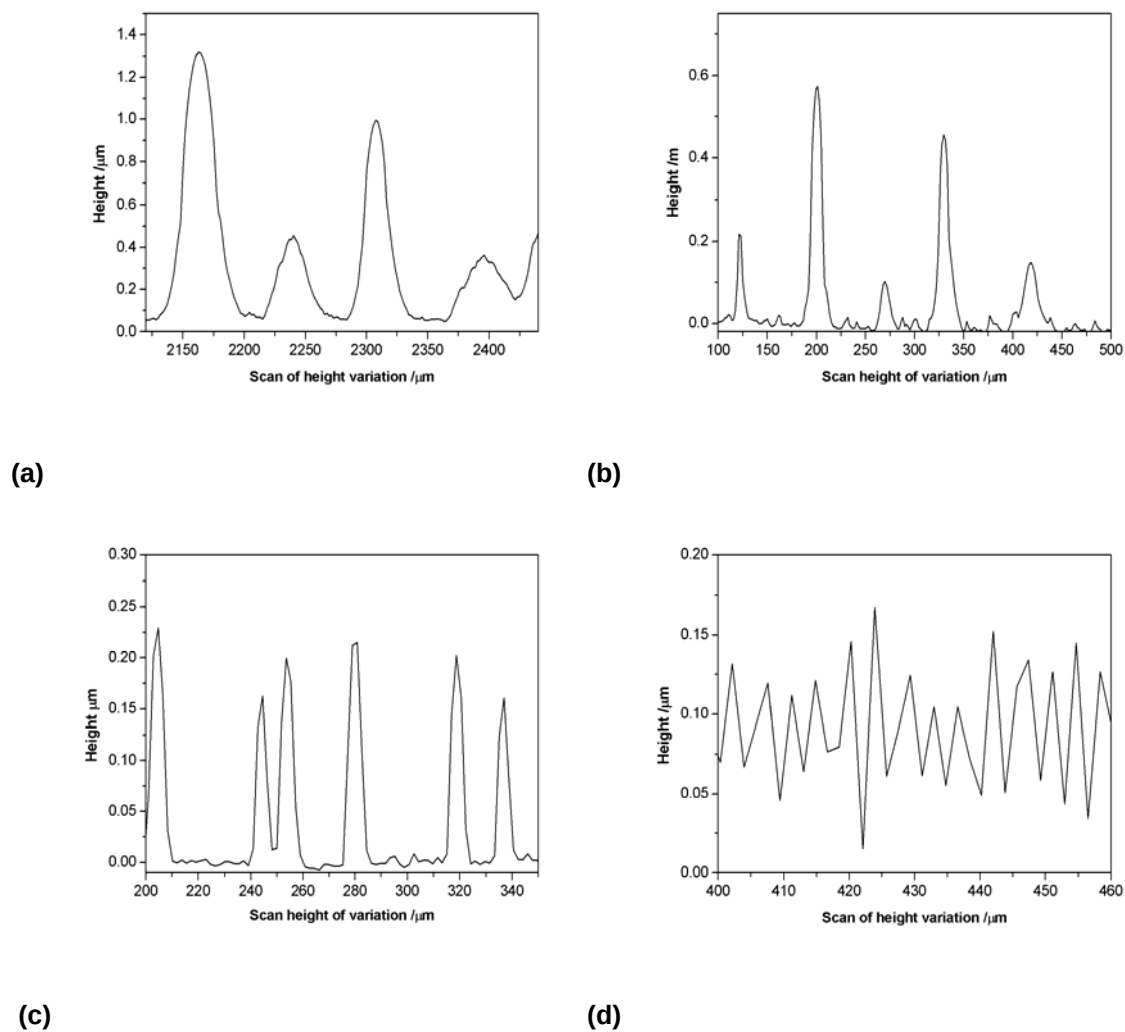


Figure 5. 7: Surface profiles of SIS films of different thickness, after heating at 120 °C (a) 0.048 μm thickness (b) 0.035 μm thickness (c) 0.029 μm thickness (d) 0.025 μm thickness

5.3.5 Surface roughness of SIS film after heating at 120 °C

The data for surface roughness after heating SIS films at 120 °C are shown in Table 5.2 for films of different thicknesses. The data shows that the bulge height and width were decreased when the film thickness decreased. The distance between bulges changed as their width changed.

Table 5. 2 SIS surface measurements for films, heated at 120 °C

Film Thickness	Average bulge Height	Average bulge Width	Average distance
/μm	/μm	/μm	between Bulge /μm
0.048	0.749	80.2	7.6
0.035	0.291	34.4	42.4
0.029	0.192	15.1	33.7

The surface profile of SIS films after heating at 120 °C (Figure 5.7) changed when compared with the profile of SIS films after heating at 45 °C (Figure 5.4). Heating films of different thickness at the same temperature will cause different surface texture as shown in Figure 5.6

5.3.6 Surface texture of SIS film after heating at 160 °C

The surface texture variation of SIS films after heating at 160 °C was investigated. The SIS films of different thickness were heated for 3 h then cooled to room temperature. Optical microscope images were captured for four films as shown in Figure 5.8. The film texture was similar to the film texture in Figure 5.6 of SIS films.

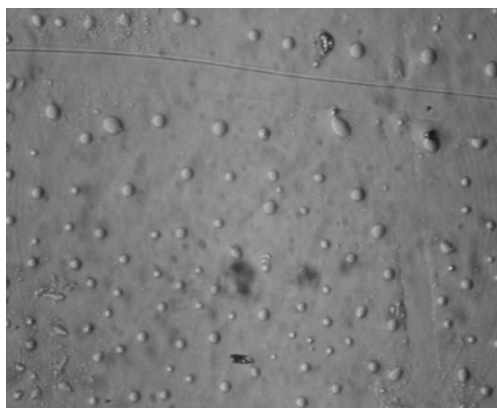
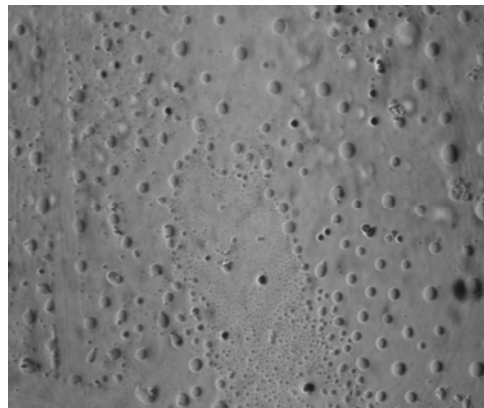
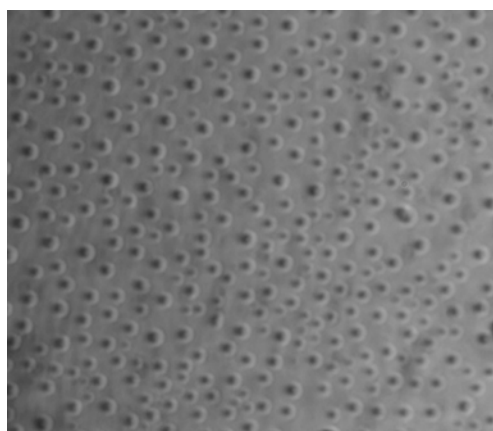
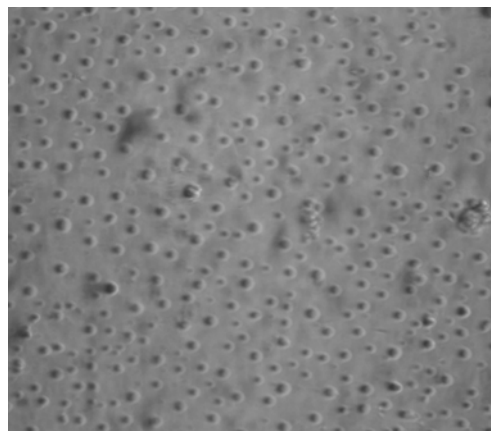
(a) film of 0.048 μm thickness(b) film of 0.035 μm thickness(c) film of 0.029 μm thickness(d) film of 0.025 μm thickness

Figure 5. 8: Images of surface texture of SIS films of different thickness, after heating at 160 °C

The surface profile of SIS films was measured using the profilometer see (Figure 5.9) and shows changes when compared with the surface profile of films after heating at 120 °C see (Figure 5.7). The profile variation of SIS films according to the film thickness was measured and investigated using films of different thicknesses and are shown in Figure 5.9 The four figures show that bulges height and width and the number of bulges per 300 μm length of scan have changed with heat treatment at 160 °C see (Figure 5.9) when compared with bulges of the same thickness of film after treated at 120 °C see (Figure 5.7).

Heating at 160 °C was similar to heating to 120 °C except the density of surface bulges was decreased. The decreased density of bulges may be close to increased miscibility of the two

phases at the higher temperature of 160 °C. Upon cooling the polystyrene phase becomes glassy at its T_g of 100 °C.

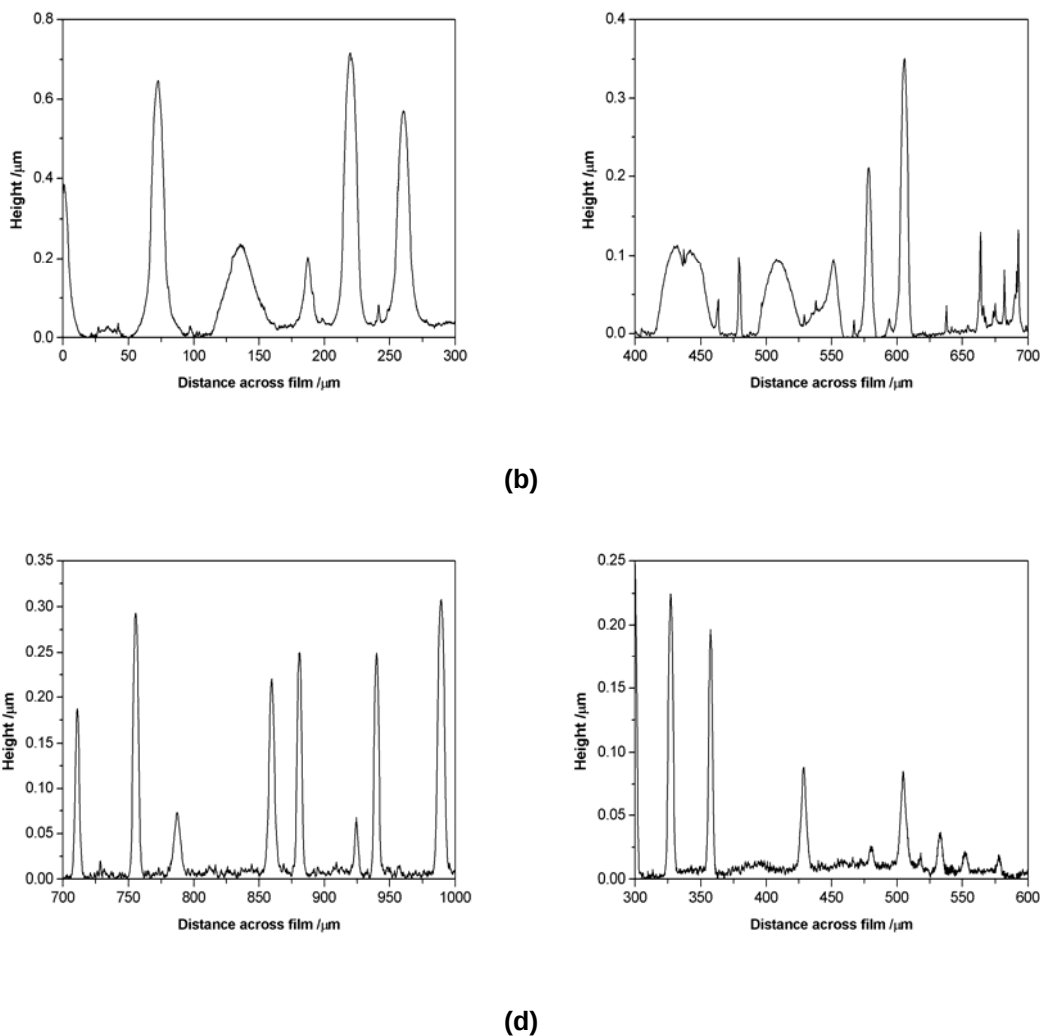


Figure 5. 9: Surface profiles of SIS films of different thickness, after heating at 160 °C.
(a) 0.048 μm thickness (b) 0.035 μm thickness (c) 0.029 μm thickness (d) 0.025 μm thickness

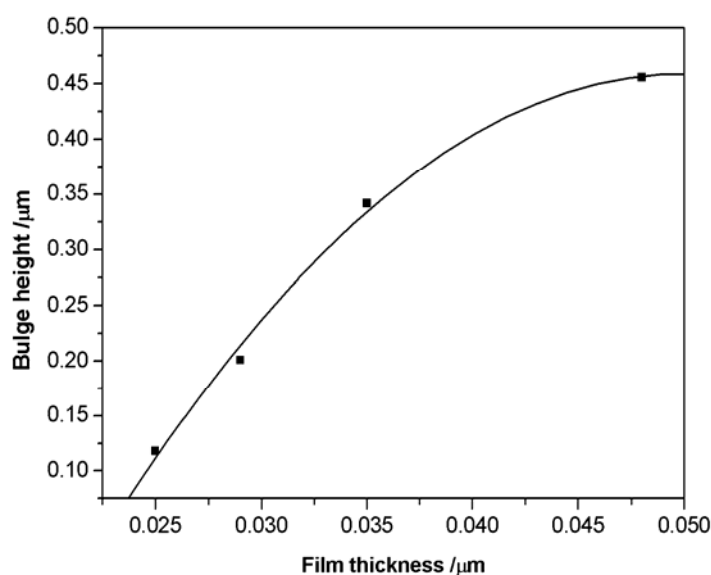
5.3.7 Surface roughness of SIS film after heating at 160 °C

Surface roughness measurements of SIS films are given in Table 5.3. The data includes bulge height, width and distances between the bulges for four different SIS film thicknesses, after heating at 160 °C.

Table 5.3 SIS roughness measurements for films of different thickness heated at 160 °C

Film Thickness	Average bulges Height	Average bulges Width	Average distance between Bulges
/μm	/μm	/μm	/μm
0.048	0.456	35.6	14.0
0.035	0.342	26.8	20.6
0.029	0.201	23.4	23.4
0.025	0.118	14.0	27.4

The relationship between bulge height and the film thickness was investigated for SIS films treated at 160 °C. The relationship is shown in Figure 5.10. The figure shows that the height of these bulges increased with film thickness. The height of bulges also increased when heating temperature increased as shown in Table 5.2 and Table 5.3 the 0.048 μm films has a bulge height of 0.749 μm after treating at 120 °C and 0.456 μm after treating at 160 °C.

**Figure 5. 10: SIS film average bulge height versus film thickness for films heated at 160 °C**

The relationship between bulge width and film thickness was investigated. The data for the surface roughness shows that the bulge width was changed when the film thickness changed. The data is plotted in Figure 5.11. Bulge width variation occurred when the temperature of the

treatment was increased for the same film thickness of $0.048\mu\text{m}$. For example; the film after heating at $120\text{ }^{\circ}\text{C}$ had a bulge width of $80.2\text{ }\mu\text{m}$ and the film after heating at $160\text{ }^{\circ}\text{C}$ had a width of $35.6\text{ }\mu\text{m}$.

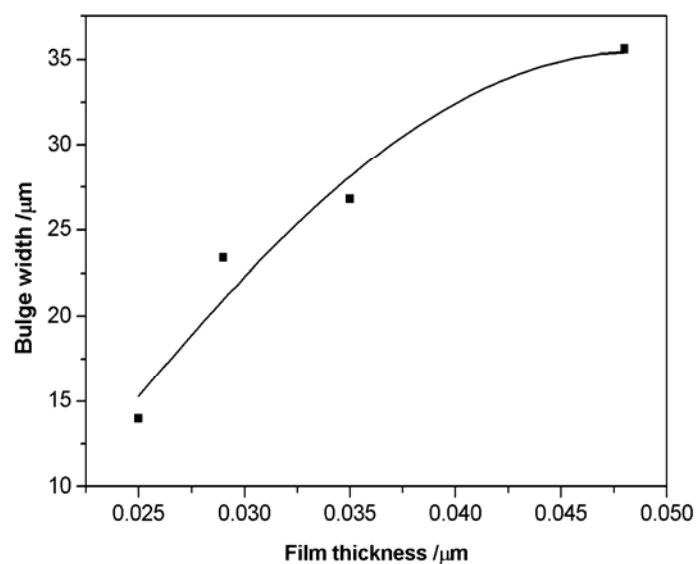


Figure 5. 11: The relation of bulge width with film thickness for SIS films after heating at $160\text{ }^{\circ}\text{C}$

The distance variation between the two close bulges versus thickness was investigated. The relationship is plotted in Figure 5.12.

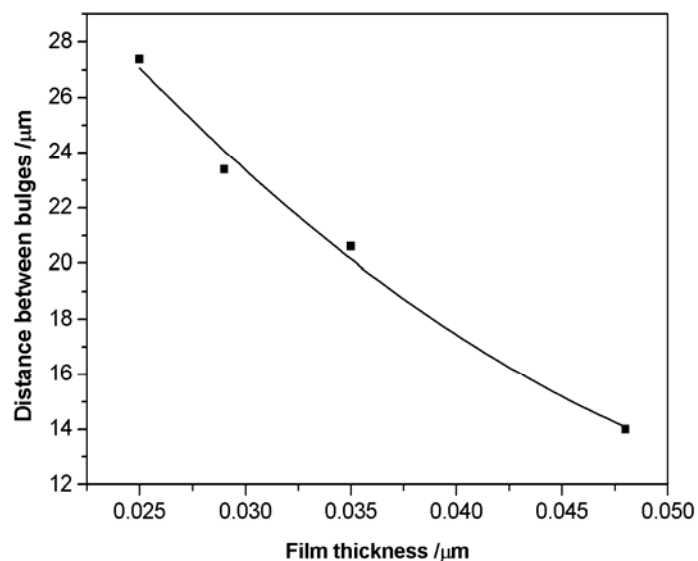


Figure 5. 12: Average Distance between bulges versus film thickness of SIS films heat treated

This relation shows that the distance between the bulges increased when film thickness decreased. The distance between centre of bulges changed with temperature variation, for example for the distance between two bulges after heating the film at 120 °C was 7.6 μm as shown in Table 5.2 and for those after heating at 160 °C the distance was 14.0 μm as shown in Table 5.3. Both SIS films have the same thickness of 0.048 μm .

5.3.8 Solvent influence on the surface texture of SIS film

Solvent influence on surface texture of block copolymer film was investigated. Film of SIS was prepared by spinning solution of SIS mixed with diethylether. The film was spun using glass substrate then was dried to room temperature for 3 h. The film surface texture was investigated using optical microscope image as shown in Figure 5.13. The surface texture was different from the texture got on the SIS film surface made from toluene



Figure 5. 13. SIS copolymer surface texture using diethylether

5.4 SBS Surface Morphology Variation

Solutions of SBS were prepared using different solvent to spin films of different thickness. Thickness measurements of SBS films were prepared to investigate surface roughness variation according to; film thickness, solvent and substrate.

5.4.1 Thickness measurement of SBS film

Thickness measurements of SBS films were carried out using an Ambios XP-2 profilometer, as shown in Figure 5.14. Thickness data are shown in Table 5.4. The four films were used for texture and profile investigation.

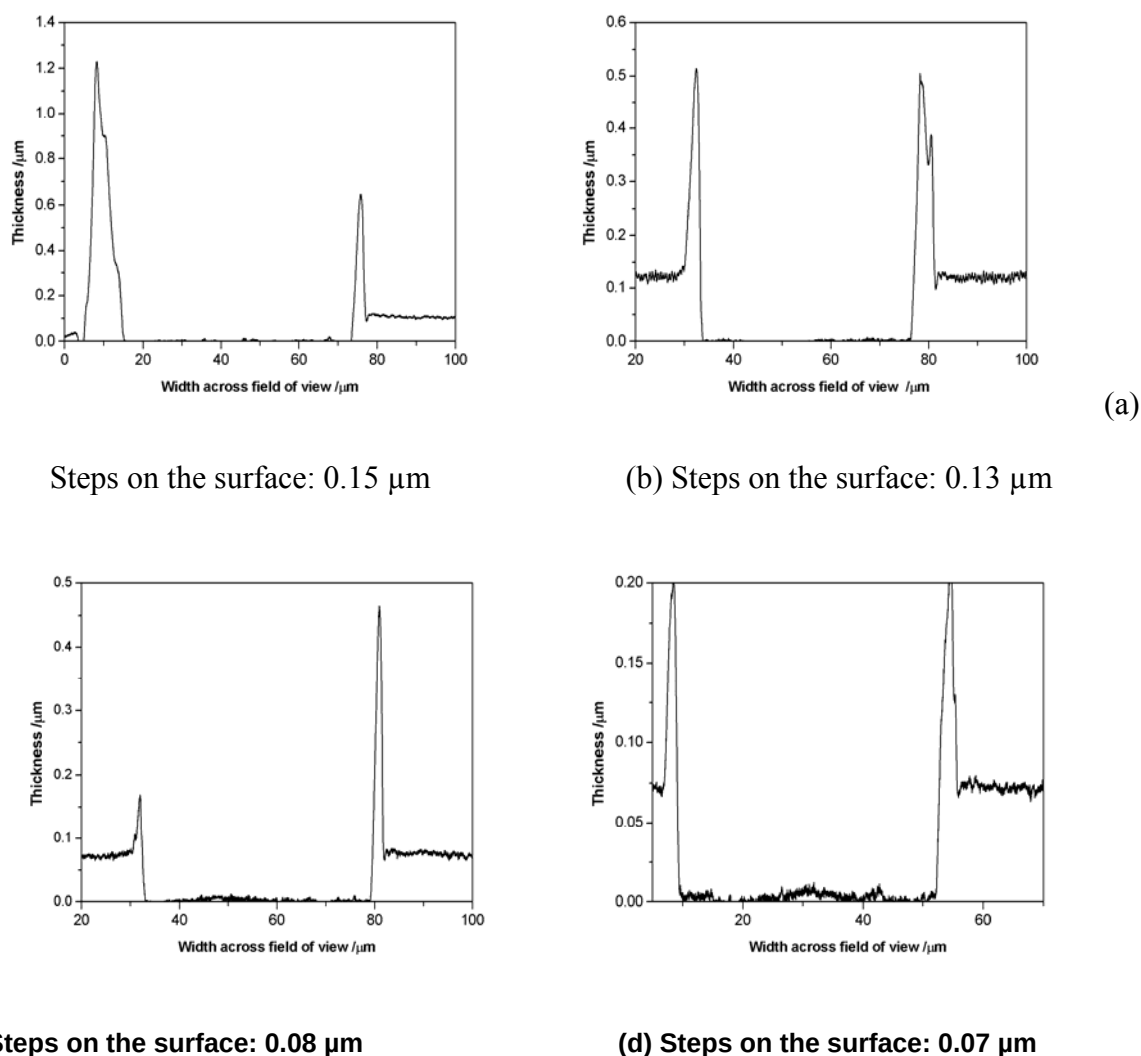
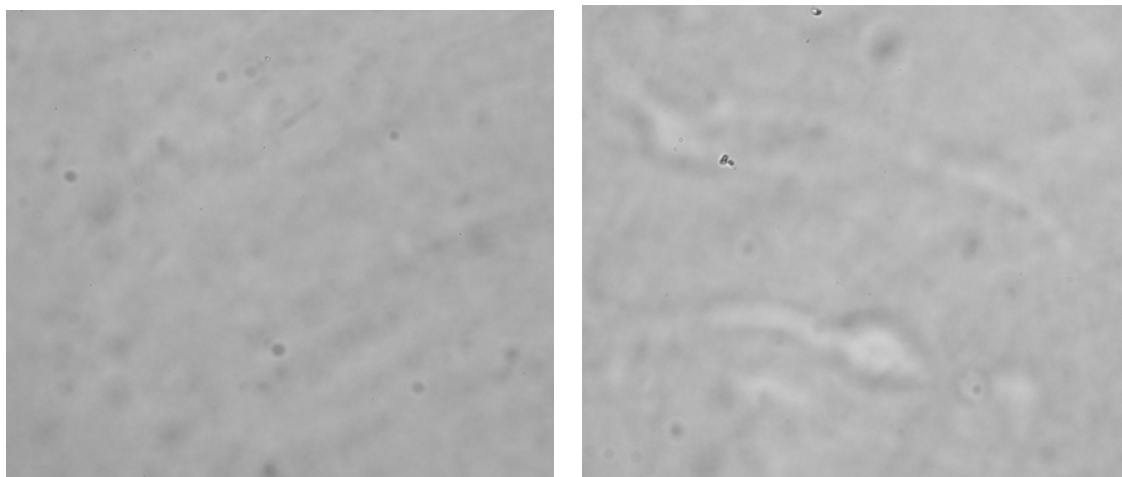


Figure 5. 14 Thickness measurements of four different SBS films

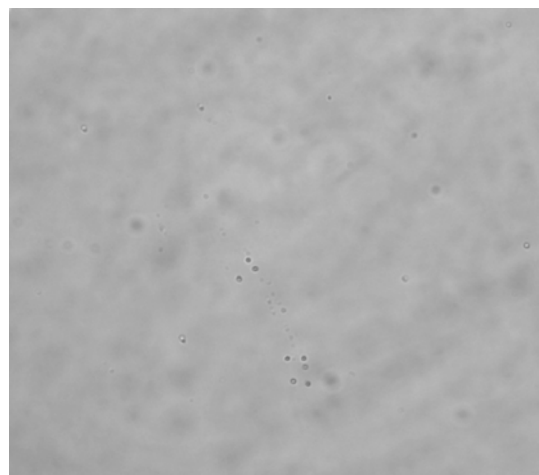
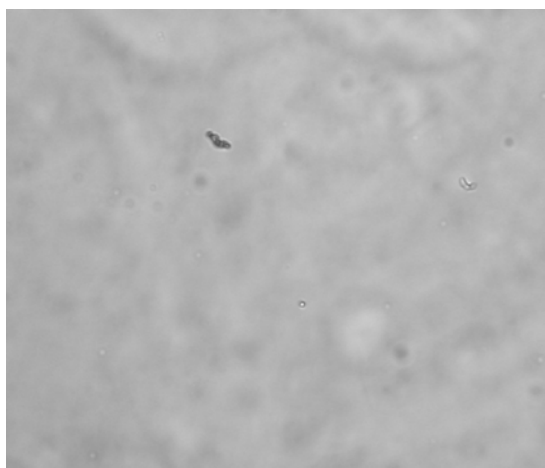
5.4.2 Surface texture of SBS using toluene as solvent

The SBS films were prepared from toluene solution. The films were cooled to room temperature. The surface texture of SBS after heating at 45 °C was investigated. The surface texture was imaged using an optical microscope, as shown in Figure 5.15. The film images of four different thicknesses show no difference in their texture as heated at 45 °C.



(a) film of 0.15 μm thickness

(b) film of 0.12 μm thickness



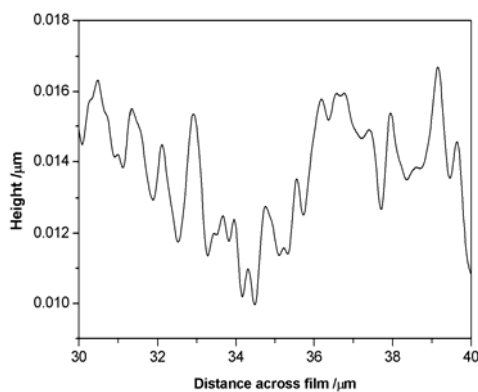
(c) film of 0.08 μm thickness

(d) film of 0.07 μm thickness

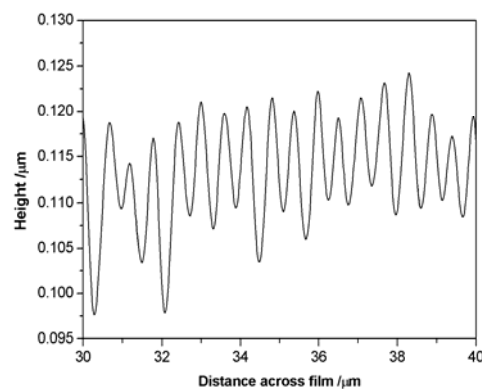
Figure 5. 15 Images of SBS surface morphology of films have different thickness, after heating at 45 °C

5.4.3 Surface roughness of SBS film

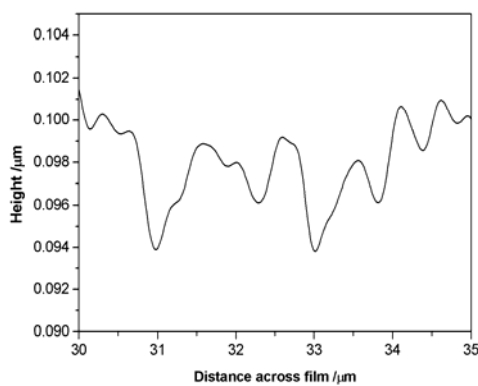
The surface roughness of four SBS films was measured using Ambois profilometer. Four plots of surface roughness are shown in Figure 5.16. The surface roughness of the film texture has shown variation to the surface according to the film thickness.



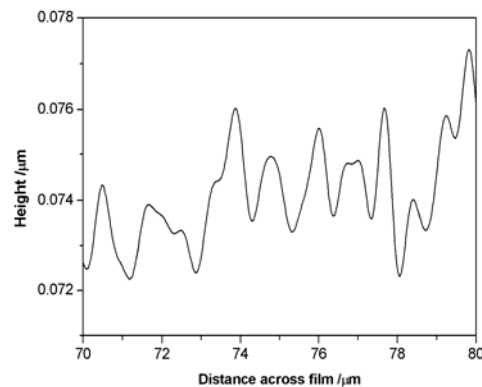
(a) Average surface roughness 0.024 μm



(b) Average surface roughness 0.021 μm



(c) Average surface roughness 0.013 μm



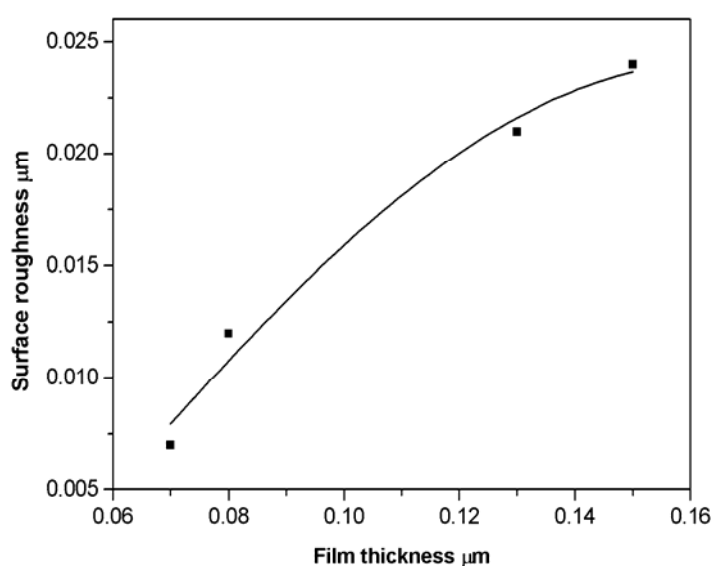
(d) Average surface roughness 0.007 μm

Figure 5. 16 Surface profiles of SBS film, used to determine surface roughness for films of four different thicknesses.

The data of thickness and roughness variations with copolymer concentration and spinning speed are shown Table 5.4. Average roughness variation versus film thickness is shown in Figure 5.17. The plot shows variation between 0.024 μm for the thickest film to 0.007 μm for the thinnest film.

Table 5. 4 Measurement data of surface roughness and film thickness for SBS films on glass substrates heated at 45 °C

Spinning speed	Film thickness	Average film roughness
/rpm	/μm	/μm
1000	0.15	0.024
2000	0.13	0.021
4000	0.08	0.012
6000	0.07	0.007

**Figure 5. 17 SBS surface roughness versus film thickness for films after heating at 45 °C**

5.4.4 Substrate effects on film thickness and surface roughness of SBS films

Surface texture of SBS films spun onto different substrates was investigated. The films were spun from solutions of SBS prepared using THF solvent and using two types of substrates, glass and silicon. The films were dried using the same conditions of a vacuum oven heating at 45 °C for 20 h, to evaporate the residual solvent. The films were then left for 6 h in a clean

room before any treatment. The thickness and surface roughness of the films were measured using the same technique (Ambios XP-2 profilometer). The measurements are given in Table 5.5.

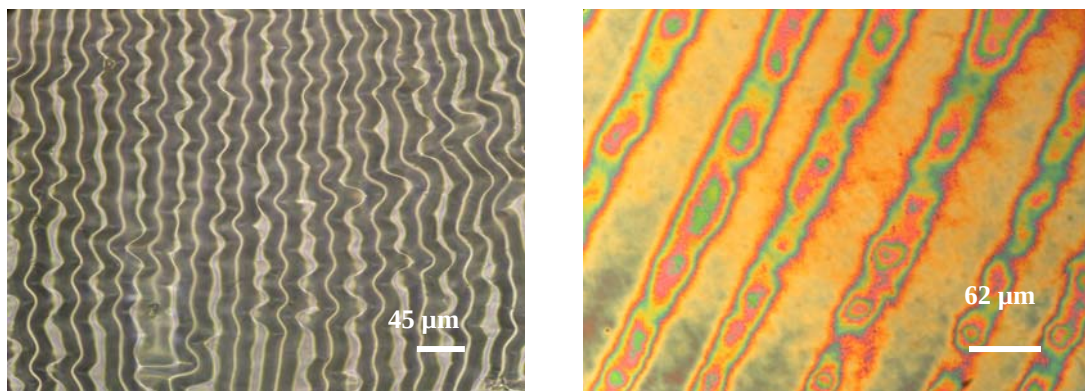
Table 5. 5 SBS film thickness and surface roughness using different substrate

Substrate	Film thickness	Surface roughness
	/μm	/μm
Glass	0.09	0.050
Silicon	0.03	0.025

The data in Table 5.5 shows the thickness of a film measured on a silicon substrate was one third that of the thickness measured for a film spun on a glass substrate. The surface roughness of the film spun on silicon was half the roughness measured for the film on a glass substrate.

5.4.5 Solvent influence on the surface texture of SBS film

The effect of solvent on the surface texture was investigated using SBS films. SBS films were prepared using THF solvent instead of toluene. The surface texture has changed as shown in Figure 5.18 when compared with the texture shown in Figure 5.15. The evaporation of THF influenced the surface roughness and surface texture. The surface texture using different substrates was investigated. Two different substrates (glass and silicon) were used to compare the surface texture using SBS prepared from solution of SBS in THF. The optical microscope images of SBS films spun onto glass and silicon are shown in Figure 5.18 (a) for a glass substrate with an SBS film thickness of 0.09 μm and (b) a silicon wafer with SBS film thickness of 0.03 μm.

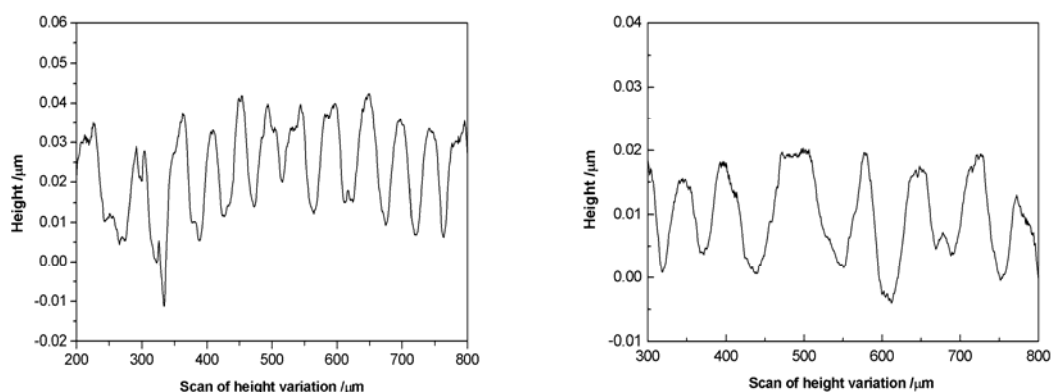


(a)

(b)

Figure 5. 18 SBS film surface (a) on a glass substrate and (b) on a silicon wafer

The surface profile of two films in Figure 5.18 (a) on glass and (b) on silicon were measured and a plot of data is shown in Figure 5.19 (a) for an SBS film spun on a glass substrate and (b) an SBS film spun on a silicon substrate.



(a)

(b)

Figure 5. 19 Surface profile of SBS film for films spun on (a) a glass substrate and (b) on a silicon wafer. The films were spun from solution of THF solvent

The two profiles of Figure 5.19 (a) and (b) show surface texture with patterns of ripples and show the number of ripples that occurred per unit length across the film surface are different. This number was 8t ripples per 500 μm using silicon as the substrate and was 11 per 500 μm for glass. The thickness measurements of the SBS film on silicon was one third the thickness of the film using a glass substrate and both films were spun from a solution of the same

concentration of SBS. Surface roughness variation of SBS film probably occurred due to copolymer block segregation to the surface and due to surface tension. The surface pattern in elastomeric structure is demaine by polybutadiene. The SBS backbone chain structure contains three segments; first is a short chain of polystyrene, second is a long chain of polybutadiene and the third segment is another short section of polystyrene as shown in Figure 5.20. The figure shows an elongated chain of SBS. [197 - 199]

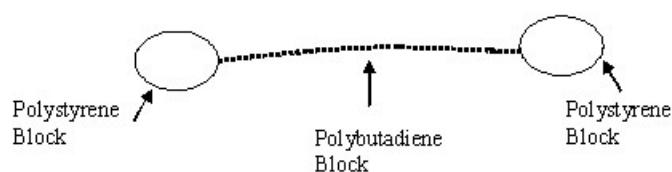


Figure 5. 20 The SBS block copolymer backbone

5.5 Conclusion

Surface property variation acquired through thermal treatment produces different textured surfaces on block copolymer films. The treatments were carried out using different thickness films. The texture variations of the surfaces were obtained when the films were spun from copolymer solutions with different solvents, toluene THF and diethylether. SIS film, after evaporation of toluene showed smooth surface at room temperature, and SIS after evaporation of diethylether showed surface with ripples. The surface texture was obtained when the film spun on different substrates, glass or silicon. SBS film, after THF evaporation showed surface with ripples and number of ripples per unit length were different when silicon used instead of glass. Surface roughness variation was obtained when two different block copolymer films (SIS or SBS) were used and for films of different thickness. The surface roughness of SIS block copolymer showed variation with film thickness and treatment at different temperatures. The films of different thickness have shown texture variation on the surface after heating to different temperatures. Three different temperatures were applied; 45 °C, 120

°C and 160 °C to the SIS films. The texture of the SIS surface heated at 45 °C was smooth but the SIS film surface that was heat treated at 120 °C and 160 °C were filled with bulges suggesting copolymer block segregation and varying surface tension. The difference between SIS images of bulges for films heated at different temperatures was the size of bulges. The graphs of roughness measurements have shown bulge profiles that were higher of larger area, with smaller distances between bulges, for films heated at 120 °C. The thickness variation resulted in different textured surfaces for films heated at the same temperature SIS films of different thickness heated at 45 °C have shown images with smooth surfaces, with no distinguishing features.

Chapter 6

Amorphous thin Polymer films

6.1 Introduction

Demands, on developing methods for making and using thin polymer films for optical devices have increased because of developments in advanced technology, such as progress in the field of electro optical devices and their applications. Modification of the optical properties of polymer materials is of interest in photonics applications of polymer films requiring control of thickness, texture and refractive index. The advent of rapid solidification techniques has established that many materials can form glasses irrespective of their physical and chemical properties [200].

There is interest in the nature of the transition from liquid to glass and the dynamics of molecular motion in glassy systems [201] in the form of thin films, such as thin polymer films used in E-O applications. Thin polymer film photoresists are used widely in semiconductor lithography, with film thickness of 100 nm [202]. The glass transition temperature is a critical material property for polymers used in chemically amplified photoresists. The selected processing and application temperature for a photoresist film is influenced by T_g . The T_g of amorphous polymers has been known to be the most important parameter determining their physical properties [203]. There are numerous technologies that focus on polymer thin films such as using thin polymer films as resists, interlayer dielectrics, alignment layers and lubricants [201, 203]. Many measurements of T_g in thin polymer films have been performed with different methods, such as ellipsometry, X-ray reflectivity, Brillouin light scattering, and positron annihilation lifetime spectroscopy [201, 202, 63]. The interaction between polymer

and substrate can have an influence the T_g , specifically the interaction may be large enough to shift the T_g [204, 205].

Polymers are used in the structure of waveguides, consisting of a polymer thin film with a raised beam where light can be directed by the polymer, such as shielding layers guiding light due to a specific refractive index. Light absorption and scattering at the interface of a shield layer can be minimized through modification of the polymer properties. Some techniques are used to minimise losses in the light quality such as smoothing and etching of the shield layer.

PMMA is considered as a suitable polymer for optical waveguide construction because of its high transparency.

6.2 Experimental

6.2.1 Materials

(a) Poly(methyl methacrylate)

PMMA [$-\text{CH}_2\text{C}(\text{CH}_3)(\text{CO}_2\text{CH}_3)-$], of $M_w = 9.96 \times 10^5 \text{ g.mol}^{-1}$, was purchased from Sigma-Aldrich Chemicals, and used to prepare thin films of different thickness. PMMA was chosen as the guiding layer because it has a higher refractive index than UV15. The refractive index of the PMMA film was measured to be 1.48 at 633 nm using a reflectance method. PMMA is made by free radical vinyl polymerization from the monomer methyl methacrylate. PMMA is an amorphous clear plastic, used as replacement for glass. PMMA has an advantage over glass in that it is more transparent than glass. Thick PMMA is a suitable material for making transparent sheets. PMMA does not dissolve in water. PMMA is known for excellent long term weatherability. PMMA is used in packaging (food handling trays and display boxes), automotive (instrument lenses, tail lamp lenses and indicator lamp lenses), house wares (jugs

and tumblers), kitchen appliance components and (baths and basins) and construction (skylights and light diffusers).

Deflection temperature under load is 75-100 °C. PMMA is resistant to most household detergents, solutions of inorganic acids, alkalies, and aliphatic acids, combustible, high tensile strength and low elongation. A NIR spectrum and its interpretation are shown in Figure 6.1 and Table 6.1

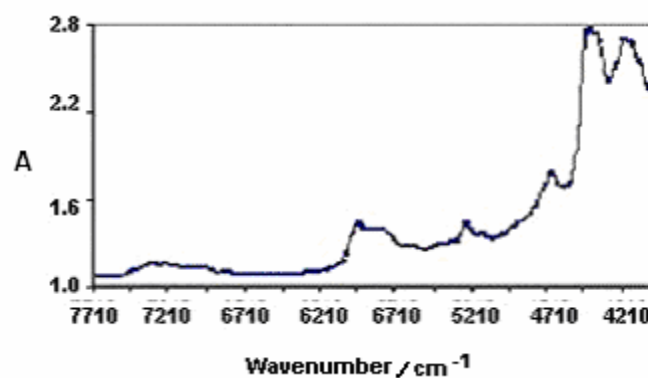


Figure 6. 1: NIR spectra of PMMA [206]

Table 6. 1: PMMA spectra analysis [206]

Wavenumber /cm ⁻¹	Assignment	Description
4681.01	CH ₂	C-H ⇒ Second overtone
5876.69	CH ₂	C-H ⇒ First combinations
5953.31	CH	C-H ⇒ First combination
8518.37	CC, CHO	C=O ⇒ First overtone

(b) Disperse Red 1

A nonlinear optical (NLO) chromophore, disperse red 1 [-(4[N-ethyl)-N-(2-hydroxyethyl)]amino-4'-nitroazobenzene) ($M_w = 314.35 \text{ g.mol}^{-1}$) was obtained from Sigma-Aldrich Chemicals. The chemical structure of the dye is shown in Figure 6.2. Purity was 95%; wavelength maximum was 502 nm. DR1 was used without any purification.

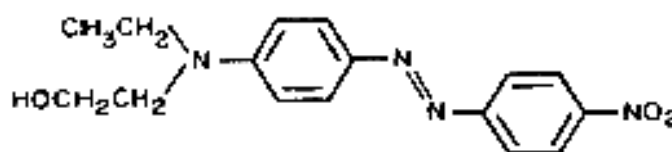


Figure 6. 2: Molecular structure of DR1 with the long molecular axis defined by the bridging group, [207]

6.2.2 Preparation of PMMA films

Solutions of PMMA (1 g.mL^{-1} and 2 g.mL^{-1}) were prepared by dissolving PMMA pellets in tetrachloroethane [$\text{Cl}_2\text{CHCHCl}_2$, Sigma-Aldrich Chemicals, $[T_b = 144\text{-}148 \text{ }^\circ\text{C}]$ at room temperature. The solutions were left for one day before being filtered through a syringe filter and applied on a substrate (glass and silicon wafer). The solutions were spin coated on silicon wafer substrate (cleaned with acetone and dried in a stream of nitrogen) to obtain films of thickness varying from $0.8 \text{ }\mu\text{m}$ to $8 \text{ }\mu\text{m}$. During the spin coating process, excess PMMA solution was spilled instantly from the substrate surface leaving a thin film of the amorphous polymer. After spinning the solution, the coated specimens were kept in a clean room while allowing the solvent to evaporate. The films were dried in a vacuum oven at $45 \text{ }^\circ\text{C}$ for 20 h for removal of residual solvent. After drying, the films were stored in a clean room before any other treatment for 3 h. Each film was scratched using a sharp blade for thickness measurement and the thickness and surface roughness were measured using a XP-2 stylus

profilometer instrument. Thickness measurements were correlated with spinning speed, concentration of solution and spinning time.

6.2.3 DR1-PMMA films preparation

A solution of a mixture of DR1 and PMMA was prepared using DR1 (0.01 g.mL^{-1}) and PMMA (0.1 g.mL^{-1}) dissolved in tetrachloroethane at room temperature. The solutions were left for one day, stirring with a magnetic stirrer and heated in an oil bath at 100°C under reflux conditions before filtering through a poly(tetrafluoroethylene) filter with a $2 \mu\text{m}$ pore size. Films of different thicknesses were spun on glass and silicon wafer substrates obtaining films of ($1.5\text{-}3.8 \mu\text{m}$) thickness. The films were dried in a vacuum oven at 40°C for 24 h for removal of residual solvent. After drying, the films were cooled in a clean room for several hours. The film thicknesses were measured using a XP-2 stylus profilometer instrument.

To make sure the particles of DR1 were completely dissolved in tetrachloroethane, images of DR1-PMMA films were tested after every 5 h during the preparation of the DR1-PMMA solution. Figure 6.3 shows images of the DR1-PMMA films during preparation of the solution (a) before and (b) after DR1 was completely dissolved. The DR1 crystals were not melted because the melting temperature was higher than the dissolving temperature. The films of DR1-PMMA of different thickness were left in clean room for 1 h before being placed in a vacuum oven at 45°C for 20 h allowing evaporation of any solvent residual.

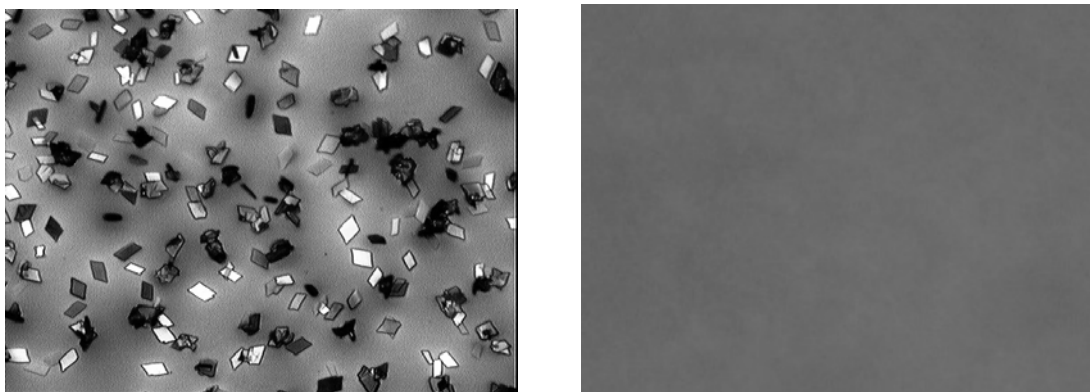


Figure 6. 3: Image (a) is obtained after the DR1 – PMMA is added in tetrachloroethane and before DR1 is dissolved completely (b) after heating at 100 °C for 1 day

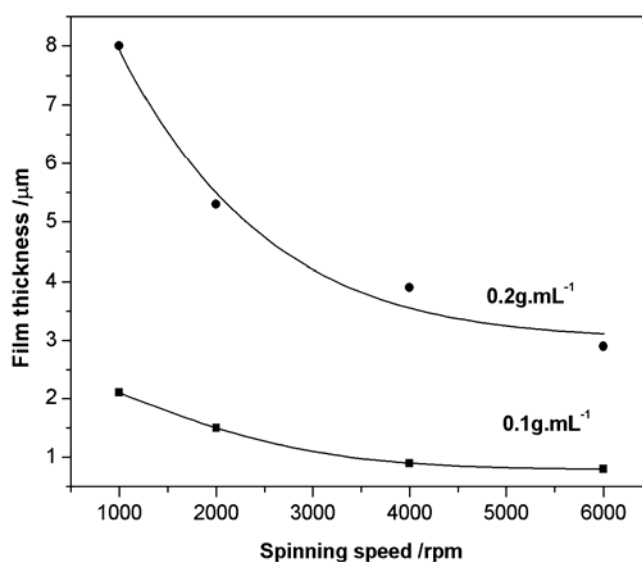
6.3 Results and Discussion

6.3.1 Film thickness of PMMA

Films of different thickness of PMMA were made using different spinning speed. Different film thicknesses were obtained from two different concentrations; 1 g.mL^{-1} and 2 g.mL^{-1} as shown in Table 6.2. The spinning time used was 60 s. The spinning speed has been shown to control the film thickness, so a range of film thickness was prepared by selection of spinning speed. PMMA film thickness variations at higher concentrations of polymer caused an increase in film thickness of approximately four times as shown in Figure 6.4. The PMMA films then were used for preparation of waveguides and investigation of optical properties.

Table 6. 2: PMMA film thickness variations with spinning speed of two different concentrations

Speed /rpm	PMMA thickness (t_1) / μm (0.1 g.mL ⁻¹)	PMMA thickness (t_2) / μm (0.2 g.mL ⁻¹)	($\Delta t = t_2 - t_1$)
1000	2.1	8.0	5.9
2000	1.5	5.3	3.8
4000	0.9	3.9	3.0
6000	0.8	2.9	2.1

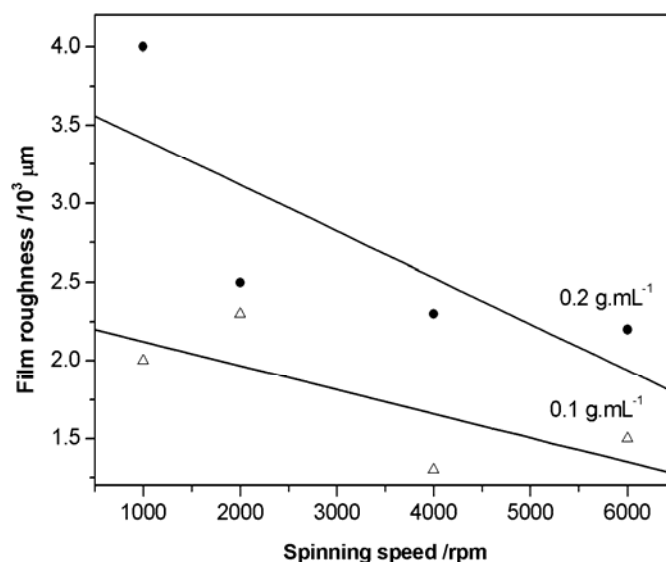
**Figure 6. 4: Correlation between film thickness and the spinning speed of two different concentrations of PMMA**

6.3.2 Influence of PMMA concentration on surface roughness

The relation between the polymer concentration and film thickness of PMMA was investigated using films prepared from solutions of two different concentrations. The surface roughness of PMMA decreased when the film thickness decreased. The surface roughness variation is shown in Table 6.3. The relation of the PMMA film roughness with polymer concentration is shown in Figure 6.5.

Table 6. 3: PMMA film roughness variations with spinning speed using two different concentrations

Speed /rpm	PMMA roughness / μm (0.1 g.10 mL ⁻¹)	PMMA roughness / μm (2 g.10 mL ⁻¹)
1000	0.0020	0.0040
2000	0.0023	0.0025
4000	0.0013	0.0023
6000	0.0015	0.0022

**Figure 6. 5: Correlation between film roughness and spinning speed of two different concentrations of PMMA**

6.3.3 Spinning time for PMMA films

Solutions of PMMA were spun on glass and silicon substrates using a selected spinning speed of 2000 rpm. Each film was made by spinning a PMMA solution for a different time. The films were dried at room temperature for 3 h then dried in a vacuum oven for 20 h at 45 °C. The films were scratched for thickness measurement. Investigation was made on film thickness versus time of spinning using two different concentrations. Thickness and

roughness versus spinning time is shown in Table 6.4. The relation between spinning time and film thickness is shown in Figure 6.6. Spinning time had little effect on the thickness and roughness of a thin polymer film. The film roughness shows some scattered results (Figure 6.7) due to differences across the film area. The rate of the solvent evaporation and the surface tension were expected to cause roughness variation.

Table 6. 4: PMMA film thickness measurements at the same speed (4000 rpm) using different times and of the same concentration (0.1 g.mL⁻¹)

Time of Spinning /s	Thickness / μm	Roughness / μm
10	1.00	0.0043
20	0.98	0.0031
40	1.00	0.0031
60	0.98	0.0029
Average	0.99	0.0034

The average values for the thickness and roughness were 0.99 μm and 0.0034 μm respectively. The difference between the maximum and minimum measurement of roughness on the PMMA film surface was 0.0041 μm .

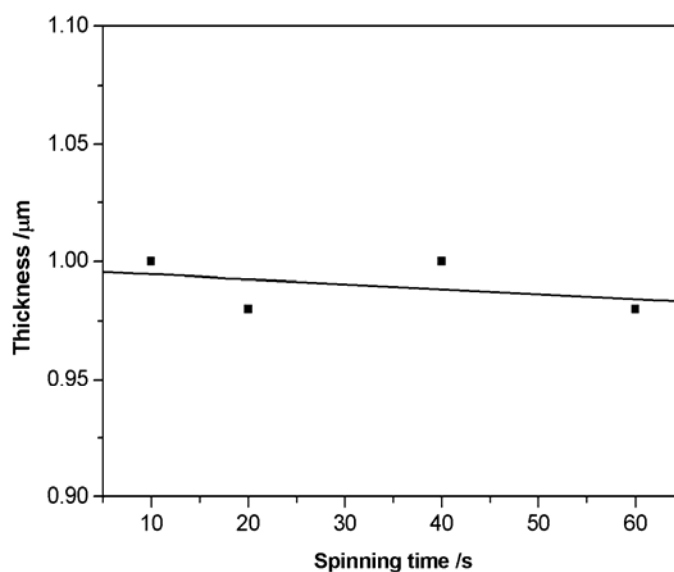


Figure 6. 6: The relation between spinning time and film thickness of PMMA

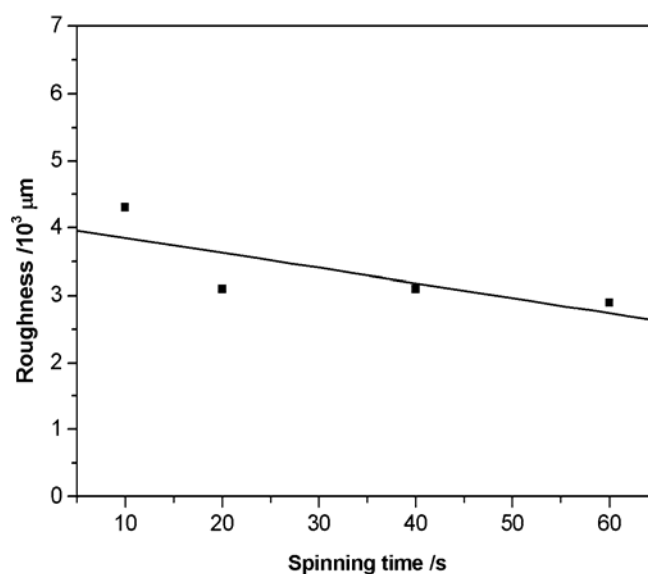


Figure 6. 7: The relation between spinning time and film roughness of PMMA

6.3.4 Film thickness of DR1-PMMA films

Films were prepared from PMMA with DR1. Solutions of DR1-PMMA mixtures were prepared with concentrations of 0.01 g.mL^{-1} DR1 and 0.1 g.mL^{-1} of PMMA. The solutions were spun to make films of different thickness by selecting suitable speeds. The DR1-PMMA

solution was spun on glass and silicone wafer. The films were left in clean room for 3 h before heating them at 45 °C in a vacuum oven for 20 h. DR1-PMMA films were left in a clean room at room temperature for 3 h before thickness measurements were made. A straight and fine scratch was made on each film. The thickness measurements were made using a XP-2 stylus profilometer. DR1-PMMA films were peeled from the substrate to obtain free standing films for further measurements.

The difference between the pure PMMA and DR1-PMMA film thickness are shown in Figure 6.8. The figure shows comparison between data of DR1-PMMA and data of pure PMMA. It can be seen in Figure 6.8 that the curve of DR1-PMMA is shifted to increased thickness when DR1 is included in the PMMA. The investigation of the thickness variation was considered for testing and modifying the transmission properties and later used in waveguide study.

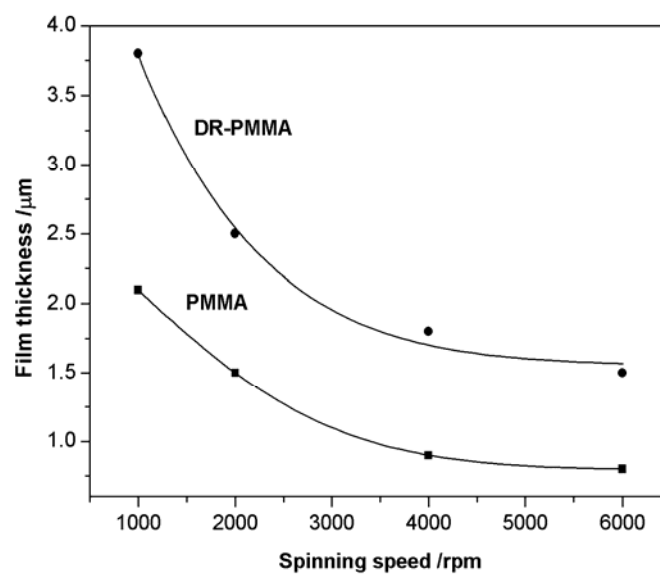


Figure 6. 8: DR1-PMMA thickness versus spinning speed per 60 s spinning period

The film surface was imaged using an optical microscope to test the film quality. Figure 6.9 shows a DR1-PMMA film with a scratch made on the surface for thickness measurement.

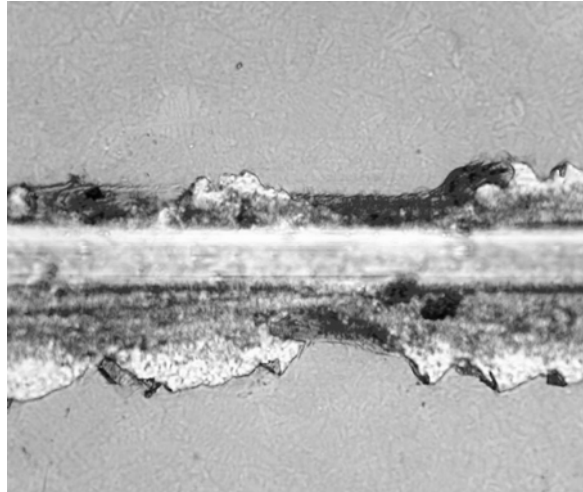


Figure 6. 9: DR1-PMMA film with scratch on the surface; used for thickness measurements

6.3.5 Surface roughness of DR1-PMMA film

Investigation of surface roughness variation with spinning speed was made. DR1-PMMA films were prepared for waveguide development and surface roughness measurements were made using films of different thickness, prepared from the same concentration of DR1 solution. The data for surface roughness is shown in Table 6.5.

It can be seen from the data in Table 6.5, that the surface roughness of the PMMA is higher ($0.0020\ \mu\text{m}$) for thicker films, and lower ($0.0015\ \mu\text{m}$) for thinner films, compared with films of DR1-PMMA where the roughness was $0.0022\ \mu\text{m}$. Probably the thinner films were more sensitive and depended on nucleation of DR1 by the substrate and surface tension. Surface roughness of spin-coated films is much smoother than solution-dipped films [9]. It has been found that films from a solution of higher polymer concentration have a rougher surface.

The relation between the surface roughness and the film thickness with spinning speed is shown in Figure 6.10. Surface roughness of DRI-PMMA films was higher than the pure PMMA films surface roughness.

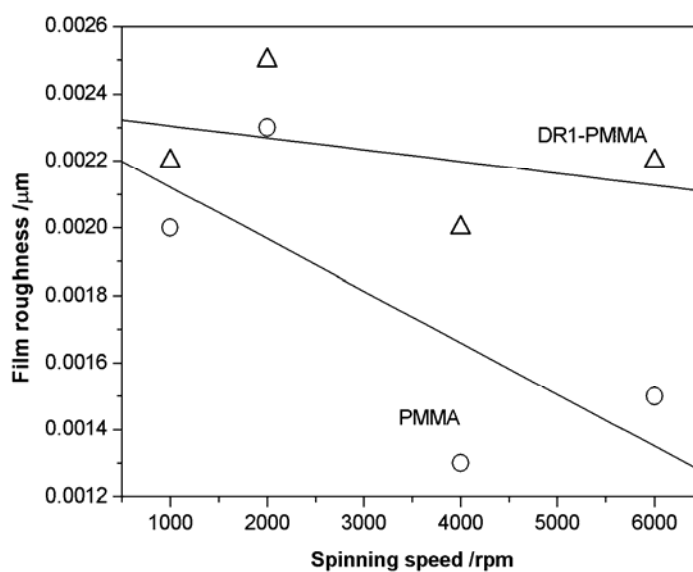


Figure 6. 10: Relation between film roughness and spinning speed of DR1-PMMA and pure PMMA

6.3.6 Thickness and roughness versus spinning time of DR1-PMMA film

Surface roughness and film thickness of DR1-PMMA were investigated using films prepared from solutions of the same concentration as shown in Table 6.5. DR1-PMMA solution was spun using the same spinning speed for different spinning times.

Table 6. 5 DR1-PMMA film thickness measurements at the same speed using different times and of the same concentration

Spinning speed 2000 rpm *		
Time of Spinning /s	Film thickness / μm	Surface roughness / μm
30	2.5	0.0030
40	2.6	0.0027
50	2.6	0.0017
60	2.6	0.0022
Average	2.6	0.0024

* Concentration [0.01 DR1+ 0.1 PMMA] g.mL⁻¹

A curve of film thickness for DR1-PMMA is plotted against spinning time in Figure 6.11. Film thickness is independent of spinning time similar to the result for pure PMMA I Figure 6.11.

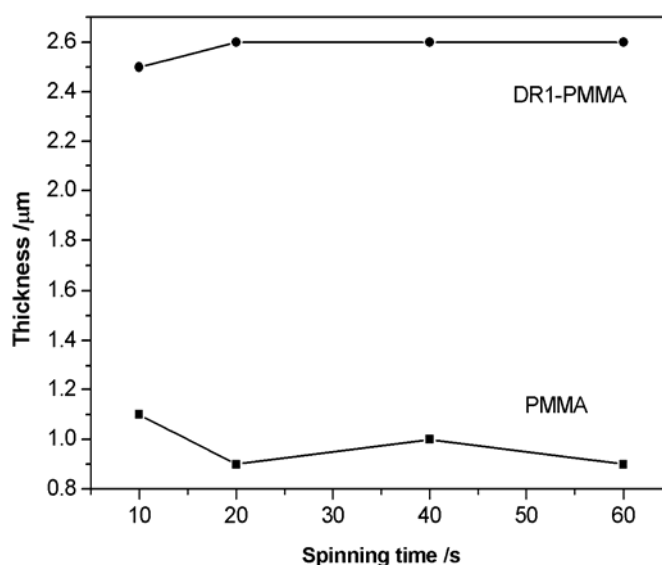


Figure 6. 11: Film thickness versus spinning time of pure PMMA and DR1-PMMA

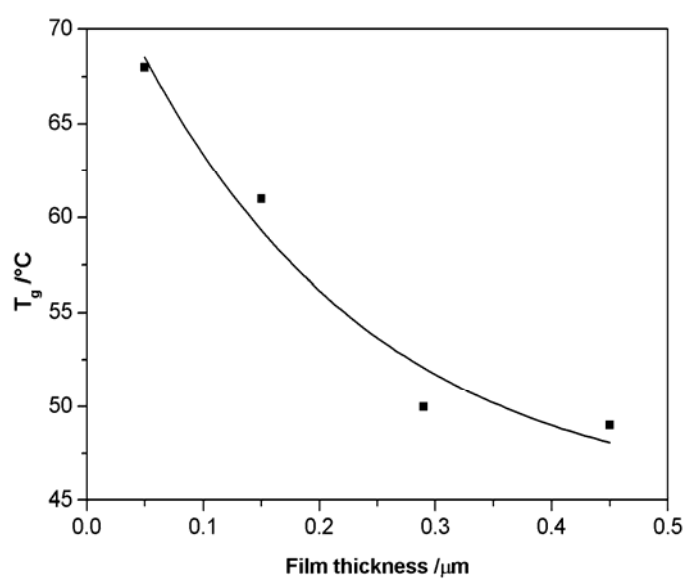
6.3.7 Glass transition temperature variation with film thickness using poly(L-lactic acid)

Solutions of PLA of molecular weight [19000 g.mol^{-1} , $T_m = 170 \text{ }^{\circ}\text{C}$], were prepared by dissolving PLA pellets in tetrachloroethane. PLA films were spun using a glass substrate to obtain different thickness of films. After drying, the films were cooled to room temperature. The PLA film thicknesses were measured using a XP-2 stylus profilometer. Tg was measured using DSC and the results are shown in Table 6.6.

Table 6. 6: PLA glass transition temperature measurement with film thickness and roughness

Film thickness	Glass transition temperature	Roughness
/μm	/°C	/μm
0.45	49	0.0075
0.29	50	0.0050
0.15	61	0.0045
0.05	68	0.0025

It can be seen from Table 6.6 and that the T_g increases as the film thickness decreases. The surface roughness decreased as the film thickness decreases.

**Figure 6. 12: PLA T_g variation according to film thickness**

6.4 Non-linear optical polymer (NLO)

The non-linear optical polymer, urethane-urea (UU) copolymer, was used in optical waveguide fabrication. The structure of UU copolymer is shown in Figure 6.13

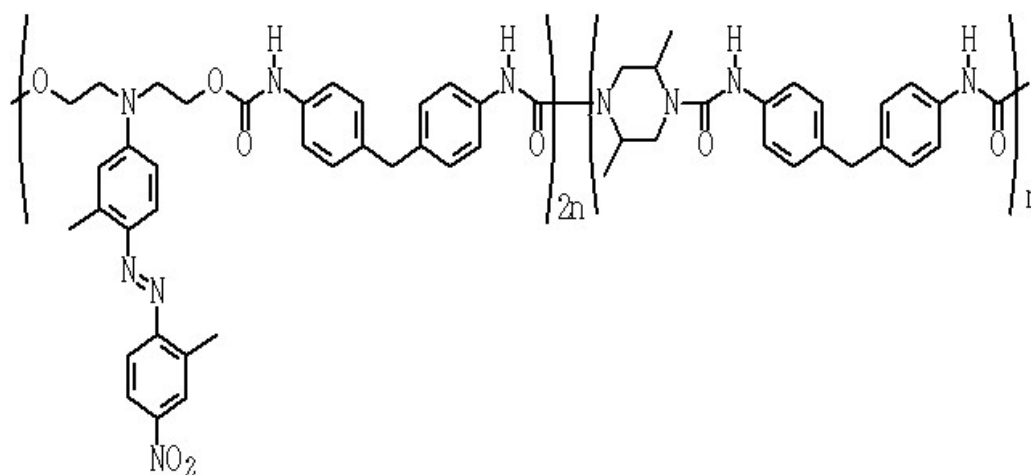


Figure 6. 13: Structure of urethane-urea copolymer (UU copolymer)

The NLO polymers used for electro-optical applications require large dipole and hyperpolarizability. Bulk properties should have acentric or non-centro symmetric arrangement, low optical loss (low refractive index), low dielectric constant and high second harmonic generation (SHG). UU copolymer has a long delocalised chromophore and satisfies all of the above requirements. Furthermore, it has electron donor (-NH) and acceptor (-NO₂) groups that can form hydrogen bonds between molecules. Other properties include amorphous, high T_g (140-150 °C), stable at high temperature due to hydrogen bonds.

6.4.1 Preparation of NLO polymer film

The procedure for UU copolymer film preparation is shown in Figure 6.14.

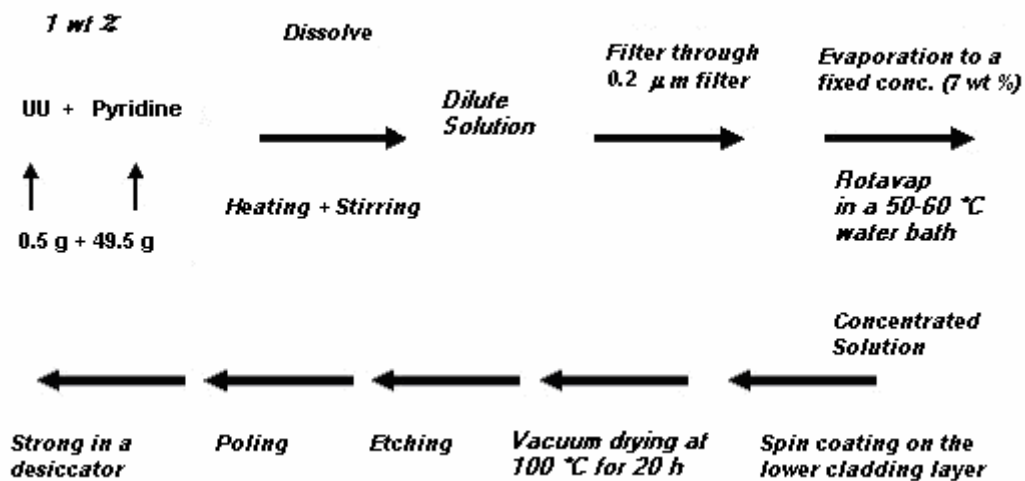


Figure 6. 14: The procedure for preparation of UU films

6.4.2 UU solution preparation and film spinning

It was found that concentrated solutions directly prepared by dissolving UU in pyridine failed to give clear films of UU polymer. Preparation of the solution was based on a dilute solution of 1 %wt at the start, increasing to a concentration that allowed film preparation within the planned range of thickness. Using a concentration of 7 % wt and spinning the solution at speeds of 500 rpm and 1000 rpm provided films of $\sim 2 \mu\text{m}$. The solubility of the polymer is shown in Table 6.7

6.4.3 Solubility table

Table 6. 7: Solubility of UU and UV15

Polymer	Solvent	Solubility	
		Yes	No
UU (T _g = 140-150 °C) ¹	Pyridine (Py, 115 °C, d = 0.978 g.mL ⁻¹ , irritant)	√	
	N-methyl-2-pyrrolidinone (NMP, d=1.028 g.mL ⁻¹ , 204 °C, irritant)	√	
	N,N-Dimethylacetoamido (DMAc, 165 °C, irritant)	√	
	N,N-Dimethylformamide (DMF, 153 °C, irritant)	√	
	DMI	√	
	Dimethyl Sulfoxide (DMSO, 189 °C, irritant)	√	
	Methanol		Partial, X
UV 15 ²	Pyridine		
	NMP		
	DMAc		
	DMF		
	DMSO		
	Methanol	√	

*1. Pyridine can be used for the spin casting of UU copolymer film, if post-cure UV15 or IPG may not be soluble in pyridine. 2. UV15 (UV-curable acrylic polymer) in methanol would be suitable to spin cast onto a silicon wafer. UV15 in methanol would be suitable to spin cast an upper cladding layer of UV15 back onto UU polymer film.

The information from Toyota recommended that 6.5 %wt solution at 500 rpm and 1000 rpm will give 1.8 μm and 0.9 μm thickness, respectively. Therefore, in order to make a ~2 μm thick film a high solution concentration was needed, such as 7-9 %wt with speeds of 500 rpm and 1000 rpm). A solution of 1 %wt was prepared by mixing 0.5 g of UU in 49.5 g of pyridine. This preparation was used to spin coat the solution at 500 rpm and 1000 rpm to obtain a film of ~2 μm thickness. To check the best concentration for making the targeted thickness of the film, different concentrations were selected and tested such as (6.0, 6.5, 7.0 %wt) and the film thickness was checked after spin coating at various speeds.

50 mL of pyridine was added to a 100 mL round bottom flask. Approximately 0.5 g of UU was added. The flask was placed on a hot plate-stirrer with an air condenser. A small magnetic stirring bar was used to stir the mixture during dissolution. The stirring speed was

set at near maximum. The solution was continuously stirred for 16 h. After 16 h no particles could be seen by eye. A glass syringe was used, to remove the solution. A 0.2 μm PTFE filter cartridge attached to the syringe was used to filter the solution over 30 min. The filtered solution was added to a 100 mL round bottom flask and sealed with a glass stopper. The flask was stored wrapped in aluminium foil to prevent light affecting the polymer solution.

6.4.4 Rotary evaporator

Acetone (50 mL) was used for cleaning the rotary evaporator. A portable rotary vacuum pump was used to obtain the necessary vacuum of ~ 90 kPa. A liquid nitrogen trap was used to condense any pyridine. The water bath used was at a temperature of 60 $^{\circ}\text{C}$ and the flask was immersed about 1 cm into the water. The equipment for UU preparation is shown in Figure 6.15 in (a) the solution preparation equipment and (b) the vacuum equipment. The evaporator was further cleaned using 10 mL of pure pyridine before UU solution preparation. The vacuum was controlled using a glass valve on the vacuum line to the pump. In this method the required amount of pyridine was evaporated in about 10-15 min. It is undesirable to heat the solution too much as this may cause decomposition of the UU polymer. The flask with 8 mL of solution of UU was fitted with a stopper and wrapped in aluminium foil.



(a)



(b)

Figure 6. 15: Equipment for UU polymer solution preparation using pyridine

UU films were examined at the Laser Physics Centre Australian National University. Films were prepared from a solution filtered with a 2 μm filter, spin coated (500 rpm for 60 s), dried in a vacuum oven at 100 $^{\circ}\text{C}$ for 20 h. An optical microscope was used as a primary test for the UU film that was spun from the copolymer solution, to indicate the purity of the solution from particles of UU powder. An image is shown in Figure 6.16.

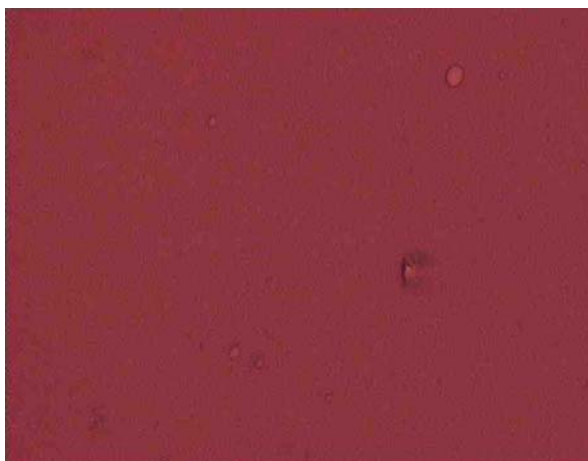


Figure 6. 16: Optical microscope image of UU copolymer film after two weeks of spinning the UU solution

Four films were spun on different substrates, such as silicon and glass and they were tested with a Metricon prism coupler, model 2010, to measure refractive indices at two wavelengths 632.8 nm and 810 nm at TE and TM beam polarizations, with film thickness, and light guiding.

6.4.5 Poling of dipoles

Poling was performed by applying an electric current, 20 kV for 2 min, after 30 min preheating time, in air at 165 °C. Selection of a temperature above the glass transition was important. At such a temperature, the arrangement of aligned dipoles could be easily brought about, and on cooling, the arrangement was frozen below the glass transition. An electrode of indium-tin oxide coated glass or an aluminium substrate was suitable.

6.5 Results and Discussion

6.5.1 Refractive index and film thickness

The measurements from the prism coupler are listed in Table 6.9. The films were $\sim 0.9 \mu\text{m}$ thick. They were birefringent and dispersive.

Table 6. 8: Refractive indices and thickness of UU films measured with a prism coupler

Sample No.	N,TE,	n,TM,	n,TE,	n,TM,	Thickness
/substrate	632.8 nm	632.8 nm	810 nm	810 nm	(μm), $\pm$ std deviation
#1 /Infrasil- silicon	1.7909	1.7670	1.7043	1.6852	0.736 ± 0.006
#2 /ITO/glass	1.8026	1.7718	1.7071	1.6847	0.910 ± 0.012
#3 /ITO/glass	1.8001	1.7689	1.7067	1.6822	0.961 ± 0.026
#4 /glass	1.7923	1.7662	1.7050	1.6826	0.893 ± 0.007

6.5.2 Absorption and transmission spectra

Maximum absorption was measured in films #1, #2, #3: $\lambda_{\text{max}} = 471 \text{ nm}$, the decay absorption coefficient α was about $3.1 \times 10^4 \text{ cm}^{-1}$. The transmission was near 670 nm. Wavelengths shorter than 670 nm were strongly absorbed in a layer of UU about 1 μm thick.

6.5.3 Waveguiding measurements

None of the films showed waveguiding of a He-Ne laser or a laser diode at 810 nm with a prism coupler. The lack of guiding of light beam at 810 nm could have been caused by absorption, because the polymer had an absorption tail at longer wavelengths, or due to strong scattering...

6.6 Conclusion

In this chapter, optical properties of PMMA, DR1-PMMA, PLA and urethane-urea polymers such as transmission and glass transition temperature have shown changes with film thickness. Surface roughness has shown changes with film thickness using optical polymers.

Solutions of PMMA, DR1-PMMA and UU were spun preparing a range of films with different thickness. PMMA films were spun from solutions of different concentration. When

concentration of a polymer was doubled, the thickness of the polymer film was changed to four times the thickness of a lower concentration. Surface roughness of PMMA was decreased when the film thickness decreased and the surface roughness of DR1-PMMA was lower than pure PMMA.

Comparison between DR1-PMMA and PMMA film thickness showed that DR1-PMMA film thickness was greater than PMMA film thickness when the concentration of PMMA was the same in both solutions such as, thickness measurement values in the thickness versus spinning speed curve of DR1-PMMA films were changed by 1.08 μm values when compared with the thickness value of the PMMA curve.

A NLO urethane-urea copolymer was used in optical waveguide fabrication. Four films were prepared using silicon and glass and refractive index measured with a Metricon prism coupler, at two wavelengths 632.8 nm and 810.0 nm at TE and TM beam polarizations. The films absorption spectra were measured with a spectrophotometer.

Absorption measurement at $\lambda_{\text{max}} = 471$ nm has shown the decay absorption coefficient α was about $3.1 \times 10^4 \text{ cm}^{-1}$. The transmission threshold came near 670 nm and the urethane-urea copolymer strongly absorbed in a layer of the material about micrometer thick.

Chapter 7

Waveguide Fabrication Using Thermosetting Polymers

7.1 Introduction

High quality thermoset polymer solutions are suitable for forming thin films for optical waveguides because of their high optical transmission, suitable refractive indices, viscosity and uniformity of films obtained by spinning solutions on a substrate such as glass or silicon. The chemical modification of well-known and extensively used polymers gives the possibility of obtaining new materials with useful properties and a new range of applications [208, 85]. The curing reactions that occur in the epoxy system have been extensively studied because of the applications of epoxy system in composites, adhesives and moulding compounds. However the curing kinetics and mechanism of the epoxy system can be affected by the molar mass of the epoxy resin, the crosslinking agent, and the resin crosslinker ratio [209, 210]. In coating applications, many factors influence the resin such as the substrate, which may cause adsorption or reaction with the resin affecting the curing reaction and the structure of the cured resin at the substrate resin interface [211]. The solvent and coating process may affect curing reactions in the epoxy system [99]. It has been shown that solvent released during curing, in an open system, caused decreased T_g , and decreased curing exotherm [99].

Master Bond UV15 is a thermoset polymer with high sensitivity to UV curing. It is an epoxy resin based polymer system for bonding, sealing and coating with high performance due to its effective cationic catalysed curing mechanism. Master Bond UV15 can be cured in short times using ultra-violet (UV) in ambient temperatures and it becomes strong and tough after curing using a source of light at wavelength of 365 nm. The adhesive and/or sealant films of cured UV15 can be employed in a wide range of temperature of $-50\text{ }^{\circ}\text{C}$ to $150\text{ }^{\circ}\text{C}$.

It is water resistant. UV15 is used as a cladding layer coating with PMMA in this study, since UV15 has a suitable refractive index (slightly lower than PMMA) and it is not dissolved in a PMMA solution. Modification of coating polymer properties will be investigated in this chapter, with variables of thickness, concentration, surface roughness and curing using UV light, followed by post-baking. The glass transition temperature will be investigated after different curing processes, as well as surface roughness, and ion beam or plasma etching for patterning the UV15 films.

The solution viscosity and the spinning speed determine the thickness of the resulting films. Modification of polymer properties is required for development and fabrication of waveguides. Modifying the polymer surface using etching treatment is essential to prepare waveguides with high transmission. The plasma etch rate of UV15 was examined for trenches of the order of 1 μm depth and 10 μm wide, suitable for photonic waveguide fabrication, produced in films from a relatively highly viscous polymer solution. Exposure to UV and heat is needed to cure the polymer through formation of a cross-linked structure. The curing process can have a significant effect on the etch rate. Different paths were taken in the curing and etching process of the spun polymer film in order to examine the relation between the cross-link density and the polymer etching process. FTIR was used to examine the polymer films and measure cross-link formation and to monitor chemical changes occurring for different levels of curing. DSC measurements were used to quantify the glass transition temperature variation with curing and baking.

A comparison of etching techniques and conditions is made to obtain optimum channel depth and wall smoothness. The waveguides described in this chapter use the pure polymers for evaluation of the preparation and etching of the polymeric layers. Further studies include a chromophore in the PMMA layer to provide an electro-optic effect.

The aim of the study in this chapter is to prepare a slab waveguide using a UV curing system as the base layer and to control film thicknesses and surface smoothness using spin coating, and using plasma etch rate of UV15 examining trenches.

7.2 Experimental

7.2.1 Materials

A photo-curable epoxy acrylate compound (UV15) was obtained from Master Bond Inc. USA. The refractive index of this polymer film was measured to be 1.45 at 633 nm. UV15 has low viscosity [uncured coating, at 24 °C is 120-150 cps], it is liquid at ambient temperature (it contains no solvents and volatiles). UV15's boiling temperature at 760 mm is >572 °C. The evaporation rate is negligible, solubility in water < 0.5 % at 24 °C [212].

7.2.2 Preparation of UV15 film

The UV15 thermoset polymer is a one-component material, so no solvent evaporation will occur after spinning. It can be spun to give films of suitable thickness for waveguides, i.e. approximately 5 to 100 µm and it is compatible with photolithography, i.e. cured UV15 does not dissolve in the solvent of the photoresist polymer.

Photo curable UV15 was used to prepare a film on a silicon wafer. The native oxide of silicon wafer was removed using a concentrated HF solution for 1 min. The substrate was cleaned using acetone and then rinsed with deionised water. UV15 was spun as supplied. A spinning speed of 4000 rpm was selected to spin a UV15 film of 6 µm thickness for 60 s spinning time on a silicon wafer.

7.2.3 Curing the film of UV15

High intensity ultraviolet radiation was used to cure the polymer film by exposure for 1 min using a Karl Suss Mask aligner UV source (10 mW.cm^{-2}). Separately, the cure of UV15 was investigated using transmission infrared spectroscopy during curing, with a lower intensity (3 mW.cm^{-2}) UV lamp at 365 nm.

UV15 exhibits a desirably fast cure rate at ambient temperatures, even in the presence of air [212]. Only the spin speed was varied to investigate and change the thickness of the deposited polymer film.

7.2.4 Photo initiator

Photoinitiated polymerisation can be described by a series of reactions, with the monomer, and then will lead to a chain reaction. A hard crosslinked resin is required for the lower polymer layer. Hardness and resistance to melting during the etching process is critical. Etching must provide a channel with smooth sides and base. A thin channel is preferred so that complex waveguides can be etched into the substrate. The layer must be crosslinked so that the next light-conducting layer can be solvent cast without dissolving or swelling the first layer. The structure and crosslinking of UV15 is shown in Figure 7.1 [213].

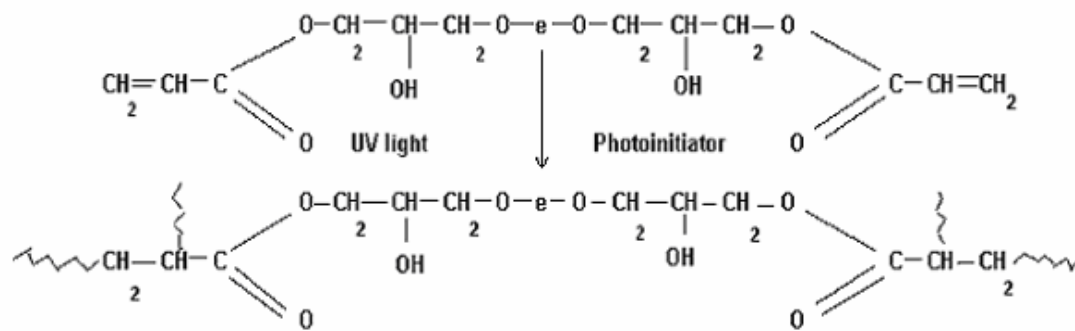


Figure 7.1 The structure and crosslinking of UV15 cured with UV light, 365 nm wavelength [213]

7.2.5 Waveguide fabrication

A silicon wafer was used as a substrate for the rib waveguide. The UV15 films were coated with AZ5206-E photoresist and patterned for waveguide testing using a standard ultra-violet exposure photolithography process. The mask pattern used for the exposure had features that were 2 to 7 μm wide and greater than 1 cm long. The photoresist waveguide pattern was 1 to 2 μm thick and suitable as an etch mask for the etch depths required in the UV15 bottom cladding layer. A film of PMMA followed by the top cladding of UV15 as shown in Figure 7.2 were then spun onto the etched layer..

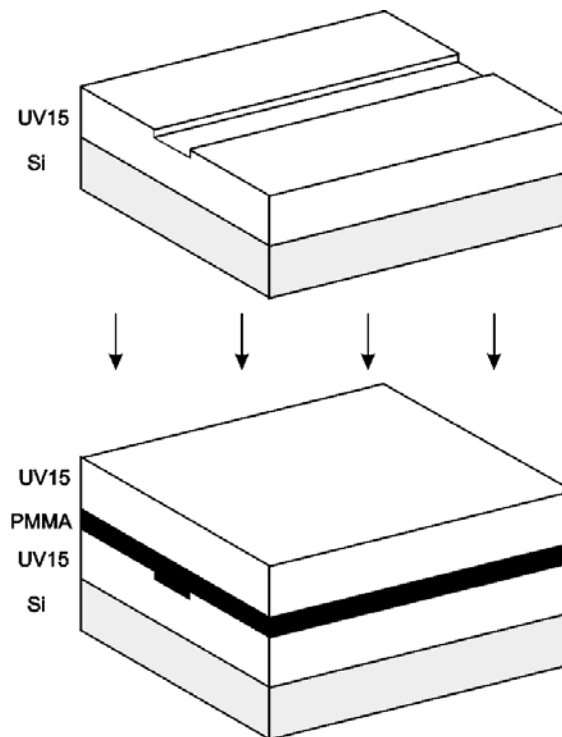


Figure.7.2: Inverted rib polymer waveguide layers; top schematic shows an etched trench in a UV15 layer; bottom schematic shows the added core and top cladding layers

7.2.6 Etching techniques

The cured resin was post baked to remove any volatiles and to relieve stresses so that etching would not lead to any thermal deformation. After post-baking the surface roughness of the films was typically $= 0.005 \mu\text{m}$ with a film thickness of $5.5 \mu\text{m}$.

Two techniques were used and compared for etching the channels on the UV15 surface:

1. Ion-beam etcher.
2. A barrel etcher.

Oxygen or argon plasma etching was performed.

1. Ion-beam etcher.

The ion-beam etcher (IBE) involves an anisotropic etching process. Two gas types were used in the IBE process;

Argon (Ar)

Ar + O₂

The IBE system consists of a 5 cm Kaufman ion gun. With Ar it was possible to maintain a steady discharge of energetic inert ions at 200 V. The highest value used was 600 V, because the photoresist may degrade due to the increase in temperature caused by ion bombardment at higher voltages. The gas flows were Ar 3.5 cm³.min⁻¹ and O₂ 2.0 cm³.min⁻¹ for all tests. A silver filled thermal paste was necessary for all voltages used so to ensure good thermal contact between the samples and the water-cooled sample table. As well as power, the other variable was the angle of the incident ion beam relative to the sample surface.

2. Barrel Etcher

Etching in a barrel etcher is an isotropic process. The etching gas used was O₂. The variables were RF power, gas flow and the presence or absence of a perforated cylinder around the sample, which reduced ion bombardment of the surface. Barrel etchers were first used in the semiconductor industry to remove photoresist and other carbonaceous materials by oxygen plasma reaction [213]. They are particularly useful when a wet chemical process cannot remove photoresist. The required time for etching was 60 min for an etching of 1 μm). This process is faster than the corresponding IBE system where the actual etching takes ≈ 10 min. but vacuum pumping the IBE system takes much longer. A perforated cylinder was used with higher power to avoid increased surface roughness. This increases the contrast to IBE where only ion bombardment occurs...

In the barrel etcher, tens of wafers can be etched simultaneously, whereas the IBE can only etch one. The photoresist and UV15 both etch at similar rates compared with non-polymer materials. The photoresist used etched slower than UV15 (≈ 50 %). However the etch rate of

UV15 and photoresist depends on the curing parameters used. Silicon does not etch in the barrel etch conditions used here.

Following the etching process it was essential to make sure that all photoresist had been removed. N-butyl acetate (BA) is a suitable solvent that can be used to dilute the photoresist. BA did not affect the UV15 surface; however it removed the photoresist when used in an ultrasonic bath. BA did appear to soften the UV15 surface and hence a further post cure step at 105 °C for 10 min was required before applying the PMMA guiding layer. Use of the BA step can be avoided by using the exact photoresist thickness (knowing the photoresist etch rate) so that no further processing was required after etching. By knowing the spin speed thickness curves, the desired thickness can be obtained. However this required varying the photolithographic process. Stubborn photoresist was left to soak overnight in BA. This soak did not seem to alter surface roughness of the UV15. Etched trenches were clear after ultrasonic bath treatment in BA, followed by a deionised water rinse and high purity nitrogen gas blow drying.

UV15 films on silicon were prepared by spinning the liquid UV15 at 4000 rpm for 30 s and immediately curing in air, using intensity greater than 10 m.W.cm⁻² for 60 s from an unfiltered mercury lamp. The UV15 films were then post cured in an oven at 110 °C for 30 min to improve stability. The surface roughness of these films was measured using an atomic force microscope (AFM) and thickness was approximately 50 µm and 35 µm respectively.

7.3 Results and Discussion

7.3.1 FTIR spectroscopy measurements

FTIR spectroscopy was used to compare the as received and cured UV15 for different curing times using a UV light of intensity 3 mW.cm⁻², under ambient conditions. The distance

between the source of UV light and the polymer surface was kept constant in all exposures. The results of treatment are shown in Figure 7.3 for spectra recorded after different irradiation times.

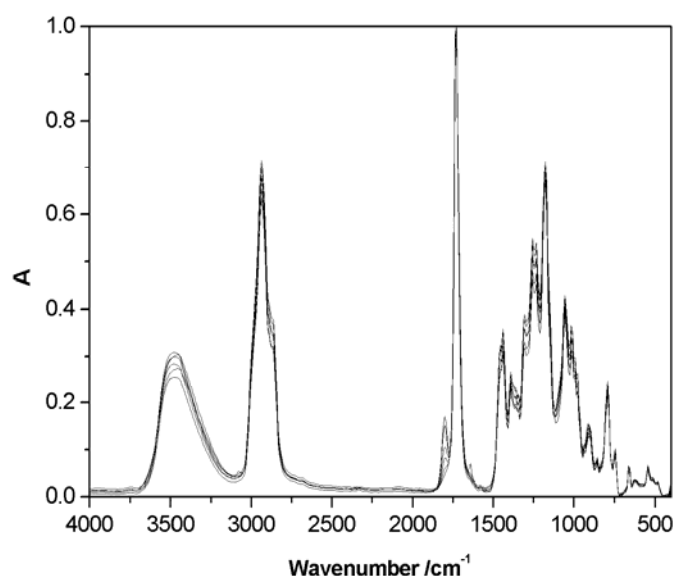


Figure.7.3: Infrared spectra of UV15 after exposure to 365 nm UV light

Figure 7.3 shows the infrared absorption spectra of UV15 after it was exposed to 365 nm UV light. The treatment was carried out on KBr plates. As a transparent substrate the polymer was cast on a KBr plate then cured for different times. The UV15 solution on KBr was mounted in the FTIR spectrometer. The spectra after incremental UV exposures were recorded for comparison of curing rate. The difference between the IR regions of ester and carbon double bond absorptions, compared with curing time is shown in Table 7.1.

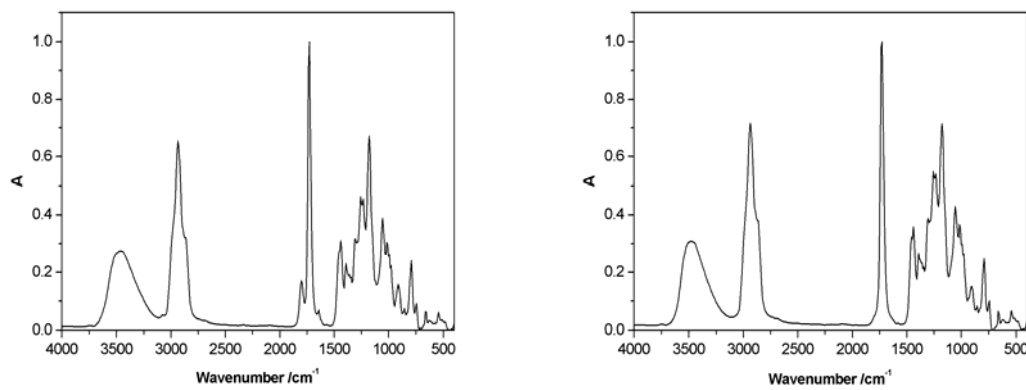
The area under the curve decreased in the unsaturated ester carbonyl and carbon-carbon double bond at 1780 and 1640 cm⁻¹ respectively. The results are shown in Table 7.1 for spectra recorded at 0-40 min.

Table.7. 1: FTIR data of curing time of UV15

Curing time /min	Difference between two consecutive spectra of cured UV15 film	Reciprocal of the difference between two consecutive spectra	$\{(A_0 - A_n) A_0^{-1}\} 100$
0	5.71	0.18	0.0
5	5.28	0.19	7.5
8	4.6	0.22	19.4
11	4.18	0.24	26.8
15	3.66	0.27	35.9
20	2.66	0.38	53.4
25	2.35	0.43	58.8
30	1.87	0.53	67.3
40	1.29	0.78	77.4

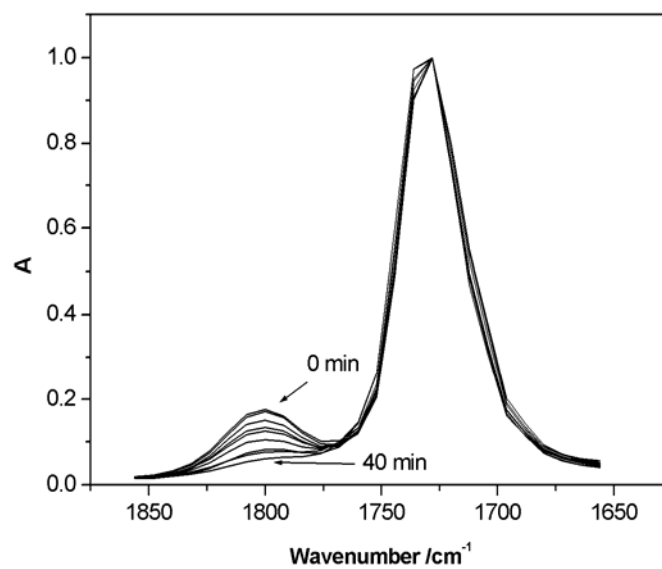
n = integer no, A_0 : is the initial value of ratio for affected area of curing UV15, A_n ; is the nth value of ratio for the affected area of curing.

Two different spectra of UV15 of minimum and maximum exposure times are shown in Figure 7.4 (a) and (b) respectively. The peaks change in intensity with curing time, over the wavenumber range between 1850 and 1700 cm^{-1} , as shown in Figure 7.4 (c)



(a)

(b)



(c)

Figure.7. 4: FTIR spectrum of UV15 (a) uncured, 0 min, (b) cured for 40 min (c) spectra of UV15 after exposure to 365 nm UV light with expanded view of ester and carbon double bond regions.

Figure 7.4(c) shows an expanded view of ester and carbon double bond regions for different periods of UV exposure.

7.3.2 Property variation of cured UV15

The property variation of the cured UV15 was investigated using UV15 cast on a KBr plate. The ratio of cured area under the peak versus curing time is shown in Figure 7.5. The relation shows the extent of curing with time and hence the UV exposure required for complete cure.

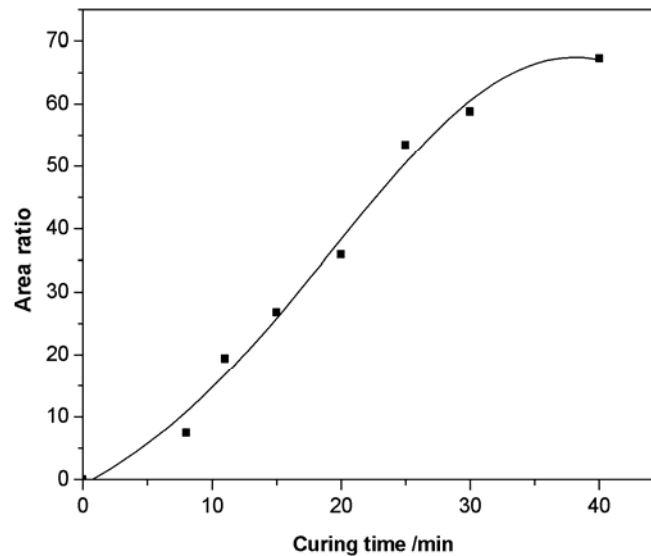


Figure.7.5: FTIR area variation under the curve ratio $[(A_0 - A_n) / A_0] \times 100$ measurements of cured UV15 using UV light for different period of time

The FTIR spectra of the EO-polymer have significant changes in the region between 1700 - 1850 cm^{-1} ranges when a comparison is made between the as-received UV15 with cured. It can be seen from the curing time that UV15 has a fast cure response to the UV light and the curing process reaches a maximum after 40 min of UV exposure where absorption was at a minimum and this was accompanied by hardening of the UV15.

7.3.3 Glass transition temperature variation of cured UV15

Tg measurements were carried out for cured films of UV15 using DSC. The investigation was based on Tg measurements of UV15 films cured for different times. A Tg increase was observed for the films. Films with low Tg during the plasma etch process exhibited surface flow giving wrinkling. This probably occurs because the plasma etching process heats the UV15 film to a temperature higher than its cured Tg value, allowing it to soften and flow. The Tg and etch rate can be optimized to obtain an optimum etch rate while preserving the smooth polymer surface during etching.

The Tg of UV15 increased with both curing time and intensity as shown in Table 7.2 and Figures 7.6 and 7.7. This means that the UV15 becomes more flexible at high temperatures. Tg gives a quantitative measure of the hardness of the polymer structure and its cross-link density. The first step in the waveguide fabrication process was curing UV15, and the second step was smoothness and etched profile properties. A highly polished silicon substrate was used to spin the UV15 film and cured from above the surface by UV radiation from a mask aligner.

Another important factor in curing UV15 is the rate of curing increased by a reflective substrate. This can be improved by spinning UV15 on KBr transparent plates and silicon wafers the reflected substrate. The etch rate achieved was low compared with films that had been cured at lower UV intensity. However, an advantage was that etching produced a sharper step for films that had moderate cure intensity. Tg variation with curing time is shown in Table 7.2. The change in Tg was ~ 5.9 °C.

Table.7.2: Glass transition temperature, versus curing time of UV15 unbacked

Curing time	Glass transition	UV intensity
/min	temperature /°C	/J.cm ⁻²
10	47.5	1.8
20	48.4	3.6
30	49.5	5.4
40	53.4	7.2

The DSC curves for UV15 films used for Tg measurement are shown in Figure 7.6. The Tg inflection temperature was higher for cures of longer periods.

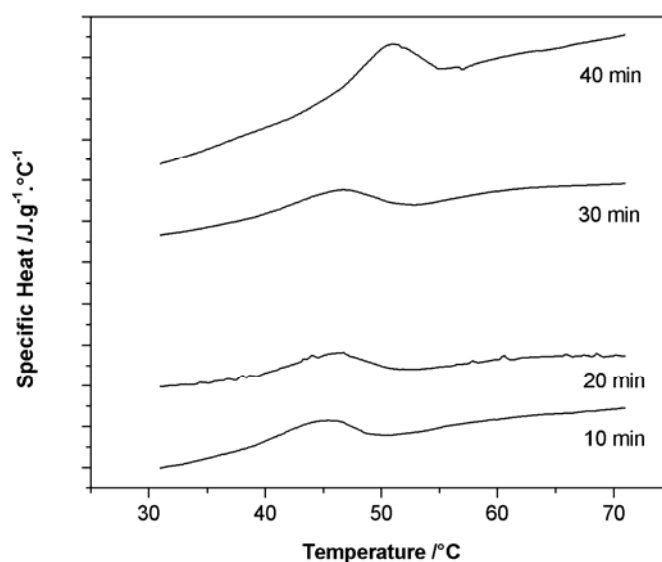


Figure.7. 6: DSC (showing Tg) for unbaked UV15 cured using UV light for different times at room temperature

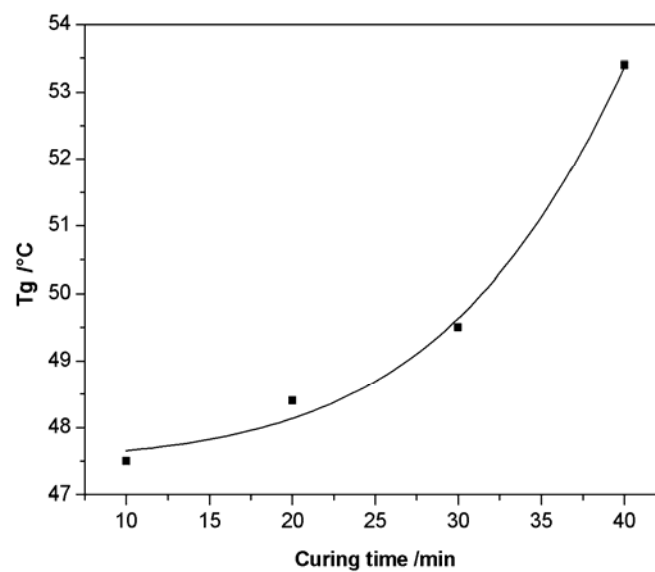


Figure.7.7: Glass transition temperatures versus curing time of UV15 cured using UV light

While investigating etching parameters, a problem arose when the masking aluminium layer appeared to wrinkle during the etching step. This was overcome by ensuring that a heating step was carried out directly after UV curing of the UV15 layer. This gave the UV15 a higher T_g and prevented the surface from flowing at substrate temperatures occurring in the plasma etching process.

7.3.4 Glass transition temperature variation of heated UV15

Aluminium thin film patterns were investigated for etching the UV15 layers. The wrinkling was not observed when heating at 80 °C for 30 min was used after the UV cure. By heating the polymer after it was cured, its T_g increased. Compared with e-beam evaporated aluminium this wrinkling phenomenon was not observed to the same extent in the films with a sputtered aluminium layer (even when the post cure process was omitted). This may be due to the hotter temperatures occurring during sputter deposition of films compared with e-beam evaporation, thus replicating the post-UV heating process to a certain extent.

DSC results in Table 7.3 show Tg values for UV15 films with heating time, curing time and UV energy intensity. These results show that the Tg of UV15 was less than 50 °C for unbaked (see Table 7.2). The UV15 films that were post baked at 80 °C for 30 min had an increased Tg from 47.5 °C to 112 °C (see Table 7.2 and 7.3). The data in Table 7.3 shows that UV15 Tg increased to more than twice the value of unbaked films. UV15 was spun on glass substrates, to prepare three different films of the polymer. Each film was cured using a 3 mW.cm⁻² UV lamp source for times of 10, 30 and 40 min, and then the films were heated for two different periods as shown in Table 7.3

Table.7. 3: Glass transition temperature of UV15 cured and the post heated films

Curing time	UV intensity	Heating	heating time	Glass transition
/min	/J.cm ⁻²	temperature /°C	/min	temperature /°C
10	1.8	80	30	112
30	3.6	80	30	115
40	7.2	100	40	120

The DSC measurements of Tg for cured and heated films of UV15 are shown in Figure 7.8.

The Tg after 40 min cure at 120 °C was increased to twice that of the unheated film of UV15.

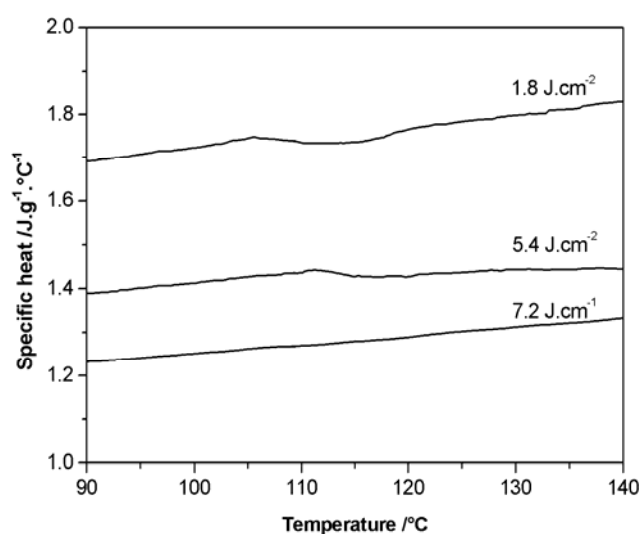


Figure.7.8 DSC curves obtained for UV15 heated at 80 °C for 30 min (top curves) and 100 °C for 40 min (bottom curve) after UV curing at various energy intensity. (Energy=intensity x time)

All interfaces should be smooth to prevent loss of light by scattering. After the UV15 polymer was cured by UV exposure, heating at 100 °C for 30 min hardened the film further before placing in the sputtering system for aluminium deposition. Heating the polymer can increase the glass transition temperature (T_g) to over 125 °C [7] and the post cure was effective in enhancing chemical resistance, particularly to solvents.

FTIR measurement of UV15 films, cured and heated, was investigated. The UV15 solution was spun on a silicon substrate making UV15 films of 17 μm thickness. The film was cured for different times using UV light. The cured film then was heated at 110 °C for 30 min. The full spectra of the cured and heated films are shown in Figure 7.9. The cure rate proved to be faster than that of the film spun on KBr plate because of the reflected UV light, as well as the incident light, adding to the curing process

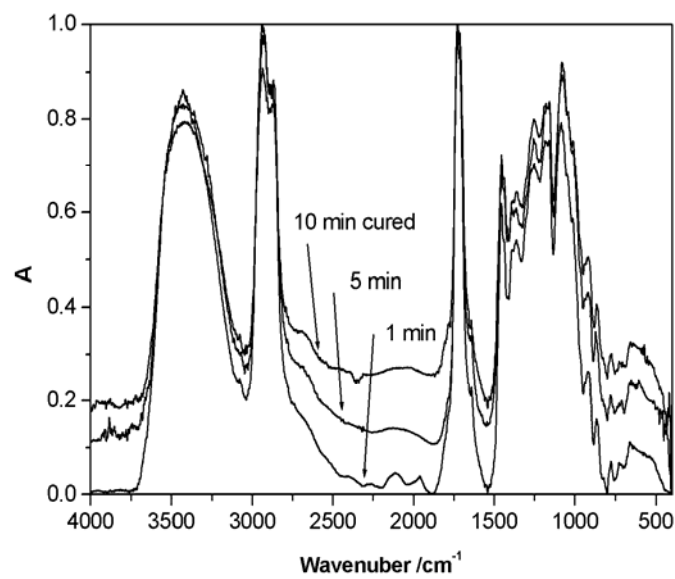


Figure.7.9: UV15 FTIR spectra of heated films for different times

The difference in cure rate can be seen by comparing the amplitude of the carbon-carbon and carbon-oxygen double bond (1800 cm^{-1} and 1640 cm^{-1}) in Figure 7.10 and Figure 7.11 with those of Figure 7.9; after 1, 5, and 10 min of curing the film.

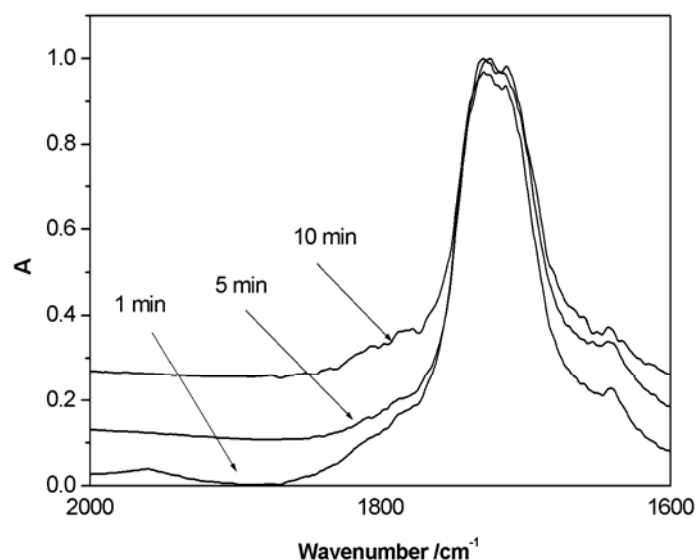


Figure.7.10: UV15 FTIR spectra of cured film for different times and heated at 110°C for 30 min showing the C=O stretching vibration at 1638 cm⁻¹

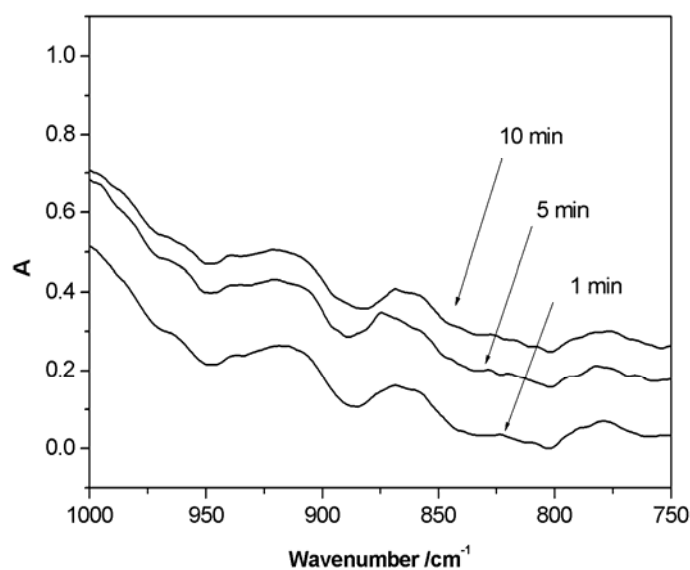


Figure.7.11: UV15 spectra of film cured for different times and post heated at 110 °C for 30 min

7.3.5 UV15 cured without post heating

UV15 films were etched after undergoing UV-curing for different times (without post UV-cure heating) were investigated. The spectra of cured UV15 films are shown in Figure 7.12.

UV15 solution was spin coated to prepare a film of thickness $\approx 17 \mu\text{m}$ on a KBr plate, and then UV-cured at $7 \text{ mW}\cdot\text{cm}^{-2}$ for different times with a Mercury broadband UV-spectrum

mask aligner. The peaks that were chosen as a reference for cross-link density were the carbon-carbon and carbon-oxygen double bond stretching absorptions at 1640 and 1800 cm^{-1} respectively.

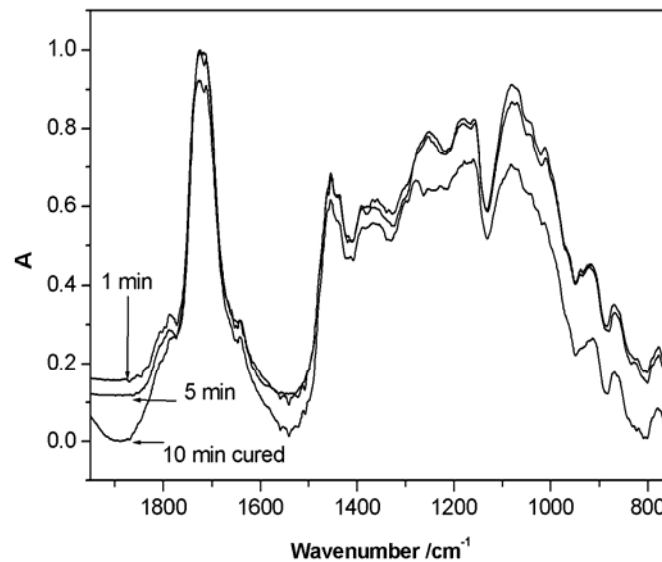
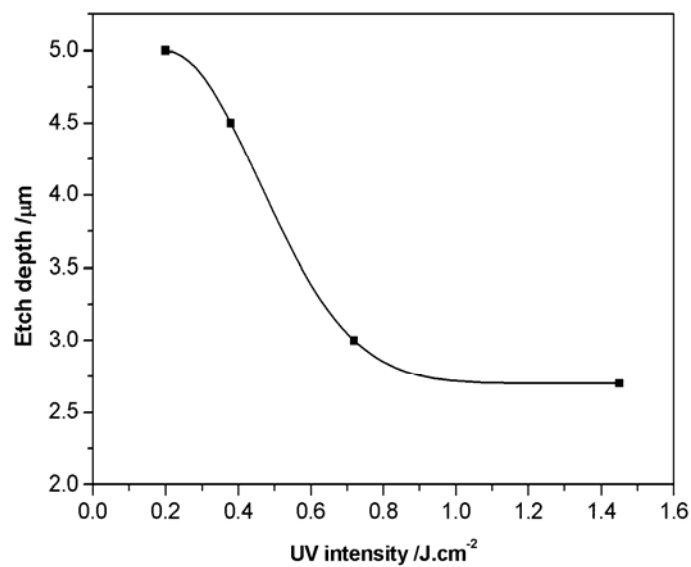


Figure.7.12: FTIR spectra of UV15 film $\sim 17\text{ }\mu\text{m}$ thick spun on a KBr plate and was cured for the times shown with a broadband spectrum mask aligner at 7 mW.cm^{-2} three different times

The FTIR spectra are shown in Figure 7.12. The film was etched to different depths and the etch depth measurements are shown in Table 7.4. The relation between the etch depth versus UV intensity for cured UV15 are shown Figure 7.13. When the film was cured at higher UV energy doses, the etch depth decreased.

Table.7. 4: Etch depth measurements for UV15 using different intensity

UV intensity /J.cm ⁻²	Etch depth /μm
1.45	2.7
0.72	3.0
0.38	4.5
0.20	5.0

**Figure.7.13: Etch depths versus UV Energy dose for cured UV15. You should give the curing conditions or refer to a table where they are listed. Also change 'Intensity' on the x-axis label**

Two different energy of UV Intensity were used to etch UV15 films. Film A (0.54 J.cm⁻²) was etched to 100 μm while film B (3.6 J.cm⁻²) was etched to 15 μm, but B was post-cured for 30 min at 110 °C. The data of etching, curing and heating for the three films are shown in Table

Table.7. 5: Etching and curing data of UV15 spun on silicon wafer

Film #	Etching depth / μm	UV-Curing intensity / J.cm^{-2}	heating time /min	Heating temperature / $^{\circ}\text{C}$
A	~ 100	0.54	-	-
B	~ 15	3.60	30	110

These films were spun from UV15 solution on silicon wafers and two groups were made with different roughness as shown in Figure 7.14 (1) and (2) on polished silicon wafer substrates. The film (A) conditions were etched at 400 W for 20 min and the films (B) conditions were etched at 200 W for 7 min intervals. Optical microscope images of etched films are shown in Figure 7.14

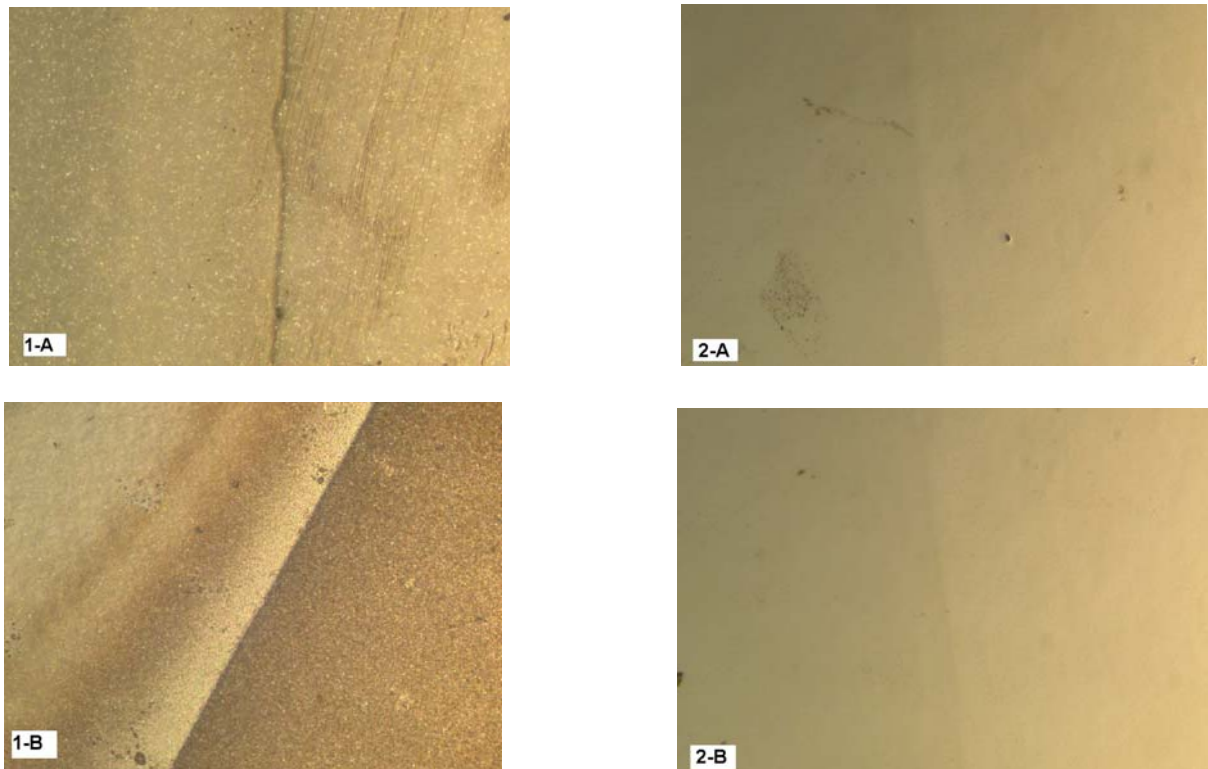


Figure.7.14: (1) Etch profiles of UV15 films spun on rough silicon wafers, (2) Etch profiles of UV15 films spun on polished silicon wafers

7.4 Waveguide Slab

7.4.1 Etching and smoothing the polymer surface to prepare a waveguide

In order to obtain suitably smooth etched trenches the un-etched UV15 surface must be smooth after curing. The spin coated polymer films were characterised using a XP-2 stylus

profilometer. This was suitable for measuring film thickness and surface roughness. The etched trench profiles were characterised using an AFM. This was more suitable for determining the cross-section shape and roughness of trenches.

The optical film thickness and roughness were measured as described in Chapter 6. It can be seen from the thickness and roughness measurements that the trends observed were as expected Figure 6.4. The thickness variation for films from high concentrations of PMMA was smaller than those at low concentrations. In general, the surface of the polymer film became smoother as the film became thinner (Figure 6.5)

7.4.2 Etching and roughness measurement

It is necessary to establish the surface roughness of the PMMA guiding layer as a function of deposition conditions, and investigate the roughness of the cladding layer and in particular the plasma etched trench that forms the mould for the optical waveguide beam. The bottom cladding layer polymer UV-15 was deposited and patterned. An AFM image of an unetched, photoresist patterned UV-15 film is shown in Figure 7.15. Roughness values of less than 0.005 μm and 0.0035 μm were obtained for the un-etched UV15 and photoresist films, respectively. These were typical values for all the films tested.

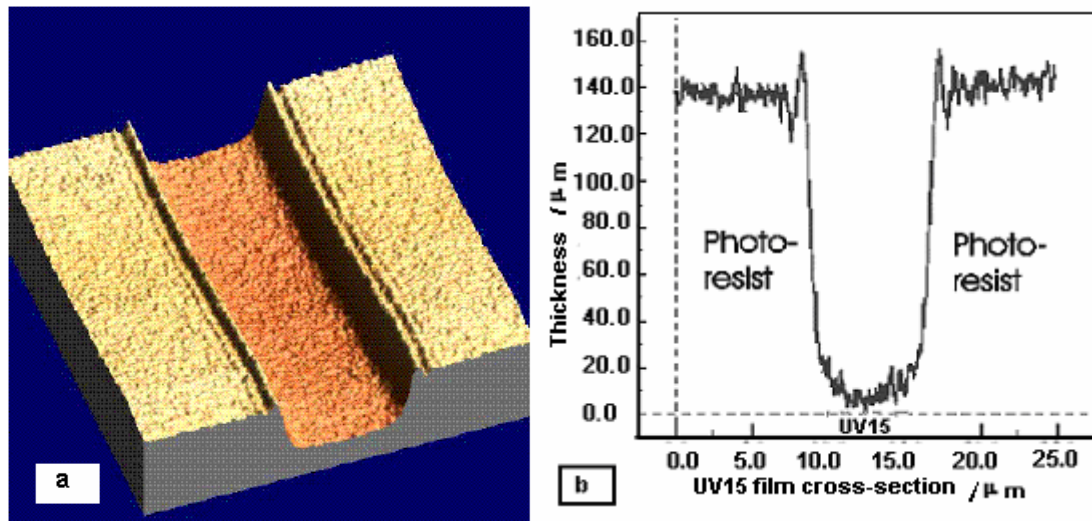


Figure.7.15: AFM of un-etched photoresist patterned UV15 (a) micrograph, (b) cross-section

7.4.3 Ion Beam treatments

IBE results (Table 7.6) show that for angles near normal incidence, the etch rate increased and roughness decreased as shown in Figures 7.16 and 7.17 respectively.

Table.7. 6: Etch depth and roughness versus angle

Angle from normal incidence /°	Etch depth /μm	Roughness /μm
10	0.18	0.014
20	0.25	0.023
30	0.20	0.029
40	0.16	0.027
50	0.17	0.044
60	0.18	0.028
70	0.12	0.020
80	0.10	0.024

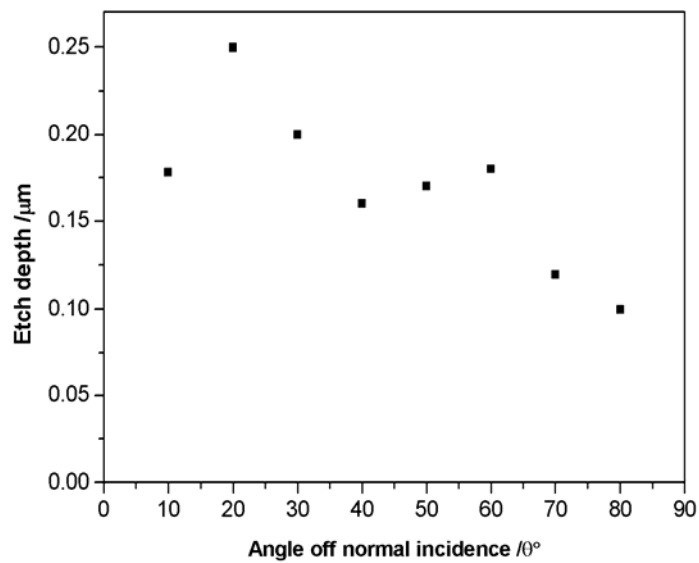


Figure.7.16: Etch depths versus beam angle for UV15 etched in IBE at 200 V in Ar $3.5 \text{ cm}^3 \cdot \text{min}^{-1}$ for 30 min

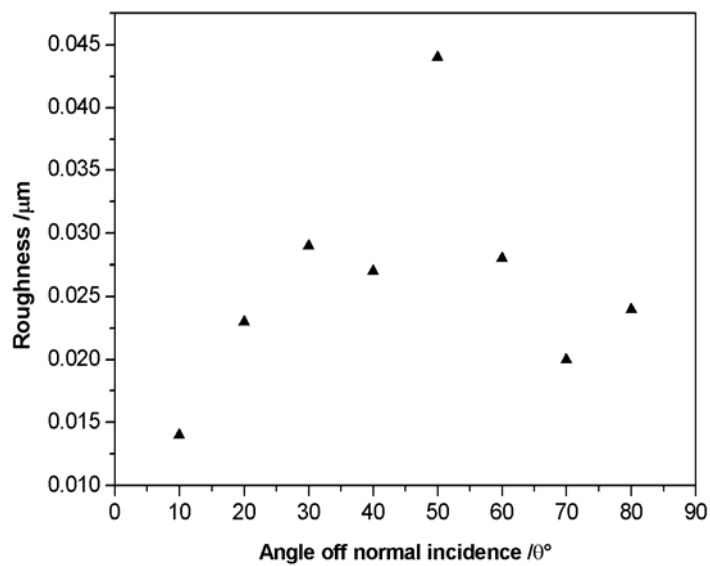


Figure.7.17 Surface roughness versus beam angle for IBE of UV15 at 200 V in Ar $3.5 \text{ cm}^3 \cdot \text{min}^{-1}$ for 30 min

Most significant is the reduction in roughness at angles near normal incidence. The introduction of O_2 did increase the etch rate though roughness was similar. This indicates that the length of time difference of plasma etching is more favourable. Increasing the gun voltage did increase the etch rate and roughness as shown in Table 7.7.

Table.7. 7: Ion beam etching of UV15 for two beam voltages. Beam at 45° (Ar 3.5 cm³.min⁻¹, O₂ cm³.min⁻¹)

Ion gas	Voltage	Time	Depth	Roughness
	/V	/min.	/μm	/μm
Ar	200	30	0.350	0.028
Ar	600	15	0.250	0.025
Ar+ O2	200	15	0.250	0.022
Ar+ O2	600	15	0.420	0.053

These results indicate that the most favourable IBE conditions are near normal incidence for the ion beam, Ar and O₂ gas mixture and a low beam voltage. The roughness values obtained were not as small as desired for low loss waveguides and therefore minimising the ion bombardment component of the etching process is desirable. Figure 7.18 shows a typical IBE etched trench in UV15. The roughness values were measured at the bottom of the trench using AFM.

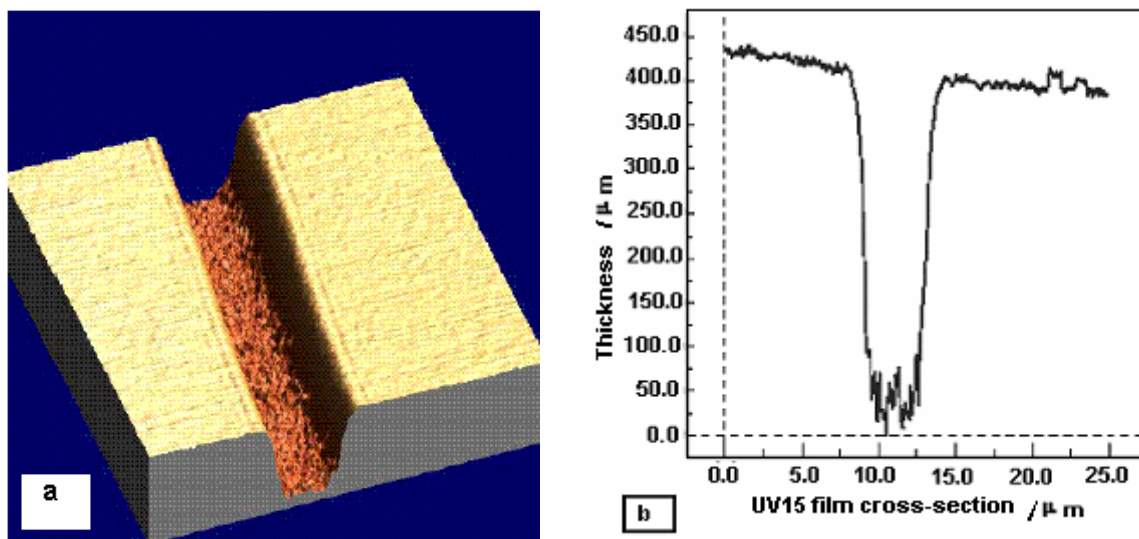


Figure.7.18: (a) AFM of UV15 trench by IBE 50° Ar, 200 V, rms bottom 18.3 nm (b) cross section
 The roughness of the bottom of the trench is more critical than the side-walls for waveguides because of its more significant surface area. Figure 7.18 shows the AFM image (a) and cross section (b) of a UV15 trench formed by IBE.

7.4.4 Barrel etcher measurements

The barrel etcher mostly etches using activated gas with energetic ions existing in the plasma generated by the RF coil. Etching of UV15 was tested with and without minimising the effect of the ion etching component. The barrel etcher process on UV15 is shown in Figure 7.19 and the barrel etch results are shown in Table 7.8.

Table 7.8: Barrel etch results for UV15 in O₂ at 400 cm³.min⁻¹, 15 min

Power /W	No ion shield		Ion shield	
	Roughness	Depth	Roughness	Depth
	/μm	/μm	/μm	/μm
100	0.009	0.2	-	-
200	0.067	0.3	0.002	0.15
400	0.055	0.5	0.007	0.50

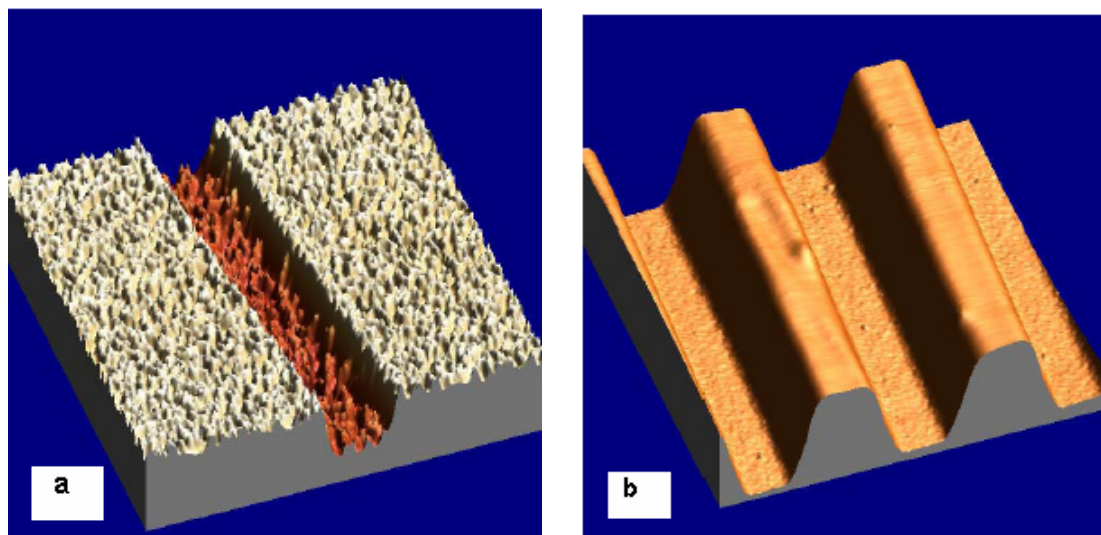


Figure 7.19: (a) barrel etcher 375 W and no ion shield, (b) 400 W with ion shield; other conditions are the same: time 15 min, O₂ gas at 400 cm³.min⁻¹, both trenches are approximately 7 μm wide and 0.5 μm deep

Figure 7.19 (a) shows UV15 etched without minimising the effect of the ions in the plasma. It can be seen from Figure 7.19 (a) that the etched surface is rough. The surface roughness increased from less than 0.005 μm before etching to 0.055 μm after etching. Figure 7.19 (b)

shows the effect of using the ion shield to minimise the ion etching for similar barrel etching conditions used for the treatment shown in Figure 7.19 (a). The roughness of the etched surface increased slightly from less than $0.005\text{ }\mu\text{m}$ before etching to $0.007\text{ }\mu\text{m}$ after etching. The presence of the ion shield significantly reduced the roughness value. Any contamination on the films probably causes damage to the surface patterns through the barrel etch process after etching. Residue on the surface has to be considered when performing barrel etching.

In the IBE system, contamination was removed during etching and is not as harmful to the etched pattern, as shown in Figure 7.20. The figure shows trenches etched in UV15 which have roughness values desired for waveguide surfaces. The roughness value for UV15 in the barrel etcher increased with RF power when ion bombardment was not minimised by using the perforated cylinder. Minimising ion bombardment led to reduced roughness, which increased much less, with increased RF power.

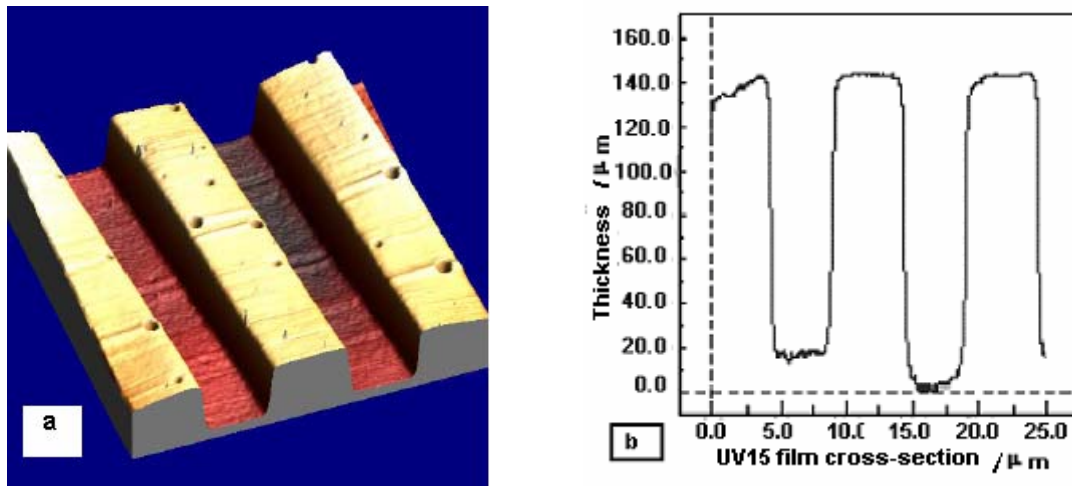


Figure.7.20: (a) AFM of UV15 trench, barrel etched, RF power of 200 W, 20 min, roughness $0.0022 \mu\text{m}$ (b) cross section, ion shield was used

7.4.5 Polishing and smoothing the end face of the waveguide

Optical microscope dark field photographs were used to investigate the roughening of the polymer surface. The technique was used to show the effect of plasma etching on the polymer surface and is a rapid method to compare the quality of different plasma etching processes.

The pattern of the polymer surface after the plasma treatment is shown in Figure 7.21 using the dark field technique. These included an image of $7 \mu\text{m}$ wide and $0.3 \mu\text{m}$ deep trenches for two different treatments; 200 W and 375 W.

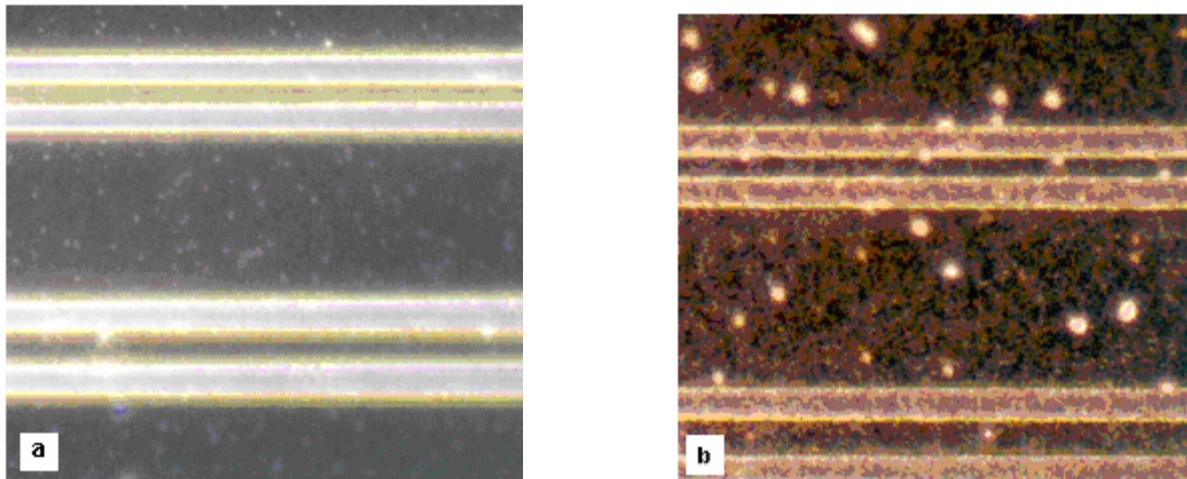


Figure.7.21 Dark field optical images of 7 μm wide and 0.3 μm deep trenches (a) 200 W ion bombardment (b) pattern with high roughness, 375 W ion bombardment

Figure 7.22 shows an AFM of a cleaved and polished end face of a waveguide. This waveguide consists of a 1 μm deep trench. As can be seen in Figure 7.22 (b) contrast in the AFM scan was obtained for each texture by polishing.

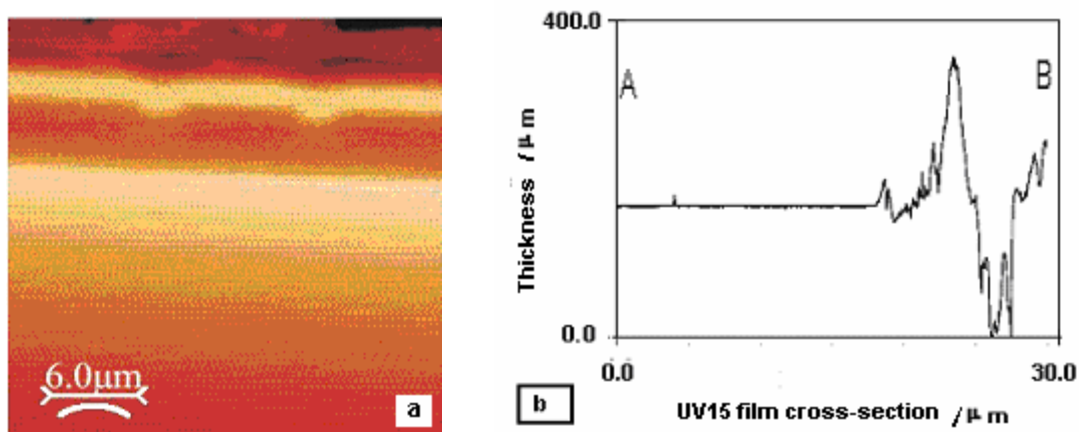


Figure.7. 22: (a) polished end face (b) cross section of the image showing raised and depressed regions.

7.5 Conclusion

The curing time variation of the UV15 was investigated. The peak changed in intensity with curing time, over the range 1850 to 1700 cm^{-1} and was improved. The ester and carbon double bond regions were cured by using different times UV curing

DSC Tg measurements were carried out for cured UV15 after different times. The Tg of UV15 increased with both curing time and UV intensity as shown in Table 7.2 and Figures 7.5 and 7.6. UV15 become more flexible at higher temperatures. The Tg was higher for longer curing periods. Tg was increased by post curing the cured polymer. When films with low Tg were treated by plasma the surface exhibited flow and gave wrinkling. Tg and etch rate were optimized while preserving the smooth polymer surface during etching.

It was observed that unheated UV15 covered with an aluminium mask layer caused a wrinkled surface during the etching process in the barrel etcher due to the heating effect of the plasma. No Wrinkles were observed when post curing was used after the UV cure. Wrinkles on the surface were not observed when an e-beam evaporated aluminium this to the same extent in the films when sputtered aluminium was used as the mask layer (even when the cure process was absent). Probably this was due to the hot sputtered films heating the films. The DSC measurements of Tg for cured and heated films of UV15 are shown in Figure 7.7. The Tg after 40 min cure varied to 120 °C, which was twice the Tg of an unheated film of UV15.

All interfaces should be smooth to prevent loss of light by scattering. The UV15 was cured by UV exposure and post curing at 100 °C for 30 min, the film hardened before placing in the sputtering system for aluminium deposition. Post cure the polymer increase the glass transition temperature (Tg) to over 125 °C [7].

The plasma etching and barrel etching are two different etching techniques that were studied; one was etching the polymer continuously while the other was to etch in intervals of 20-30 min. It was found using the barrel etcher that the etch rate was significantly higher and more linear over a longer time, this may be due to exposure to air after each etching interval. Various issues relating to the processing steps were investigated and discussed, such as: post curing the polymer, after spinning and before the photolithography step; different methods of

etching (such as continuous etching and etching in increments of 20 min); factors affecting the polymer etch rate and the substrate. Using a glass cover slip to seal the channels provided a clear view of fluorescent macro-particles flowing through them, allowing for the flow through specifically designed channel layouts.

Optical waveguides have been prepared and optimised using plasma etching of a photocured UV15 cladding layer on a silicon substrate. The guiding layer of PMMA was spin coated onto the cladding layer and into the etched channels. Relatively smooth interfaces were produced using a barrel etcher with oxygen plasma at 100-200 W and $400 \text{ cm}^3 \cdot \text{min}^{-1}$ intensity for 15 min. Surface roughness of $0.002 \text{ }\mu\text{m}$ has been obtained for a $0.15 \text{ }\mu\text{m}$ deep trench and $0.007 \text{ }\mu\text{m}$ for a $0.500 \text{ }\mu\text{m}$ deep trench. The RF power used to generate the plasma was not critical to the process as in table 7.8. Results for ion-beam etching suggest that the more energetic ions produce unfavourable surface roughness.

Ion beam etching was preferable at near normal beam incidence angle where the etch rate increased and the surface roughness was less. Introduction of oxygen increased the etch rate compared with argon. A typical trench in UV15 was measured using an AFM. The surface roughness was determined from the trench.

Etching of UV15 was carried out with minimum effect of the ion-etching. UV15 etched without more effect of the ions in the plasma, where surface roughness increased from less than $0.005 \text{ }\mu\text{m}$ to $0.055 \text{ }\mu\text{m}$ was obtained. Surface contamination of films prior to etching has significant influence on damage patterns after etching in the barrel etcher. With ion beam etching the contamination was removed during the etching process, so it was not detrimental to the final trench.

Low loss optical waveguide transmission is only possible in a polymer beam centre layer if the beam surfaces are smooth. The suitability of low surface roughness of etched patterned

waveguiding beams can be obtained using reactive plasma etching. Ions generated using barrel etcher, had effect on the surface roughness and was reduced significantly by a perforated shield around the film. Plasma was a suitable process of etching with desired roughness values obtained. Surface roughness of $0.002\text{ }\mu\text{m}$ was obtained for a $0.150\text{ }\mu\text{m}$ deep trench and $0.005\text{ }\mu\text{m}$ for a $0.700\text{ }\mu\text{m}$ deep trench.

Chapter 8

Conclusion and Future Research

8.1 Conclusion

Surface properties of crystalline and amorphous polymer and copolymer thin films have been investigated. Thickness variation has been shown to influence surface roughness and surface texture. The glass transition temperature and crystal size have shown variation with film thickness, although melting temperature remained unchanged. Solvent and substrate variation have been shown to influence copolymer surface texture.

Polycaprolactone (PCL) films of different thickness were prepared and used in this study and their investigation is presented in Chapter 4. The thickness range covered was from 0.07-16 μm . Different spinning speeds were used to prepare different thickness of films. Solutions of different concentration of the PCL were useful for preparing films of a wide range of thickness. The concentration range was 0.2-2.0 g.mL^{-1} . Comparison between the PCL surface roughness and film thickness has shown that the surface roughness significantly decreased when the film thickness was decreased. It was shown that the surface roughness increased when the concentration of PCL was increased. Spinning time apparently, covered no change in the film thickness... DSC measurements were made to investigate melting variation with film thickness. Melting temperature showed no change with film thickness. It was found that the thinner films had higher melting enthalpy. Crystal size showed significant variation with film thickness. It was found that thinner films had larger crystal size. Thin crystalline polymer films were found to have smooth surfaces and thicker films were found to have rougher surfaces.

Surface property variation has been investigated in Chapter 5 using thermal treatment of block copolymers. Two different block copolymers were used in this investigation, poly(styrene-*b*-butadiene-*b*-styrene) (SBS) and poly(styrene-*b*-isoprene-*b*-styrene) (SIS). The surface roughness of the SIS significantly varied when the copolymer film thickness was varied. It was shown that the thinner film of SIS had low roughness. The surface texture of the SIS changed when the heating temperature of the surface was changed. It was found that films heated at high temperatures such as at 120 and 160 °C, showed bulges when compared with films heated at a low temperature of 45 °C. The surface of SIS heated at 45 °C was smooth. Surface tension variation of SIS heated at 120 and 160 °C caused bulges of different size, height and width to fill the surface, which related to polymer separation. Different size, height and width of the bulges were found on SIS films of different thickness. Thicker films have shown bulges with increased height and with larger size.

SBS surface roughness variation was observed according to film thickness e.g. thinner film had lower roughness. Surface roughness of SBS films significantly varied with film thickness and heating treatment. Different surface textures of SBS films were obtained with films spun from solutions of different solvent (toluene and THF). A film that was spun from a toluene solution of SBS gave a smooth surface and a film spun from a THF solution of SBS gave a surface that was filled with ripples. SBS films spun on different substrates such as glass and silicon showed different texture and different thickness. Optical microscopy images of surface texture of different substrates containing the same copolymer showed different numbers of ripples per unit length. The number of ripples that was observed on the glass substrate was greater than the number that was obtained when using silicon as the substrate. It was found that the ripples in the films on the silicon were lower than those on the glass substrate.

In Chapter 6, amorphous polymer optical properties of absorption, transmission and glass transition temperature showed changes with film thickness. Surface roughness improved using films of optical polymers such as poly(methyl methacrylate) (PMMA), Disperse Red 1-poly(methyl methacrylate) (DR1-PMMA) and Urethane-Urea (UU).

An increase in PMMA concentration caused an increase in film thickness. The surface roughness of PMMA was found to vary with film thickness and decreased when the film thickness was decreased. Time of spinning showed no effect on the thickness and roughness of PMMA film. A thicker film of DR1-PMMA was obtained when compared with pure PMMA, because of DR1 concentration. The surface roughness of PMMA consistently decreased with film thickness but the DR1-PMMA films surface roughness showed less variation with film thickness. Poly(L-Lactic acid) (PLA) was used to investigate glass transition temperature variations with film thickness. The T_g measurements using DSC was shown to increase when film thickness was decreased.

NLO polymer, UU, was selected to be used in optical wave guide fabrication. Films of UU were prepared using substrates of silicon and glass. Light transmission quality using UU films was examined with a Metricon prism coupler, by measuring refractive indices at two different wavelengths, 632.8 nm and 810 nm at transverse electric mode and transverse magnetic mode beam polarizations. The films transmission and absorption spectra were measured with a spectrophotometer in the wavelength range 200-3200 nm. A maximum absorption at $\lambda_{\text{max}} = 471 \text{ nm}$ was observed and decay absorption coefficient was found to be equal to $3.1 \times 10^4 \text{ cm}^{-1}$ and this wavelength was strongly absorbed in the layer of the material up to several micron thick. The UU films failed to show waveguiding of He-Ne laser or the laser diode at 810 nm of the prism coupler. A lack of guiding of the light beam was caused by absorption at 810 nm

A fabrication technique was investigated for microchannels. A suitable technique for fabricating microfluidic channels in a thermoset polymer was developed and characterised to observe nano-particle flow. The fabrication technique included photolithography, plasma etching in a barrel reactor and thin film deposition using sputtering and evaporation. The technique developed was flexible with etch depths from 1 μm to 100 μm . The technique could be adapted for etching the microchannels for producing polymer optical waveguides.

The curing property variations of UV15, cured under different conditions, were investigated. The FTIR peak variation was obtained with curing time, over the range 1850-1700 cm^{-1} . The ester and carbon double bond regions were cured by changing the time of UV curing.

The Tg variation of UV15 films was obtained. The Tg of UV15 increased with both curing time and UV intensity and has shown an influence on the polymer optical property at high temperatures. The Tg had a higher value when exposed to the UV light for a longer period of curing. It was found that the Tg of UV15 cured with UV increased when heated and increased after was cured for 40 min at 120 $^{\circ}\text{C}$.

The UV15 exhibited surface flow giving wrinkled surfaces for films with low Tg when treated by plasma. It was observed that unheated UV15 (covered with a mask layer for pattern etching the film) caused a wrinkled surface to occur during the etching process in the barrel etcher, due to the heating effect of the plasma. The polymer etch mask was a thin film of aluminium deposited onto the polymer surface. As well as the plasma etch wrinkling of the unheated polymer; the aluminium deposition process could also cause wrinkling. This depended on whether the deposition process heated the polymer. There were no wrinkles on the surface with e-beam evaporated aluminium. When the aluminium was deposited by sputtering, then wrinkling could occur if the substrate became hot (depending on the intensity of the plasma). It was found when the UV15 was cured and then heated at 100 $^{\circ}\text{C}$ for 30 min;

this makes the film harder before placing it in the sputtering system for aluminium deposition, and it was found that the post cure was effective in enhancing chemical resistance, particularly to solvents.

Optical waveguides have been prepared and optimised using plasma etching of a photocured UV15 cladding layer on a silicon substrate. The guiding layer of PMMA was spin coated onto the cladding layer and into the etched channels. Relatively smooth interfaces were produced using a barrel etcher with oxygen plasma at 100-200 W and $400 \text{ cm}^3 \cdot \text{min}^{-1}$ for 15 min. Surface roughness of $0.002 \text{ }\mu\text{m}$ was obtained for a $0.15 \text{ }\mu\text{m}$ deep trench and $0.007 \text{ }\mu\text{m}$ for a $0.500 \text{ }\mu\text{m}$ deep trench. Introduction of oxygen increased the etch rate compared with argon. A typical trench in UV15 was measured using an AFM. The surface roughness of etched UV15 films increased with the effect of the ions of the plasma treatment.

8.2 Recommendations for Future Research

The technique of scanning the surface of polymer films using a stylus profilometer was beneficial, in terms of exploring a specific area on the polymer thin film. Films of block copolymers need more attention when phase separation phenomena shows on the surface. The stylus could be used to explore the block copolymer surface when it is covered with bulges or wrinkles that happen because of segregation of polymer to the surface. Wrinkles on the surface of block copolymers have been found to be of different sizes when the copolymer film was spun on different substrates. The surface of the copolymer after heating sometimes shows dark spots because of polymer separation. The diameter of dark spot, bulge or wrinkle could be measured in two dimensions using the stylus and by multiple measurements across the spot can help to create a topographic shape of the spot.

Surface roughness is crucial in optical polymer preparation and their applications such as waveguides. The optical microscope would be unable to explore the surface roughness of optical polymers because of their glassy behaviour. The profilometer technique is useful for measurement of the surface roughness of polymers. The roughness of optical polymers needs further attention. The measurements using the stylus could be cooperative in measuring the roughness variation after each stage of treatment of the optical thin film and after modification of T_g since these aspects have been published.

The glass transition of amorphous thermoset polymers needs further consideration and more work is needed on their modification to suit making optical devices. The T_g variation and control of this variation could be achieved. This will be important for the optical polymers, PMMA-DR, the urethane-urea and UV15, NOA73 and NOA61 for which T_g modification has not been reported.

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